Physics-Informed Neural Quantum States for Quantum Dots: Tangent-Space Geometry, Message Passing, and Wigner Molecules

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Abstract

How far can a compact, well-conditioned neural network go as a variational wavefunction for interacting electrons; what does letting the particles talk to one another actually buy; and can such a state be trained without a Markov chain? We take up these three questions for two-dimensional harmonic quantum dots—from the compact, kinetic-energy-dominated droplet at strong confinement to the dilute Wigner molecule at weak confinement—for $N \in \{2, 6, 12, 20\}$ electrons and trap frequencies ω from 1 down to 10^{-3}.

The ansatz is a Slater determinant dressed by a neural Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Both are made *safe by construction* for Laplacian-based training: coordinates are scaled to trap units, pair features are soft-core, and the electron–electron cusp is imposed analytically. Each learnable object comes in two forms—one that pools per-particle and pairwise features in a single pass, and one that passes messages between particles on a graph—and we read the trained states through the geometry of their log-derivatives, the quantum geometric tensor and the neural tangent kernel. Training is by residual pretraining followed by stochastic reconfiguration (SR), and, separately, by importance-sampled collocation that uses no Markov chain.

Message passing, it turns out, buys one thing: a *relational channel*. Seen through the tangent kernel it compresses the correlator onto roughly half as many effective directions as a single-pass network, and it keeps the backflow from collapsing: at the Wigner crystal a conventional backflow loses accuracy—its displacement field collapsing onto a single collective mode—while the message-passing backflow stays near reference quality. Underneath both is a single mechanism, and it is visible even at strong confinement where the two networks are identical in energy: message passing keeps the correlator's dominant tangent directions aligned with the physical collective modes of the Hamiltonian, whereas a separable network reaches the same energy while leaving much of its tangent capacity along directions a compact physical dictionary does not resolve—a deficit merely disguised at strong confinement and sharpening at weak. The correlator's loss of that alignment and the backflow's rank collapse set in at one confinement, pointing to a common relational bottleneck. The same geometry governs the rest. SR helps in proportion to how ill-conditioned the tangent metric is, but only while that metric can still be estimated reliably; once the sampling that builds it collapses, inverting it does harm. And importance-sampled collocation matches SR–VMC while its sampling stays healthy, reaching a genuine Wigner molecule at $\omega = 0.0035$ with no Markov chain in the optimisation. SR–VMC carries the $N = 20$ energies, while the Markov-chain-free route is demonstrated for $N \leq 12$. Energies match diffusion Monte Carlo within uncertainty for $N = 2$ and to $0.008-0.05\%$ for $N \in \{6, 12\}$ wherever a DMC reference exists ($\omega \geq 0.1$); at $N = 20$, beyond the published DMC grid, they match the tabulated reference energies of an earlier master's thesis to the same order. Below $\omega = 0.1$ no external benchmark exists for $N \geq 6$, and that boundary is marked wherever it occurs.

On the physics, the trained states let us map the Fermi-liquid-to-Wigner crossover from the inside. We give topology-resolved shell, bond-orientational, and stiffness diagnostics, with shell-mixture reconstructions of $g(r)$ matching the measured pair distribution to cosine similarity $0.989-1.000$: at $\omega = 10^{-3}$ the $N = 6$ dot is a single ring with a robust $(1, 5) \leftrightarrow (0, 6)$ near-degeneracy (radial mode $\approx 114\, a_0$), the $N = 12$ dot a two-shell molecule with a modal $(3, 9)$ split, and the $N = 20$ dot a three-shell molecule that collapses toward a single shell within one decade in ω. One thread runs through architecture, optimiser, and training paradigm alike: what matters is not raw capacity but staying aligned with the low-dimensional, physically collective structure of the problem—the same structure that, in real space, is the Wigner molecule.

To go wrong in one's own way is better than to go right in someone else's.

FYODOR DOSTOEVSKY
Crime and Punishment

Life must be understood backwards. But they forget the other proposition, that it must be lived forwards.

Søren Kierkegaard
Journals

Preface

When I came to UiO, Morten read my background and said: "You know PINNs, you know many-body quantum systems. I want you to do PINNs on quantum dots." I said "sure" and then asked the only sensible follow-up question I had: "Where do I start?"

I did not realise then that this was the question I had been assigned to answer. This thesis is the long answer. The short answer is: start by arranging to be wrong in a way that can be measured.

That is what variational methods make possible. The variational principle does not say how to improve a trial wavefunction, but it does guarantee that its energy lies above the ground-state energy. Improvement therefore has a direction: lower energy is better. For a project with few fixed landmarks, that was enough to begin. The harder task was learning what a disappointing result meant.

It was a solitary problem, not for want of literature—neural quantum states, FermiNet, PauliNet, and neural backflow offered much to learn from—but because the particular route I was trying to take had little precedent. When an experiment failed, it was rarely clear whether the method was unsound, the architecture inadequate, the optimiser poorly matched to the problem, or the code quietly broken. For long stretches, the honest answer was that I did not know. The work became an attempt to construct tests that could tell these explanations apart.

The most important lessons came from failures. An error in the importance-sampling code went undetected for months because its signature—loss of accuracy at larger particle numbers and weaker confinement—was exactly what an inadequate architecture might have produced. In Nd dimensions, a small error in a log-density grows linearly with dimension and is amplified in the normalised weights; the bug was invisible where experiments were cheap and ruinous where they mattered. Discovering it damaged my confidence in everything else. Re-establishing that confidence, rather than assuming it, became a substantial part of the later work.

Other failures taught the same lesson from different directions. A Langevin refinement step introduced bias into the gradient. A second-order optimiser became least reliable when its estimated metric was too noisy to resolve. Neither result was useful merely because it was negative. Each exposed an assumption that had been invisible until the computation violated it. Negative results of this kind are not a concession or an embarrassment; they are part of the method, and I have written them up as such.

That diagnostic stance also changed what I thought the network was for. I began by asking whether a machine-learning construction could solve the many-body Schrödinger equation. I ended up using the trained state as an apparatus: something to examine for the structure it had captured and the directions of variation that mattered. That shift, from treating a network as an answer to treating it as an object of study, is the thread running through this thesis.

The work also made a connection harder to ignore. Its obstacles have a double character: each is at once a physics problem and an optimisation problem. Stochastic reconfiguration, developed for variational wavefunctions, is natural gradient descent on the variational manifold. The REINFORCE estimator from reinforcement learning is the score-function gradient of the variational energy. The Fisher information matrix of statistics is the quantum geometric tensor of physics. The same mathematics appears under different names in different communities. Recognising that felt less like discovering a bridge between fields than like realising that the two fields had been approaching the same problem from opposite sides.

Looking back, it is easy to make a catalogue of what I would now do differently. Most of that knowledge, however, was purchased by the failures and could not have been read off in advance. The regret and the result are the same object seen from two sides. That is worth remembering, and easy to forget.

The most consistent source of inspiration, motivation, and steadiness through all of it has been my girlfriend, Elida. Her support made it possible to begin this work and to stay with it when it became difficult. I am more grateful than I can put down here.

I owe a warm and sincere thanks to my family, who have supported me in every undertaking I have had and have simply always been there; and to my friends, inside CCSE and out, who supplied the joy, the laughter, and a good deal of the sanity this project required.

To Morten, again, and properly: thank you for the assignment, for the two years, and above all for a genuine and undisguised interest in what I was finding at every stage—including the stages where what I was finding was that I had been wrong. That interest is most of what encouragement actually consists of.

Which is, at last, the answer to the question I asked on the first day. You start by being willing to be wrong on the record, and then you keep the record.

Contents

Part I

Introduction

The world was to me a secret which I desired to divine.

MARY SHELLEY

Frankenstein; or, The Modern Prometheus

Introduction

A system of many interacting electrons is described completely by its wavefunction. The same object puts the system almost out of reach, because a complete description of N interacting particles is a function of all their coordinates at once, and the space of those coordinates grows with every particle added. The equation that fixes this wavefunction is known exactly; the entire difficulty is solving it.

The competition that makes it hard is physical. Kinetic energy tends to delocalise the particles; interactions and external potentials drive localisation and correlation; and the balance between them plays out in a configuration space far too large to enumerate. Yet the quantities we need—energies, correlation functions, phase boundaries—are precisely the ones that depend on getting that balance right. This is the *many-body problem*, the point at which quantum mechanics turns from a compact set of principles into one of the hardest computational challenges in physics.

A wide range of electronic-structure methods has been developed to address this. Mean-field approaches such as Hartree–Fock replace the full many-body problem with an effective single-particle picture, where each electron moves in an average field from the others. This captures Fermi statistics and some exchange effects but leaves most correlation physics untreated. More systematic methods such as Full Configuration Interaction (FCI) are, in principle, exact within a finite basis, but their cost grows combinatorially with system size. Coupled Cluster (CC) theory provides a more efficient and size-extensive hierarchy of approximations, but it remains computationally demanding and can struggle in strongly correlated regimes or near degeneracies.

Quantum Monte Carlo (QMC) methods, including Variational Monte Carlo (VMC) and Diffusion Monte Carlo (DMC), take a different approach: they represent the wavefunction implicitly through stochastic sampling. For suitable trial states, they can produce highly accurate benchmark energies and correlations for continuum systems. However, QMC is not free of difficulties. The fermion sign problem limits direct simulations, fixed-node approximations inherit bias from the chosen trial nodes, and constructing high-quality trial wavefunctions is increasingly difficult as systems become more strongly correlated or inhomogeneous. In practice, then, much of the many-body problem reduces to a narrower one: choosing an expressive, well-conditioned ansatz and optimising it reliably.

Machine learning and many-body wavefunctions. In parallel with these developments, machine learning—especially deep learning—has become a standard tool for representing complex, high-dimensional functions and fitting them to data. Neural networks now play a central role in domains such as computer vision, language modelling and reinforcement learning. They combine flexible parametric families with inductive biases (e.g. symmetry and locality), normalisation, and optimisation procedures that scale well with dimension.

In physics, this has led to two broad uses of neural networks. In the first, they act as *surrogates*: they learn mappings from parameters or boundary conditions to observables, or approximate solution operators of partial differential equations. Physics-informed neural networks (PINNs) and operator-learning approaches fall into this category by embedding differential operators and constraints directly into the training objective. In the second, neural networks are used as *variational ansätze* for quantum states (neural quantum states, NQS), where the network outputs amplitudes, phases or log-probabilities for many-body configurations and is trained by variational Monte Carlo.

One argument for this second use is routinely overstated, and disposing of it first clarifies what remains. Universal approximation results guarantee that a sufficiently large network can *represent* any suitably regular function to arbitrary accuracy. They say nothing about whether gradient-based optimisation will *find* that representation, nor at what sample cost. This thesis measures that gap directly: the same ansatz, demonstrably able to express the ground state, is placed in training schemes that do and do not allow it to get there. Representability is cheap; trainability is the difficulty.

A weaker but sounder argument survives that objection. Established methods pay for accuracy in one of two currencies. FCI is exact within a finite basis, but its cost grows combinatorially with that basis, so a modest two-dimensional dot of twenty electrons is already far beyond reach. Cheaper methods such as coupled cluster buy their favourable scaling instead by deciding in advance which correlations may be described at all.

A variational neural ansatz pays in neither currency. It is not an expansion whose size must grow with the space it covers, and it fixes no prior truncation on which correlations are available. It is a parameterisation free, in principle, to concentrate its capacity wherever the solution actually varies. Whether it does so in practice is an empirical question, and this thesis treats it as one.

That question also suggests a different way to use a solver. Nearly every established method for the many-body Schrödinger equation is constructed outward from physical principles: an approximation is chosen, justified physically, and then implemented. A trained network inverts the order. It is obtained largely on machine-learning terms, and only afterwards interrogated as to what it has represented—and unlike an implicit stochastic representation, it is a concrete parameterised object whose internal structure can be examined directly. This motivates a question that the size of the problem does not by itself settle. A twenty-electron dot is a function of forty coordinates, and any basis expansion of it grows combinatorially—but it does not follow that the solution itself is forty-dimensional in any useful sense. How many directions does the *solved state* actually vary along? Not the dimension of the equation, but the dimension of the answer.

We make that question quantitative through the geometry of the wavefunction's log-derivatives. Perturbing the parameters of a trained ansatz over an ensemble of configurations and collecting the resulting sensitivities into a covariance matrix yields, in the simplest reading, a principal-component analysis of the state: the eigenvalues report how much variation each independent direction carries, and their distribution counts how many directions are doing genuine work. The physically appropriate version of this covariance is the quantum geometric tensor (equivalently, the Fisher metric), and the corresponding count is the participation ratio of its spectrum—an effective dimension rather than a hard rank, insensitive to the raw parameter count. Chapter 7 develops this instrument and applies it to architecture, optimiser, and training paradigm alike; the questions it answers are how this effective dimension behaves as particles are added and as the trap is opened, and whether different architectures arrive at the same solution by different routes.

In principle, then, neural networks provide highly expressive ansätze for quantum problems. In practice, several issues arise. Objectives based on local energies involve second derivatives and can be numerically delicate. Coordinate parameterisations that do not respect known structure can lead to large gradients and Laplacian spikes near electron coalescence. Generic architectures can waste capacity on degrees of freedom that are irrelevant from a physical point of view. These challenges motivate hybrid approaches in which as much known structure as possible—antisymmetry, cusp conditions, scaling, and symmetries—is built into the ansatz and its coordinates, and the network is used to learn the remaining correlation corrections.

A further ingredient concerns not *what* structure is imposed but *how* the network represents the electrons. Correlation is intrinsically relational—what one electron should do depends on where the others are—so architectures that let particles exchange information (message passing), rather than processing each coordinate independently, are a natural candidate for the correlator and for the backflow that deforms the nodes. How much this relational capacity actually buys, and why, is one of the questions this thesis takes up.

Where this work sits. Representing a quantum state by a neural network and optimising it variationally was introduced by Carleo and Troyer [1]; for continuum fermions the idea has since matured into highly accurate *ab initio* solvers such as FermiNet [2] and PauliNet [3], and into explicit neural *backflow* transformations [4]—the direct ancestor of the coordinate backflow used here. Message passing, the relational mechanism at the centre of this thesis, has recently been built into neural quantum states for extended fermion systems [5, 6]. Against that backdrop the present work is deliberately narrow and complementary. Rather than pushing a new architecture to ever larger systems, it asks *why* the relational channel helps, and uses the tangent-kernel geometry to make the answer precise; it focuses on the two-dimensional quantum-dot Wigner regime, where correlation is strongest and where topology-resolved structural diagnostics remain sparse; and it treats conditioning, the optimiser, and Markov-chain-free training as objects of study in their own right rather than fixed choices. To our knowledge these questions have not previously been addressed together for parabolic quantum dots.

Quantum dots as a controlled testbed. Two-dimensional parabolic quantum dots provide a clean and tunable setting in which to explore such ideas. They offer a simple route from Fermi-liquid–like behaviour to Wigner-molecule physics. As the trap frequency ω decreases, the single-particle level spacing collapses while Coulomb repulsion becomes dominant. Electrons then tend to localize into ring or shell patterns. In the lab frame, rotational symmetry and quantum fluctuations smear these into smooth densities; in a co-rotating frame, one recovers crystalline-like structures whose topology depends on particle number. For $N{=}6$ and $N{=}12$, classical and semiclassical studies have identified preferred shellings such as $(1,5)$ and $(3,9)$ and mapped parts of the crossover [7–10].

From the perspective of this thesis, quantum dots are attractive for two reasons. First, they are genuinely non-trivial: Coulomb singularities, strong correlations and an actual Fermi-liquid–to–Wigner crossover appear already for moderate N. Second, they are still simple enough that accurate DMC references exist in many regimes and that detailed structural diagnostics (shell occupancies, bond order, virial ratios) are computationally feasible. For fully quantum dots, however, quantitative topology-resolved diagnostics (occupancy fractions, bond-orientational order, shell "stiffness") and a systematic picture of the Wigner regime remain relatively sparse. In addition, it is not well understood how modern neural ansätze internally represent these states.

Scope and questions. In this thesis we revisit two-dimensional harmonic quantum dots with a Slater–Jastrow(+backflow) ansatz implemented as a physics-informed neural network. The design is meant to be *safe by construction* for Laplacian-based objectives: coordinates are scaled to trap units ($\tilde{R} = \sqrt{\omega}R$), learned pair features are soft-core (avoiding spurious $1/r$ behaviour in derivatives), and the electron–electron cusp is enforced analytically rather than learned. On top of a Slater determinant of harmonic-oscillator orbitals, two learnable objects carry the correlation: a Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Each can be built either without message passing—encoding and pooling per-particle and pairwise features in a single pass, as in a DeepSet—or as a *message-passing* network in which the particles exchange and update information through explicit, transported edge state before a quantity is read out.

Once such an ansatz reaches reference accuracy—which, as we will see, a surprisingly compact one does—the questions worth a thesis are *why* its parts behave as they do, and what it can tell us about the electrons. We pursue four. The first three we read through a single lens: the geometry of the wavefunction's log-derivatives, captured by the quantum geometric tensor and the neural tangent kernel.

1. **Architecture.** What does message passing buy that a separable network cannot? We find it supplies a *relational channel*—a way to condition each particle on where the others are—that compresses the correlator onto fewer effective directions, keeps the correlator's leading tangent modes aligned with the physical collective modes of the Hamiltonian, and keeps the backflow from collapsing to a single collective mode in the strongly correlated limit. A separable network, by contrast, *estimates* the correlation energy without *representing* it—a deficit already measurable at strong confinement, where the energies are identical. The correlator's loss of physical alignment and the backflow's rank collapse set in at the same confinement and shift together when the correlator is swapped, supporting a common relational bottleneck in the amplitude and nodal channels.

2. **Optimiser.** When does natural-gradient (stochastic reconfiguration) preconditioning actually change the result? Its value tracks the conditioning of the tangent metric: negligible for well-conditioned, $|\Psi|^2$-sampled VMC, decisive for the ill-conditioned collocation objective.

3. **Paradigm.** Can an accurate wavefunction be trained *without* Markov-chain Monte Carlo, using importance-sampled collocation instead—and how far into the correlated regime does that reach before the sampling loses the target?

4. **Physics.** Can the trained wavefunction serve as a scientific instrument for Wigner-molecule physics—shell topologies, angular order, and melting—and does its internal representation reflect that physics?

These are not four separate stories. The same low-dimensional, physically collective structure that message passing captures is what natural gradients exploit, what $|\Psi|^2$ sampling stays aligned with, and what ultimately crystallises into the Wigner molecule itself.

Diagnostics and relation to prior work. To address these questions, we go beyond total energies. Alongside DMC benchmarks, we use:

- automatic shell detection and shell-resolved reconstructions of the pair distribution $g(r)$, testing whether the global $g(r)$ is explained by shell geometry and combinatorics;
- bond-orientational order parameters $|\Phi_m|$ and angular Lindemann-like ratios, adapted from studies of classical Wigner and Yukawa layers [11], to quantify ring order and angular stiffness;
- the density parameter r_s, estimated from the first peak of $g(r)$ following Egger *et al.* [7], to locate our dots along the Fermi-liquid–to–Wigner scale;
- representation-focused diagnostics for the neural ansatz: entropy effective ranks of correlator and backflow features, principal component ablations, and simple linear probes that relate latent axes to physical observables (global size, radial variance, near-origin mass);
- tangent-kernel diagnostics of the ansatz itself: the quantum geometric tensor and neural tangent kernel, the effective dimension of the variational tangent space, and symmetric state overlaps used to compare architectures and training paradigms on an invariant footing.

All of these are applied as post-processing to $|\Psi|^2$ samples, so the same analysis could, in principle, be used to study classical, quantum-chemical, or other neural wavefunctions on the same footing.

Main contributions. The main contributions of this thesis are:

1. **Compact, conditioned accuracy.** A small Slater–Jastrow(+backflow) ansatz in trap units reaches DMC-level energies for N=2 and holds $\sim 10^{-2}$–10^{-1}% relative errors for $N \in \{6, 12, 20\}$ across $\omega \in \{0.1, 0.5, 1.0\}$, extrapolating smoothly to ultra-weak traps ($\omega = 10^{-3}$). Conditioning—trap units, analytic cusp, safe pair features—matters at least as much as network size for stability and accuracy in these dots.
2. **Message passing as a relational channel, made precise by the tangent kernel.** Read through the quantum geometric tensor, a message-passing correlator represents the ground state in roughly half as many effective tangent directions as a separable one, and a message-passing backflow retains full rank exactly where a conventional, single-round-message backflow collapses to a single collective mode at the Wigner crystal and loses accuracy. The two effects point to a common relational mechanism—a channel the physics requires—and both are invisible in the total energy until weak confinement turns the geometric deficit into an energetic one.
3. **Optimiser and paradigm, governed by the same geometry.** Natural-gradient (SR) preconditioning helps precisely when the tangent metric is ill-conditioned: negligibly for $|\Psi|^2$-VMC, decisively for collocation. And an MCMC-free, importance-sampled collocation objective matches VMC when the sampling is healthy and, with a physics-informed proposal built from the Wigner shell, reaches a genuine Wigner molecule at $\omega = 0.0035$ with no Markov chain in the gradient training (a Metropolis probe is used only for checkpoint selection and final certification).
4. **Topology-resolved Wigner diagnostics.** Across $\omega \in [1, 10^{-3}]$ we quantify the crossover using two-shell probabilities and inner-ring occupancy histograms, shell-mixture reconstructions of $g(r)$ with cosine similarity 0.989–1.000 to global data, and sector-resolved bond order and Lindemann indices. At $\omega = 10^{-3}$ the N=6 dot forms a persistent single-ring Wigner molecule with a clear $(1, 5) \leftrightarrow (0, 6)$ near-degeneracy, while the N=12 dot forms a two-shell molecule with a commensurate $(3, 9)$ split as the dominant topology, complementing classical and quantum studies of parabolic dots [7–9, 12].
5. **Representation insight, with its gauge explicitly drawn.** The correlator features occupy a low-dimensional manifold (one to three principal directions) whose leading axes act as collective coordinates for global size and radial fluctuations, with a $\phi \leftrightarrow \psi$ recomposition across regimes, while backflow spans many local modes. We take care to separate the parts of this picture that are *gauge-like*—fixed by how correlation is divided between the Jastrow and the backflow, two substitutable channels—from the invariants (energy, state overlap, and the energetic price of ablating a component) on which the architectural conclusions actually rest.

Thesis structure. The remainder of the thesis is organised as follows. The next chapter sets out the theoretical background: the many-body Schrödinger equation in harmonic traps, the electronic-structure methods we benchmark against, the physics of Wigner molecules, and the machine-learning tools used here—neural function approximators, physics-informed objectives, neural quantum states, and the tangent-kernel geometry (quantum geometric tensor and neural tangent kernel) that serves as the analytical lens for the results. A method chapter then presents the concrete Slater–Jastrow(+backflow) PINN ansatz in its separable and message-passing forms, the optimisation pipeline (residual pretraining, stochastic reconfiguration, and importance-sampled collocation), and the diagnostics used throughout.

The results are given in two chapters that do different jobs. The first establishes what we obtained: the energies against every available reference, the Wigner-molecule physics the trained states let us see, and the internal representation that carries it. The second asks why those choices worked and where they stop, reading architecture, conditioning and sampling through the geometry of the wavefunction's log-derivatives, and taking the Markov-chain-free training programme from its results to its two walls. A discussion then asks what all of it establishes, and a conclusion closes the loop from the physics back to the method. The appendices develop the conditioning theory (Laplacian and Coulomb), the collocation–backflow catch-22 that motivated the weak-form objective, and the full structural diagnostic tables.

Part II

Theory

Men do not know how what is at variance agrees with itself. It is an attunement of opposite tensions, like that of the bow and the lyre.

HERACLITUS
Fragments

Chapter 1

Quantum Theory

This chapter lays the foundation on which the rest of the thesis rests. We first introduce the mathematical framework of quantum mechanics, then discuss the challenges of many-body systems and how second quantization provides a more tractable approach. We then introduce the model systems used throughout this thesis to benchmark and test our methods, and present the Hartree–Fock method, a fundamental mean-field approach to approximating many-body wavefunctions. Finally, we discuss Full Configuration Interaction (FCI), which provides an exact solution to the many-body problem within a finite basis set, and Variational Monte Carlo (VMC), a stochastic method for approximating ground-state properties. The following chapter then shifts to machine learning—statistical learning theory, optimisation, neural networks, and physics-informed neural networks.

1.1 Mathematical Foundations

At the centre of quantum mechanics lies a rich mathematical framework built upon linear algebra and functional analysis. This section introduces the essential mathematical concepts and notations that underpin quantum theory, including Dirac notation, operators, Hilbert spaces, and the representation of quantum states and observables.

1.1.1 Hilbert Spaces and Function Representations

In essence, quantum mechanics is formulated in complex vector spaces—specifically *Hilbert spaces*: complex inner-product spaces that are *complete* in the norm induced by their inner product, so that every Cauchy sequence converges within the space and limits and basis expansions are well defined [13]. For square-integrable functions the inner product is

$$\langle\phi|\psi\rangle = \int \phi^*(x)\,\psi(x)\,dx,$$

which, in the physics convention (linear in the second slot), obeys conjugate symmetry $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$, linearity in the ket $\langle\phi|\alpha\psi_1 + \beta\psi_2\rangle = \alpha\,\langle\phi|\psi_1\rangle + \beta\,\langle\phi|\psi_2\rangle$, and positivity $\langle\psi|\psi\rangle \geq 0$ (with equality iff $|\psi\rangle = 0$); it induces the norm $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$. An orthonormal basis $\{\psi_i\}$ satisfies $\langle\psi_i|\psi_j\rangle = \delta_{ij}$. A function $a(x)$ expands in a basis $\{\psi_i(x)\}$:

$$a(x) = \sum_i \psi_i(x)a_i, \quad a_i = \int \psi_i^*(x)a(x)dx.$$

The Dirac delta $\delta(x-y)$ replaces the Kronecker delta in continuous spaces. A vector $|a\rangle$ in an n-dimensional Hilbert space $\mathcal{H}$ can be decomposed into a complete orthonormal basis $\{|i\rangle\}$:

$$|a\rangle = \sum_i |i\rangle \langle i|a\rangle = \sum_i a_i |i\rangle ,$$

where $a_i = \langle i|a\rangle$ are complex coefficients. The completeness relation,

$$I = \sum_i |i\rangle \langle i| ,$$

ensures the basis spans $\mathcal{H}$. Vectors in Dirac notation have dual "bra" counterparts $\langle a| = |a\rangle^\dagger$, enabling inner products $\langle a|b\rangle$ and outer products $|a\rangle \langle b|$.

1.1.2 Operators and Matrix Representations

Linear operators $\hat{\mathcal{O}}$ map vectors to vectors: $\hat{\mathcal{O}} |a\rangle = |b\rangle$. Their adjoints $\hat{\mathcal{O}}^\dagger$ satisfy $\langle a| \hat{\mathcal{O}}^\dagger = \langle b|$. Hermitian operators ($\hat{\mathcal{O}}^\dagger = \hat{\mathcal{O}}$) represent observables, while unitary operators ($\hat{\mathcal{O}}^\dagger = \hat{\mathcal{O}}^{-1}$) preserve inner products [14, 15]. In a basis $\{|i\rangle\}$, operators are represented by matrices:

$$O_{ij} = \langle i| \hat{\mathcal{O}} |j\rangle .$$

Non-commutativity of operators is quantified by the commutator:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

1.1.3 States in Hilbert Space

A (pure) quantum state is a normalised vector $|\psi\rangle \in \mathcal{H}$ (global phase is physically irrelevant). Computations proceed by choosing an orthonormal (ON) basis $\{|i\rangle\}_{i=1}^{\infty}$ and expanding

$$|\psi\rangle = \sum_i c_i |i\rangle , \qquad c_i = \langle i|\psi\rangle , \qquad \sum_i |c_i|^2 = \|\psi\|^2.$$

Completeness is the resolution of the identity, $\sum_i |i\rangle\langle i| = \mathbb{I}$. For continuous ON bases (e.g. position), $\int |x\rangle\langle x|\, dx = \mathbb{I}$ and the wavefunction is the coordinate representation $\psi(x) = \langle x|\psi\rangle$.

Different ON bases are related by a unitary change of basis. Given $\{|i\rangle\}$ and $\{|\alpha\rangle\}$,

$$|\alpha\rangle = \sum_i U_{\alpha i} |i\rangle , \qquad U_{\alpha i} = \langle \alpha|i\rangle , \qquad U^\dagger U = \mathbb{I}.$$

This freedom is exploited to match the physics (symmetry, locality, boundary conditions), e.g. harmonic-oscillator orbitals for confined charges or plane waves for periodic systems, leading to sparser representations and faster convergence.

For N particles the Hilbert space factorizes as $\mathcal{H}_N = \mathcal{H}_1^{\otimes N}$. With a single-particle ON set $\{|\varphi_p\rangle\}$, a convenient fermionic ON basis is the set of Slater determinants $\{|\Phi_I\rangle\}$ built from N distinct orbitals; any N-body state expands as

$$|\Psi\rangle = \sum_I C_I |\Phi_I\rangle , \qquad \sum_I |C_I|^2 = \|\Psi\|^2.$$

In practice one truncates to a finite active space guided by energy scales and symmetry quantum numbers, which makes many-body calculations tractable while retaining the relevant physics.

1.2 Quantum Mechanics

1.2.1 The Schrödinger Equation

In a Hilbert space $\mathcal{H}$, the time evolution of a normalised state $|\Psi(t)\rangle$ is postulated to be unitary and generated by the Hamiltonian H [14, 15]:

$$i\, \partial_t |\Psi(t)\rangle = H |\Psi(t)\rangle , \qquad (\hbar = 1).$$

If H is time–independent, solutions separate as

$$|\Psi(t)\rangle = \sum_n c_n \, e^{-iE_n t} \, |\psi_n\rangle \, , \qquad H \, |\psi_n\rangle = E_n \, |\psi_n\rangle \, ,$$

so $\{|\psi_n\rangle\}$ forms an orthonormal basis of stationary states.

For an N-particle system in d spatial dimensions, with coordinates $R = (\mathbf{r}_1, \ldots, \mathbf{r}_N)$, we use the many-body Hamiltonian

$$H = -\frac{1}{2} \sum_{i=1}^{N} \nabla^2_{\mathbf{r}_i} \; + \; \sum_{i=1}^{N} V_{\text{ext}}(\mathbf{r}_i) \; + \; \sum_{i<j} V_{\text{int}}(|\mathbf{r}_i - \mathbf{r}_j|),$$

and the time-independent Schrödinger equation (TISE)

$$H \, \Psi(R) = E \, \Psi(R), \qquad \int |\Psi(R)|^2 \, dR = 1.$$

Identical particles restrict Ψ to the appropriate symmetry sector (symmetric for bosons, antisymmetric for fermions).

Throughout we work in atomic units ($\hbar = m_e = e = 1$) and focus on stationary problems. In the quantum-dot applications below we take $V_{\text{ext}}(\mathbf{r}) = \frac{1}{2}\omega^2 r^2$ and $V_{\text{int}}(r) = 1/r$ (in 2D this is understood with the usual Laplacian in two dimensions). Our computations therefore amount to approximating low-lying eigenpairs (E, Ψ) of the many-body Hamiltonian above using compact basis expansions and variational ansätze.

1.3 Second Quantization

1.3.1 Motivation: Why first-quantized wavefunctions don't scale

In first quantization an N-fermion state is an antisymmetric function $\Psi(x_1, \ldots, x_N)$ expanded in Slater determinants of M one-particle orbitals. The number of configurations grows as $\binom{M}{N}$ (and permanents for bosons grow even faster), making storage and operator application increasingly intractable. We therefore move to the occupation-number formalism, which encodes antisymmetry and particle indistinguishability algebraically.

1.3.2 Fock space and occupation representation

Let $\{|p\rangle\}$ be an orthonormal set of one-particle *spin-orbitals* (spatial orbital + spin). The fermionic Fock space is

$$\mathcal{F}_- \; = \; \bigoplus_{n=0}^{\infty} \wedge^n \mathcal{H}_1,$$

with vacuum $|0\rangle$ and occupation-number basis $|n_1 n_2 \ldots\rangle$, $n_p \in \{0, 1\}$. Creation/annihilation operators $a_p^\dagger, a_p$ act by adding/removing a particle in $|p\rangle$ (subject to Pauli) and obey the canonical anticommutation relations (CAR)

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \qquad \{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0,$$

with number operator $\hat{n}_p = a_p^\dagger a_p$. A Slater determinant occupying the set $I = \{p_1 < \cdots < p_N\}$ is $|\Phi_I\rangle = a_{p_1}^\dagger \cdots a_{p_N}^\dagger \, |0\rangle$ and is represented by the bitstring $|n_1 n_2 \ldots\rangle$ with $n_p = \mathbf{1}_{p \in I}$. Unitary changes of one-particle basis $|p'\rangle = \sum_p U_{p'p} \, |p\rangle$ lift to $a_{p'}^\dagger = \sum_p U_{p'p} a_p^\dagger$, so physics is basis independent.

1.3.3 Operators in second quantization

Any *one-body* operator $O = \sum_i O(i)$ has matrix elements $O_{pq} = \langle p| \, O \, |q\rangle$ and becomes

$$O \; = \; \sum_{pq} O_{pq} \, a_p^\dagger a_q.$$

A *two-body* interaction $V = \frac{1}{2}\sum_{i\neq j} v(i,j)$ has

$$V \;=\; \frac{1}{2}\sum_{pqrs}\langle pq|v|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r \;=\; \frac{1}{4}\sum_{pqrs}\langle pq\|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r,$$

where $\langle pq|v|rs\rangle$ are two-electron integrals over spin-orbitals and $\langle pq\|rs\rangle = \langle pq|v|rs\rangle - \langle pq|v|sr\rangle$ denotes the antisymmetrized integral. In this form, operator algebra and particle indistinguishability are handled by the CAR, independent of N.

1.4 Hartree–Fock (HF)

Hartree–Fock (HF) approximates the N-fermion ground state by a single Slater determinant built from orthonormal spin–orbitals $\{|p\rangle\}_{p=1}^{M}$. In second quantization the Hamiltonian is

$$H = \sum_{pq} h_{pq}\, a_p^\dagger a_q + \frac{1}{4}\sum_{pqrs}(pq\|rs)\, a_p^\dagger a_q^\dagger a_s a_r,$$

with one-body matrix elements $h_{pq} = \langle p|\hat{h}_0|q\rangle$ and antisymmetrized two-body integrals $(pq\|rs) = (pq|rs) - (pq|sr)$ over spin–orbitals.

1.4.1 HF energy functional and the Fock operator

Let $|\Phi\rangle = \prod_{i=1}^{N} a_i^\dagger\,|0\rangle$ be a Slater determinant (occupied spin–orbitals $i, j, \ldots$). Its energy is

$$E[\Phi] = \sum_i h_{ii} + \frac{1}{2}\sum_{i,j}(ij\|ij).$$

Stationarity of $E[\Phi]$ under unitary rotations in the occupied+virtual subspace gives the HF equations

$$F\,|\phi_p\rangle = \varepsilon_p\,|\phi_p\rangle\,, \qquad F = \hat{h}_0 + \sum_{j\in\mathrm{occ}}\Big(J_j - K_j\Big),$$

where J_j and K_j are the Coulomb and exchange operators generated by orbital j. In a fixed orthonormal one-particle basis $\{|\mu\rangle\}$ these become the matrix *Roothaan–Hall* equations

$$\mathbf{F}\,\mathbf{C} = \mathbf{C}\,\boldsymbol{\varepsilon}, \qquad F_{\mu\nu} = h_{\mu\nu} + \sum_{\lambda\sigma} P_{\lambda\sigma}\Big[(\mu\nu|\lambda\sigma) - \tfrac{1}{2}(\mu\lambda|\nu\sigma)\Big],$$

with the closed-shell density matrix $P_{\lambda\sigma} = 2\sum_{i\in\mathrm{occ}} C_{\lambda i}C_{\sigma i}^{*}$ (sum over occupied *spatial* orbitals; the factor 2 is the double occupancy that makes the $\frac{1}{2}$ exchange coefficient correct). (For a non-orthogonal basis S, use $\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon}$.)

Self-consistency (SCF). HF is solved iteratively: initialise $P^{(0)}$, build $\mathbf{F}[P^{(n)}]$, diagonalize to get $\mathbf{C}^{(n+1)}$, update $P^{(n+1)}$, and iterate until the total energy and/or P change is below tolerance. Damping or DIIS mixing stabilizes convergence. The dominant costs are building two-electron contributions ($\mathcal{O}(M^4)$) and diagonalization ($\mathcal{O}(M^3)$).

Remarks for quantum dots. For parabolic confinement $V_{\mathrm{ext}} = \frac{1}{2}\omega^2 r^2$ we typically choose 2D harmonic-oscillator orbitals as the one-particle basis (the Cartesian orbitals used in our implementation), which respect the shell structure and accelerate convergence. Closed shells admit restricted HF (RHF); spin-polarized cases may require UHF. Koopmans' theorem (orbital energies as removal/addition estimates) can provide a rough diagnostic but is not used for final energetics here.

1.5 Full Configuration Interaction (FCI)

FCI gives the exact ground state *within the chosen one-particle basis* by expanding in *all* N-electron Slater determinants,

$$|\Psi\rangle = \sum_I C_I \, |\Phi_I\rangle \, , \qquad H\mathbf{C} = E\,\mathbf{C}.$$

The determinant space has dimension $\binom{M}{N}$ for fixed spin–orbitals M and particles N (or smaller with symmetry constraints). FCI thus captures the full correlation energy $\Delta E = E_{\mathrm{FCI}} - E_{\mathrm{HF}}$ in that basis and serves as the gold-standard benchmark.

1.5.1 Slater–Condon rules and sparsity

Matrix elements $\langle \Phi_I | H | \Phi_J \rangle$ are nonzero only if Φ_I and Φ_J differ by at most two spin–orbitals:

- *Diagonal:* $H_{II} = \sum_{i \in I} h_{ii} + \frac{1}{2} \sum_{i,j \in I} (ij \| ij)$.
- *Singles* $i \to a$*:* $H_{0,i^a} = F_{ia}$; in canonical HF orbitals, $F_{ia} = 0$ (Brillouin's theorem), so the HF determinant does not couple to singles.
- *Doubles* $ij \to ab$*:* $H_{0,ij^{ab}} = (ij \| ab)$.

This locality yields a very sparse Hamiltonian in the determinant basis and enables iterative eigensolvers without forming $\mathbf{H}$ explicitly.

1.5.2 Diagonalization and symmetry reduction

In practice one targets the lowest eigenpairs with Davidson or Lanczos iterations. Symmetries (N, S_z, total spin S if spin-adapted CSFs are used, and L_z in an angular-momentum-adapted basis) block-diagonalize the problem and dramatically shrink the working space and memory. For few-electron quantum dots (e.g. $N = 2, 6$) this makes high-accuracy FCI feasible for moderate M; for larger N the determinant growth becomes the bottleneck.

1.5.3 Role in this thesis

These methods set the vocabulary rather than supply the numbers. HF fixes what *correlation energy* means—the part of the exact energy a single determinant cannot reach—and FCI names the exact answer within a finite one-particle basis, so that "0.05% from the reference" has a scale attached to it. We compute neither here. Our determinant is built from harmonic-oscillator orbitals rather than HF orbitals, and our external benchmark is diffusion Monte Carlo [16], which for these systems is both more accurate than an affordable FCI and available at the particle numbers we care about. The reader should carry forward the definitions, not expect the tables.

1.6 Quantum Dots

1.6.1 Model and units

We consider N spin-$\frac{1}{2}$ fermions in d=2 with harmonic confinement and Coulomb repulsion,

$$H \;=\; \sum_{i=1}^{N} \Big(-\tfrac{1}{2}\nabla^2_{\mathbf{r}_i} + \tfrac{1}{2}\omega^2 r_i^2 \Big) \;+\; \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \qquad (\hbar = m_e = e = 1).$$

Spin does not appear in H (spin–independent), so the many-body eigenstates can be chosen as products of a spatial wavefunction and a spin function with well-defined S and S_z.

1.6.2 One–particle structure and shell filling

We use two-dimensional harmonic-oscillator spin–orbitals. In the implementation these are the *Cartesian* 2D-HO eigenstates $\{\phi_{n_x n_y}(\mathbf{r})\chi_\sigma\}$ with energies $E_{n_x n_y} = \omega(n_x+n_y+1)$ and spin $\sigma \in \{\uparrow,\downarrow\}$. The principal index $k = n_x + n_y$ labels a shell of $(k+1)$ degenerate spatial states, each twofold spin-degenerate; for the closed shells studied here this Cartesian basis spans the same filled subspace as any complete basis of those shells, so the reference determinant is basis-independent. Filling from the bottom yields the *closed-shell* electron numbers

$$N_{\text{closed}} = (K+1)(K+2), \quad K = 0, 1, 2, \dots \quad (2, 6, 12, 20, 30, \dots).$$

All other N correspond to a *partially filled* top shell (open shell), with a degenerate noninteracting manifold.

1.6.3 Spin structure and Slater form

Write single-particle spin-orbitals as $\varepsilon_\alpha(x) = \phi_\alpha(\mathbf{r})\,\chi_\sigma(s)$, with $x = (\mathbf{r}, s)$. Because $\chi_\uparrow$ and $\chi_\downarrow$ are orthogonal and H is spin-independent, the Slater determinant simplifies:

Closed shells (RHF). For even N with doubly occupied spatial orbitals $\{\phi_1, \dots, \phi_{N/2}\}$, the determinant factorizes into two identical spatial determinants, one for each spin:

$$\Psi_{\text{closed}}(X) = \frac{1}{(N/2)!}\ \det[\phi_a(\mathbf{r}_i)]_{i\in\uparrow, a=1..N/2}\ \det[\phi_a(\mathbf{r}_j)]_{j\in\downarrow, a=1..N/2}\ \times\ \Xi_{S=0,S_z=0},$$

where Ξ is the singlet spin function. This is the *restricted* HF (RHF) reference and a natural starting point for correlated ansätze.

1.6.4 Kato's Cusp Conditions (2D Coulomb systems)

For Coulomb interactions the potential diverges as $1/r$ at coalescence. Finiteness of the local energy

$$E_L(R) = \frac{(H\Psi)(R)}{\Psi(R)}$$

requires that the kinetic singularity cancels the $1/r$ divergence. Kato's theorem, first showed in [17], gives the necessary short–distance behaviour of the (spherically averaged) wavefunction. Let r_{ij} be the interparticle distance, $\mu_{ij} = m_i m_j/(m_i + m_j)$ the reduced mass (in a.u.), Z_i the charge number (electron $Z = -1$, nucleus $Z = +Z_A$), and D the spatial dimension. Then

$$\frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\Big|_{r_{ij}\to 0} = \frac{2\,\mu_{ij}\,Z_i Z_j}{D-1} \quad \text{(s-wave / opposite-spin channel).}$$

For *identical fermions* the s-wave is excluded; the leading channel is p-wave ($\ell = 1$), and the wavefunction carries a linear node at coalescence. The general like-spin cusp is then

$$\frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\Big|_{r_{ij}\to 0} = \frac{2\,\mu_{ij}\,Z_i Z_j}{2\ell + D - 1} = \frac{2\,\mu_{ij}\,Z_i Z_j}{D+1} \quad \text{(like-spin, } \ell = 1\text{).}$$

Specializing to 2D electrons (atomic units). With $m_e = 1$ and $\mu_{ee} = 1/2$:

$$\boxed{a_{ee}^{\uparrow\downarrow} = \frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\Big|_0 = 1\ , \qquad a_{ee}^{\uparrow\uparrow} = \tfrac{1}{3}\ .}$$

The like-spin value follows from the p-wave (linear-node) analysis carried out in full in Appendix B.2, the canonical derivation used throughout this thesis; it gives the familiar $\frac{1}{4}$ in three dimensions and $\frac{1}{3}$ here in two. The 2D like-spin cusp is therefore *not* simply half the opposite-spin value—a coincidence that holds only in 3D—and the implemented value is $1/(D+1) = \frac{1}{3}$. For electron–nucleus coalescence ($\mu_{eA} \approx 1$, $Z_i Z_j = -Z_A$):

$$\boxed{a_{eA} = \frac{\partial}{\partial r_{iA}} \ln \overline{\Psi}\Big|_0 = -2Z_A\ .}$$

Why this matters. These conditions ensure the kinetic energy's r^{-1} singularity cancels the Coulomb r^{-1} term so that E_L remains finite as particles coalesce. Enforcing the cusp explicitly improves conditioning.

1.7 Structural diagnostics and Wigner markers

A Wigner molecule is not defined by a single number, so the crossover has to be read from several markers at once. This section says what each one measures and why it responds to Wigner order; the estimators, thresholds and stability protocol are defined once, in the Analysis chapter (§5.1), and the measurements themselves are in Chapter 6.

Radial localisation. The pair distribution $g(r)$ and the radial probability $P(r)$ say how far apart the electrons sit and how tightly that distance is fixed. The informative quantity is not the typical radius—which simply grows as the trap weakens—but the *ratio* of the radial width to it. A distribution that becomes relatively narrower around a growing radius is a shell stiffening; one that merely dilates is a droplet expanding. This is the finite-system analogue of a Lindemann criterion.

Shell topology. In a parabolic trap the classical minimum energy configuration is a set of concentric rings with definite occupancies—$(1, 5)$ at N=6, $(3, 9)$ at N=12 [8, 9]. Detecting those occupancies frame by frame turns a qualitative picture into a distribution over topologies, which is what makes it possible to say that a state is *mostly* $(3, 9)$ with real weight on $(2, 10)$—a rotating molecule fluctuating between near-degenerate shellings rather than a single frozen crystal.

Orientational order. Radial structure alone cannot distinguish a ring of electrons at fixed angular spacing from a ring free to slide. The bond-orientational order parameter Φ_m measures m-fold angular order on a detected ring after removing global rotation, and an angular Lindemann number measures how stiff those spacings are. Both are standard in two-dimensional Coulomb and Yukawa systems [11, 12], and together with the radial markers they separate angular from radial melting—the two need not happen at the same ω.

Classical scaling. Finally, the strong-coupling limit makes a parameter-free prediction: inter-particle distances should approach $r \propto \omega^{-2/3}$. Fitting the measured exponent and watching it approach $-2/3$ from above is an external check on the whole picture, since nothing in the ansatz knows about that number.

How to read them together. As ω falls, the width ratios and angular Lindemann numbers shrink, shell gaps sharpen, the shell-resolved $|\Phi_m|$ become nonzero after phase alignment, and the radial exponent approaches its classical value. No one of these is decisive—a narrow $P(r)$ can be an artefact of poor sampling, and a large $|\Phi_m|$ can be a fluctuation on a short chain—but they fail in different ways, which is why Chapter 6 reports them together and Chapter 5 states the stability checks each one is subjected to. Quantum Wigner markers in parabolic dots are reviewed in Refs. [7, 10–12].

1.8 Variational Methods

Quantum Monte Carlo (QMC) methods offer powerful tools for solving the many-body Schrödinger equation through stochastic sampling. These methods yield accurate ground-state properties by estimating expectation values of observables with controlled approximations. Among the various QMC techniques, Variational Monte Carlo (VMC) and Diffusion Monte Carlo (DMC) are particularly notable; VMC optimizes a trial wavefunction based on the variational principle, while DMC projects out the ground state through imaginary-time evolution [18, 19].

1.8.1 Variational Monte Carlo (VMC)

The Variational Principle

The variational principle guarantees that for any normalised trial wavefunction $\Psi_T(\mathbf{R};\boldsymbol{\alpha})$, the expectation value

$$E[\Psi_T] = \frac{\langle \Psi_T|\hat{H}|\Psi_T\rangle}{\langle \Psi_T|\Psi_T\rangle}$$

provides an upper bound to the true ground-state energy E_0. Here, $\mathbf{R} = (\mathbf{r}_1, \ldots, \mathbf{r}_N)$ and $\boldsymbol{\alpha}$ represents the set of variational parameters. Optimising $\boldsymbol{\alpha}$ minimizes $E[\Psi_T]$.

Sampling

In VMC, the probability density is defined as

$$P(\mathbf{R}) = \frac{|\Psi_T(\mathbf{R})|^2}{\int |\Psi_T(\mathbf{R})|^2 \, d\mathbf{R}},$$

and configurations are sampled using the Metropolis-Hastings algorithm, initially described in [20], with acceptance probability

$$A(\mathbf{R} \to \mathbf{R}') = \min\left(1, \frac{|\Psi_T(\mathbf{R}')|^2}{|\Psi_T(\mathbf{R})|^2} \frac{T(\mathbf{R}' \to \mathbf{R})}{T(\mathbf{R} \to \mathbf{R}')}\right),$$

where T is the proposal density (for a symmetric proposal the ratio is unity and the acceptance reduces to the Metropolis form). When a drift is added, a *quantum force*

$$\mathbf{F}(\mathbf{R}) = 2\,\nabla \ln|\Psi_T(\mathbf{R})|,$$

further guides the sampling toward regions of high probability. This leads to what is known as importance sampling. A more precise discussion of this is found in [21, 22].

Local Energy and Parameter Optimisation

The local energy is defined as

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi_T(\mathbf{R})}{\Psi_T(\mathbf{R})}.$$

If Ψ_T were exact, $E_L(\mathbf{R})$ would be constant. In practice, the variational energy is approximated by averaging E_L over many configurations:

$$E[\Psi_T] \approx \frac{1}{N_{\mathrm{MC}}} \sum_{i=1}^{N_{\mathrm{MC}}} E_L(\mathbf{R}_i).$$

Variational parameters are then optimised (using gradient descent or stochastic reconfiguration) to reduce both the energy and its variance.

1.8.2 Diffusion Monte Carlo (DMC)

Imaginary-Time Evolution and Projection

DMC refines the trial wavefunction by projecting out the ground state through imaginary-time evolution. The Schrödinger equation becomes

$$-\frac{\partial \Psi(\mathbf{R},\tau)}{\partial \tau} = \left(\hat{H} - E_T\right)\Psi(\mathbf{R},\tau),$$

where $\tau = it$ and E_T is a trial energy. As $\tau \to \infty$, $\Psi(\mathbf{R},\tau)$ converges to the ground-state wavefunction provided the initial state has nonzero overlap with it.

Stochastic Implementation

DMC simulates the imaginary-time evolution using a combination of diffusion, drift, and branching:

1. **Diffusion and Drift:** Particles execute random walks, with drift guided by the quantum force $\mathbf{F}(\mathbf{R})$, steering them toward regions where Ψ_T is large.

2. **Branching/Death:** Walkers are assigned a weight,

$$w(\mathbf{R}, \Delta\tau) = \exp\Big[-\Big(E_L(\mathbf{R}) - E_T\Big)\Delta\tau\Big],$$

which determines whether they are replicated or removed, ensuring that the walker distribution evolves toward the ground-state probability density.

1.8.3 Applications and Numerical Considerations

Model Systems

QMC methods have been successfully applied to various quantum systems:

- **Hydrogen Atom:** A simple trial wavefunction $\Psi_T(r) = e^{-\alpha r}$ yields the exact ground state for $\alpha = 1$, illustrating zero variance.

- **Harmonic Traps:** In systems confined by harmonic potentials, Jastrow factors [23] like $\exp[-ar_{ij}/(1 + \beta r_{ij})]$ model inter-particle repulsion while Slater determinants maintain the proper fermionic symmetry.

Computational Efficiency

Key factors in achieving efficient QMC simulations include:

- **Efficient Determinant Updates:** For a single-particle move, the Sherman–Morrison formula updates the inverse Slater matrix in $\mathcal{O}(N^2)$ rather than recomputing the determinant from scratch at $\mathcal{O}(N^3)$.

- **Optimised Gradient and Laplacian Calculations:** Recursive methods and efficient numerical routines for evaluating $\nabla\Psi_T/\Psi_T$ and $\nabla^2\Psi_T/\Psi_T$ are essential.

1.8.4 Conclusion

Variational and Diffusion Monte Carlo methods together provide a robust framework for studying quantum systems. VMC offers a flexible approach for optimising trial wavefunctions, while DMC projects toward the ground state through imaginary-time evolution—though for fermions the practical fixed-node variant is variational only within the imposed nodal surface, and retains a nodal bias unless those nodes are exact. These methods, working in concert, enable accurate and efficient investigations of systems from simple atoms to strongly correlated materials. Ongoing improvements in trial wavefunction design and stochastic algorithms continue to extend the reach of QMC in modern computational physics.

Chapter 2

Machine Learning

2.1 Machine Learning as Statistical Learning

Machine learning (ML) is best understood as the study of algorithms that approximate functions from data. At its core, it is not about "intelligent machines," but about developing systematic ways to extract patterns from observations and to generalise these patterns to unseen situations. From a mathematical perspective, ML is a branch of *statistical learning theory*, which views the learning process as the minimization of an expected error (or *risk*) under uncertainty.

Formally, consider an input space $\mathcal{X} \subseteq \mathbb{R}^d$ and an output space $\mathcal{Y}$, together with an unknown joint distribution $P(X, Y)$ over pairs (x, y). The aim of learning is to construct a function $f_\theta : \mathcal{X} \to \mathcal{Y}$, parameterised by $\theta \in \Theta$, such that $f_\theta(x)$ predicts y with small error. A loss function

$$\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$$

quantifies the discrepancy between prediction $f_\theta(x)$ and truth y. The central object of learning is then the **population risk**

$$\mathcal{R}(\theta) = \mathbb{E}_{(x,y)\sim P}\big[\ell(f_\theta(x), y)\big]. \tag{2.1}$$

Since the distribution P is unknown, we only have access to a finite dataset $D = \{(x_i, y_i)\}_{i=1}^n$. The corresponding **empirical risk** is

$$\hat{\mathcal{R}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(f_\theta(x_i), y_i). \tag{2.2}$$

This setup is known as *empirical risk minimization (ERM)*, and it underlies virtually all modern machine learning methods [24].

2.1.1 Types of learning

Depending on the information provided in the dataset, learning problems are traditionally categorized as:

- **Supervised learning.** We are given labeled pairs (x_i, y_i), and the goal is to approximate the conditional expectation $\mathbb{E}[Y|X = x]$. Regression (continuous Y) and classification (discrete Y) are canonical examples.
- **Unsupervised learning.** Only input data $\{x_i\}$ is observed, without labels. The aim is to uncover structure, such as clustering, dimensionality reduction, or latent-variable representations.
- **Reinforcement learning.** An agent interacts with an environment, receiving observations and rewards. The task is to learn a policy that maximizes expected cumulative reward.

Both of the first two categories appear in this thesis, and it is worth saying where. The training objectives that carry the bulk of the work—the Schrödinger residual and the variational energy—are *unsupervised*: there are no labelled targets anywhere, and the loss is supplied entirely by the physics the network is asked to satisfy. The one genuinely *supervised* ingredient is the second phase of residual pretraining, in which a published reference energy is used as an explicit target; the first phase, where the network sets its own target as a stabiliser, is better described as self-supervised. Reinforcement learning enters only obliquely—the REINFORCE estimator used for the collocation gradient is borrowed from it—and is not otherwise used.

2.1.2 Generalisation and the bias–variance tradeoff

The essential challenge in ML is not just minimizing the empirical risk $\hat{\mathcal{R}}(\theta)$, but ensuring that this leads to a low population risk $\mathcal{R}(\theta)$ on unseen data. A model that simply memorizes the training set achieves low empirical risk but poor generalisation — a phenomenon known as *overfitting*.

Statistical learning theory formalizes this challenge via the *bias–variance tradeoff*. Given a data-generating process and a learning algorithm, the expected squared error can be decomposed into

$$\mathbb{E}\big[(f_\theta(x)-y)^2\big] = \underbrace{(\mathbb{E}[f_\theta(x)]-f^*(x))^2}_{\text{bias}^2} + \underbrace{\mathbb{E}[(f_\theta(x)-\mathbb{E}[f_\theta(x)])^2]}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible noise}}, \tag{2.3}$$

where $f^*(x)$ is the true regression function and σ^2 the intrinsic noise of the data. Complex models tend to have low bias but high variance, while simple models have the opposite. Successful learning algorithms balance these two contributions. This is discussed in more detail in [25].

2.2 Optimisation

The minimization of empirical risk,

$$\hat{\mathcal{R}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(f_\theta(x_i), y_i),$$

is an *optimisation problem.* The hypothesis class $\{f_\theta\}$ is usually nonlinear and high-dimensional, so closed-form solutions are not available. Instead, optimisation is performed iteratively by updating parameters $\theta \in \mathbb{R}^p$ according to the geometry of the loss landscape.

2.2.1 Gradient descent

The most basic algorithm is *gradient descent.* Starting from an initial guess $\theta^{(0)}$, parameters are updated by

$$\theta^{(t+1)} = \theta^{(t)} - \eta\, \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}), \tag{2.4}$$

where $\eta > 0$ is the *learning rate.* Intuitively, the gradient points in the direction of steepest increase of the loss, so subtracting it decreases the loss most rapidly in a local sense.

The choice of learning rate is crucial: if too small, convergence is slow; if too large, the algorithm may diverge.

2.2.2 Stochastic gradient descent (SGD)

For large datasets, evaluating the full gradient at every step is prohibitively expensive. Instead, one uses *stochastic gradient descent (SGD)*, which approximates the gradient on a randomly sampled mini-batch $B \subset D$:

$$\nabla_\theta \hat{\mathcal{R}}_B(\theta) = \frac{1}{|B|}\sum_{(x_i,y_i)\in B} \nabla_\theta \ell(f_\theta(x_i), y_i). \tag{2.5}$$

The update rule is then

$$\theta^{(t+1)} = \theta^{(t)} - \eta\, \nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)}). \tag{2.6}$$

SGD introduces randomness into the optimisation trajectory. Paradoxically, this noise often *helps*: it prevents the algorithm from getting trapped in poor local minima and can improve generalisation through an implicit regularization effect [24, 26].

2.2.3 Momentum and adaptive methods

Plain gradient descent treats all directions in parameter space equally. To accelerate convergence, one may use *momentum*, maintaining a velocity vector v_t:

$$v_{t+1} = \beta v_t + (1-\beta)\,\nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)}), \tag{2.7}$$
$$\theta^{(t+1)} = \theta^{(t)} - \eta v_{t+1}, \tag{2.8}$$

where $\beta \in (0,1)$ controls the contribution of past gradients. Momentum smooths oscillations and allows the optimisation to traverse valleys in the loss landscape more effectively.

Beyond momentum, *adaptive methods* such as RMSProp and Adam rescale learning rates per parameter based on historical gradient magnitudes [27]. For instance, Adam combines momentum with an adaptive normalisation:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \tag{2.9}$$
$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \tag{2.10}$$
$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{2.11}$$

where $g_t = \nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)})$ and $\hat{m}_t, \hat{v}_t$ are bias-corrected versions of m_t, v_t. Adam is widely used in practice for its robustness across diverse problems [28].

2.2.4 Second-order methods and natural gradient

First-order methods only use gradient information. In principle, convergence can be accelerated by incorporating curvature via the Hessian $H(\theta) = \nabla_\theta^2 \hat{\mathcal{R}}(\theta)$. Newton's method updates parameters as

$$\theta^{(t+1)} = \theta^{(t)} - H(\theta^{(t)})^{-1}\,\nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}). \tag{2.12}$$

However, computing and inverting the Hessian is usually infeasible for large models.

An attractive alternative is the *natural gradient* [29, 30], which replaces the Euclidean metric with the geometry of probability distributions. The key object is the *Fisher information matrix* (FIM),

$$F(\theta) = \mathbb{E}\big[\nabla_\theta \log p_\theta(x)\,\nabla_\theta \log p_\theta(x)^\top\big], \tag{2.13}$$

where p_θ denotes the model distribution. The natural gradient update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta\, F(\theta^{(t)})^{-1}\,\nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}). \tag{2.14}$$

This preconditioning makes updates invariant to reparameterizations of θ and often improves conditioning of the optimisation. In practice, computing F^{-1} exactly is intractable, but approximations (e.g. stochastic reconfiguration in variational Monte Carlo, Kronecker-factored approximations in deep learning) make the method viable [2].

2.2.5 The variational tangent space: geometric tensor, tangent kernel, and effective dimension

For a variational quantum state the Fisher metric takes a particularly concrete form, and that form is the analytical lens used throughout the results of this thesis. Write the per-sample log-derivatives—the *score*—of an ansatz Ψ_θ as

$$O_{ki} = \frac{\partial \log|\Psi_\theta(\mathbf{x}_k)|}{\partial \theta_i}, \qquad k = 1,\dots,B, \;\; i = 1,\dots,P, \tag{2.15}$$

for a batch of B configurations drawn from $|\Psi_\theta|^2$ and P parameters θ (here θ collects *all* variational parameters). Its centred Gram matrices define two dual objects. The *quantum geometric tensor* (QGT),

$$S = \tfrac{1}{B}\, O^\top O \in \mathbb{R}^{P\times P}, \tag{2.16}$$

is the Fisher–Rao metric of the state written in terms of the amplitude score. It is proportional to the Fisher matrix $F(\theta)$ of the previous subsection specialised to $p_\theta = |\Psi_\theta|^2$: because $\nabla_\theta \log|\Psi_\theta|^2 =$

$2\,\nabla_\theta \log|\Psi_\theta|$, one has $F = 4S$, the factor of four being a convention absorbed into the learning rate. Stochastic reconfiguration is natural-gradient descent under this metric, $\dot{\theta} = -S^{-1}\nabla_\theta E$ [31, 32]. Its Gram dual, the *neural tangent kernel* (NTK) [33],

$$K = OO^\top \in \mathbb{R}^{B\times B}, \tag{2.17}$$

shares the nonzero spectrum of S up to the overall factor B (its nonzero eigenvalues are $B\lambda_a$), and is cheaper to diagonalise when $B < P$.

Two scalars read from that spectrum organise the analysis. The *condition number* $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ measures how anisotropic the tangent metric is, and therefore how far a natural-gradient step departs from a Euclidean one—the quantity that decides when SR is worth its cost. The *effective dimension*

$$d_{\text{eff}}(S) = \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2} \tag{2.18}$$

is the participation ratio of the spectrum: it counts how many independent tangent directions carry appreciable Fisher weight at the solution, regardless of the raw parameter count P. A small d_{eff} means the state is reached along a few collective directions; a large one means the same solution is spread across many. Because these are properties of the tangent geometry rather than of the raw parameter count, and are stable across seeds and sampling measures for a given ansatz, they let us compare architectures, optimisers, and training paradigms on a common footing— which is exactly what the second results chapter (Chapter 7) does.

One caveat must be kept in view. The participation ratio d_{eff} is computed from the QGT eigenvalues in the ordinary Euclidean parameter coordinates, and those eigenvalues are *not* invariant under an arbitrary reparameterization: rescaling a single parameter $\theta_1 \mapsto c\,\theta_1$ leaves the physical state untouched but changes d_{eff} (for a diagonal metric it moves from 2 toward 1 as $c \to \infty$). d_{eff} is therefore *not* a coordinate-free intrinsic dimension of the wavefunction; it is a property of the (architecture, parameterisation, normalisation) triple. We use it accordingly—as a comparison between fixed architectures under matched conventions and a common probe measure—and read a smaller d_{eff} as "this architecture concentrates its Fisher weight into fewer active directions," not as an absolute manifold dimension.

2.2.6 Optimisation challenges in deep learning

Optimisation in deep neural networks poses unique challenges:

- **Non-convexity.** Loss landscapes are highly non-convex, with many local minima and saddle points. Empirically, most local minima reached by SGD perform similarly well.
- **Vanishing and exploding gradients.** In deep networks, gradients can shrink or blow up during backpropagation. Proper initialisation and activation choice are essential to mitigate this problem.
- **Flat vs. sharp minima.** Empirical evidence suggests that flat minima (regions where the loss remains low under perturbations of θ) lead to better generalisation. SGD's noise implicitly biases optimisation toward flatter minima.

Optimisation is therefore more than a means of fitting: the choice of update also shapes the inductive biases and generalisation behaviour of the trained network. The sections that follow examine the other ingredients that determine those biases—network expressivity, activation functions, and initialisation.

2.3 Feed-forward neural networks

Feed-forward neural networks are the basic building block from which the architectures of this thesis are assembled. They compose affine maps with elementwise nonlinearities to represent functions that linear models cannot. We first define the feed-forward map, and then discuss the two design choices that most affect its behaviour in physics-informed settings, where gradients and Laplacians of the network enter the loss: the activation function and the weight initialisation.

2.3.1 From linear models to nonlinear function classes

Classical regression models are linear in their parameters:

$$f_\theta(x) = \theta^\top x.$$

While simple and interpretable, linear models are limited to representing affine functions. Introducing basis expansions (e.g. polynomials, Fourier series, kernels) enlarges the hypothesis space, but requires manual design.

Neural networks automate this process. A feed-forward neural network with L layers is a composition of affine maps and nonlinear activation functions:

$$h^{(0)} = x, \tag{2.19}$$

$$h^{(\ell)} = \sigma\Big(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}\Big), \quad \ell = 1, \dots, L-1, \tag{2.20}$$

$$f_\theta(x) = W^{(L)} h^{(L-1)} + b^{(L)}. \tag{2.21}$$

Here $\sigma : \mathbb{R} \to \mathbb{R}$ is a nonlinear activation. Without nonlinearity, the network collapses to a single linear transformation; with it, the space of representable functions expands dramatically.

2.3.2 Activation functions

Classical Activation Functions

The earliest activation functions were smooth, saturating nonlinearities:

$$\sigma_{\text{sigmoid}}(x) = \frac{1}{1+e^{-x}}, \tag{2.22}$$

$$\sigma_{\tanh}(x) = \tanh(x). \tag{2.23}$$

Both map $\mathbb{R}$ to a bounded range and are differentiable. The hyperbolic tangent is zero-centred, which often improves optimisation.

Despite their smoothness, these functions suffer from the *vanishing gradient problem*. For large $|x|$, their derivatives approach zero, causing gradients to vanish during backpropagation in deep networks. As a result, learning becomes prohibitively slow.

To address vanishing gradients, the *Rectified Linear Unit* (ReLU) was introduced:

$$\sigma_{\text{ReLU}}(x) = \max(0, x). \tag{2.24}$$

ReLU is unbounded above and has derivative 1 for positive inputs, avoiding gradient saturation. This makes deep networks with ReLU trainable even with many layers.

However, ReLU introduces its own drawbacks:

- Non-smoothness at $x = 0$, which can lead to optimisation instability in contexts where derivatives or Laplacians are central (e.g. physics-informed models).
- The *dying ReLU* problem, where neurons with negative inputs become permanently inactive.

Modern activations

To combine the benefits of ReLU with smoothness, a number of *smoothed rectifiers* have been proposed:

$$\sigma_{\text{SiLU}}(x) = x \cdot \sigma_{\text{sigmoid}}(x), \qquad \text{(Sigmoid Linear Unit)} \tag{2.25}$$

$$\sigma_{\text{Swish}}(x) = x \cdot \frac{1}{1+e^{-\beta x}}, \qquad \text{(generalised SiLU with } \beta > 0\text{)} \tag{2.26}$$

$$\sigma_{\text{GELU}}(x) = x \cdot \Phi(x), \qquad \text{(Gaussian Error Linear Unit)} \tag{2.27}$$

$$\sigma_{\text{Mish}}(x) = x \cdot \tanh(\log(1+e^x)). \tag{2.28}$$

Here $\Phi(x)$ is the standard Gaussian cumulative distribution function.

These activations are smooth (C^∞) and non-saturating, preserving gradient flow while avoiding non-differentiable kinks. Empirically, GELU, SiLU/Swish, and Mish improve optimisation stability and generalisation across a wide range of tasks.

2.3.3 Initialisation

The initialisation of neural network weights has a critical impact on optimisation. At the start of training, parameter values determine how signals propagate forward and backward through the network. Poor initialisation can lead to exploding or vanishing activations and gradients, making optimisation unstable or ineffective. Proper schemes balance signal magnitudes across layers and preserve gradient flow, especially in deep networks.

The role of initialisation

Consider a feed-forward network with layers

$$h^{(\ell)} = \sigma\Big(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)}\Big),$$

where $W^{(\ell)}$ has entries drawn from some distribution. If weights are too large, activations grow exponentially with depth, leading to exploding gradients. If too small, activations shrink toward zero, causing vanishing gradients. Both phenomena impede learning.

Initialisation therefore aims to preserve the variance of activations and gradients across layers, such that

$$\mathrm{Var}[h^{(\ell)}] \approx \mathrm{Var}[h^{(\ell-1)}].$$

Xavier/Glorot initialisation

Glorot and Bengio (2010) proposed initialising weights with variance scaled by the number of input and output units:

$$W_{ij}^{(\ell)} \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}},\ \sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}\right). \tag{2.29}$$

Here n_{in} and n_{out} denote the fan-in and fan-out of a layer. This choice balances variance across layers for activations with approximately unit derivative (e.g. tanh).

Kaiming/He initialisation

For rectified activations such as ReLU, He et al. (2015) proposed *Kaiming initialisation*, scaling variance by $2/n_{\text{in}}$:

$$W_{ij}^{(\ell)} \sim \mathcal{N}\left(0,\ \frac{2}{n_{\text{in}}}\right). \tag{2.30}$$

This accounts for the fact that ReLU activations output zero half of the time, requiring larger variance to maintain signal propagation.

Gain and activation dependence

Both Glorot and Kaiming schemes can be generalised with a *gain* factor g that depends on the activation function:

$$\mathrm{Var}[W_{ij}^{(\ell)}] = \frac{g^2}{n_{\text{in}}}.$$

For example, $g = \sqrt{2}$ is recommended for ReLU, while $g = 1$ is typical for tanh and smooth activations. Choosing the correct gain ensures that variance neither explodes nor decays through the network.

Orthogonal and random normal initialisation

An alternative approach is to initialise each weight matrix as an orthogonal matrix scaled by a variance factor. Orthogonal initialisation preserves the norm of activations across layers, which can stabilize very deep networks.

Simpler schemes such as sampling from a standard normal distribution without scaling are generally inferior: they do not respect the scaling needs of different architectures or activation functions, and often result in unstable training.

2.3.4 Summary

Initialisation is particularly delicate in deep networks. Even small imbalances in variance can compound exponentially with depth. Smooth activations (SiLU, GELU, Mish) alleviate gradient instability but require careful scaling, as their effective slope differs from ReLU or tanh.

Key takeaways include:

- Proper scaling of variance prevents vanishing/exploding signals.
- Initialisation must match the activation function (Glorot for tanh/smooth, Kaiming for ReLU).
- Orthogonal schemes can improve stability in deep networks.
- For PINNs and scientific ML, initialisation is not merely a convenience but essential for stable computation of derivatives.

Initialisation and activation choice thus form an inseparable pair: together they determine whether information propagates stably through the network. This is taken up again for physics-informed models in Section 2.6.

2.4 Architectures for many-body wavefunctions

The feed-forward network of the previous section maps a fixed-size input vector to an output. A many-body wavefunction is not a generic function of such a vector: identical particles impose a permutation symmetry, and correlation is intrinsically *relational*—what one particle should do depends on where the others are. A plain multilayer perceptron acting on the concatenated coordinates $(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ respects neither property; it would have to learn the symmetry from data, and its parameter count would grow with N. This section introduces the two architecture classes that build the required structure in—permutation-invariant pooling and message passing—and situates the specific message-passing network used in this thesis as one instance of the latter.

2.4.1 Permutation symmetry and equivariance

For identical particles the modulus $|\Psi|$ is invariant under exchange (the sign is carried by the antisymmetric Slater reference of Chapter 3). The two learnable objects of the ansatz therefore have distinct symmetry requirements. A scalar *correlator* f must be permutation-*invariant*,

$$f(P_\sigma R) = f(R),$$

while a per-particle *backflow* displacement field $\mathbf{g} = (\mathbf{g}_1, \ldots, \mathbf{g}_N)$ must be permutation-*equivariant*,

$$\mathbf{g}(P_\sigma R) = P_\sigma\, \mathbf{g}(R),$$

for any permutation σ (here P_σ relabels the particles). Enforcing these by construction is both more data-efficient than learning them and yields a parameter count independent of N, because the same shared maps are applied to every particle and pair.

2.4.2 Permutation-invariant pooling: DeepSets

The simplest way to build an invariant function is to embed each particle independently and pool the embeddings symmetrically. This is the DeepSets construction [34]: any suitably regular permutation-invariant function admits the form

$$f(R) = \rho\Big(\textstyle\sum_i \phi(\mathbf{x}_i) \Big),$$

with a per-particle embedding ϕ, a symmetric pool $\sum_i$, and a readout ρ. A two-body extension encodes pairs, $\phi(\mathbf{x}_i, \mathbf{x}_j)$, and pools over them, capturing pairwise correlation. What characterises this class is that each particle—or pair—is embedded *independently* and only then pooled: no information is exchanged between particles before the pool. We refer to such networks as *separable*, and they serve as the non-message-passing baseline for the correlator.

2.4.3 Message passing and graph networks

To let a particle's representation depend on the others' in an iterated, higher-order way, one places the particles on a graph—nodes are particles, edges are pairs—and passes learned messages along the edges [35, 36]. In a generic round ℓ, each edge gathers information from its endpoint nodes and each node aggregates its incoming edge messages,

$$\mathbf{m}_{ij}^{(\ell)} = \phi_e\big(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}^{(\ell)}\big), \qquad \mathbf{h}_i^{(\ell+1)} = \phi_v\Big(\mathbf{h}_i^{(\ell)},\ \mathrm{Agg}_{j\neq i}\, \mathbf{m}_{ij}^{(\ell)}\Big),$$

with shared maps ϕ_e, ϕ_v and a symmetric aggregation Agg, so that permutation equivariance holds by construction; a final readout produces an invariant scalar (correlator) or an equivariant field (backflow). Both DeepSets and message passing use pairwise information, but only message passing lets that information *propagate*: after several rounds a particle's representation reflects not just its neighbours but their neighbours' responses. This iterated propagation is what we call the *relational channel*, and asking what it buys—in the correlator and in the backflow—is one of the central questions of this thesis. Message passing has been used in exactly this spirit for neural quantum states of extended fermion systems [5, 6].

2.4.4 The CTNN: one message-passing instance

The message-passing correlator and backflow studied here are realised by a particular network we refer to as the *CTNN*: a copresheaf-style message-passing network that maintains explicit *node and edge* features and transports information between them through learned maps, rather than recomputing and discarding one-shot messages. The correlator iterates this exchange over a down/up "V-cycle"; the backflow applies a single such copresheaf round. Its concrete update equations, hidden dimensions, and parameter counts are specified in the Methods chapter (§3.5.2, Eqs. (3.13)–(3.15)). Two points of framing matter for the rest of the thesis. First, the CTNN is *one* concrete realisation of the message-passing class defined above; the conclusions we draw about the value of message passing concern the class, with the CTNN as the instance actually studied. Wherever we contrast a "message-passing" network with a "separable" one, the separable member is a DeepSets-type network and the message-passing member is the CTNN. Second, we write "copresheaf-*style*" deliberately: the name reflects the node/edge transport structure of the updates, not a claim to a formal sheaf-theoretic construction.

2.4.5 Relevance to the ansatz

Both learnable objects of the ansatz—the Jastrow correlator and the coordinate backflow—can be built in either form: as a separable DeepSets-type network, or as the message-passing CTNN. Holding everything else fixed and varying only this choice isolates the effect of the relational channel, which Chapter 7 analyses through the geometry of the tangent kernel introduced in §2.2.5.

2.5 Expressivity, approximation, and generalisation

The central reason neural networks have become the dominant model class in machine learning is their *expressivity*. However, in contrast with what classical statistical theory suggests, neural networks have been found to mitigate the curse of dimensionality, generalise well despite overparameterization, and optimise effectively in highly non-convex landscapes. Understanding these phenomena has been a major focus of recent theoretical research. In this section, we review key results on the approximation capabilities of neural networks, with an emphasis on implications for physical modelling.

2.5.1 Universal approximation theorems

Foundational results established that neural networks are *universal approximators*. It has been proved that a feed-forward network with at least one hidden layer and a suitable nonlinear activation (e.g. sigmoid, ReLU, tanh) can approximate any continuous function on a compact set to arbitrary accuracy [37, 38]. Formally:

Theorem 2.1 (Universal Approximation, informal)**.** *Let f^* be continuous on a compact $K \subset \mathbb{R}^d$. Then for any $\epsilon > 0$, there exists a neural network f_θ with a single hidden layer such that*

$$\sup_{x\in K} |f_\theta(x) - f^*(x)| < \epsilon.$$

This result guarantees existence but says nothing about how many neurons are required, how they should be trained, or how generalisation occurs.

2.5.2 Depth versus width

Cybenko's theorem establishes that a *single* hidden layer with a nonpolynomial sigmoidal activation is a *universal approximator*: for every continuous f on a compact domain and every $\varepsilon > 0$ there exists a one–hidden–layer network whose output g satisfies $\|f - g\|_\infty < \varepsilon$ [37]. The result concerns *existence* of an approximant and does not provide constructive bounds on the number of units required, rates of approximation, or generalisation guarantees.

In practice, depth is introduced for efficiency, not for universality. By composing intermediate features, additional layers often reduce the number of units required to attain a target accuracy. For certain target classes, there are known separations where a one-hidden-layer network needs exponentially many neurons to match the accuracy achievable with a deep architecture of polynomial size. Hence modern models leverage depth primarily for parameter efficiency and a stronger inductive bias (via hierarchical feature reuse), while width remains the lever for raw capacity [24].

2.5.3 Curse of dimensionality

Classical approximation theory predicts that uniformly approximating a generic $f : \mathbb{R}^d \to \mathbb{R}$ requires resources that grow exponentially in d—the *curse of dimensionality* (CoD). This perspective cannot explain the strong empirical performance of deep networks on very high–dimensional tasks, prompting a search for structural conditions under which the CoD does *not* apply.

A first resolution is *low effective dimension*: if the data (and loss) live on a $d' \ll d$ manifold M and error is measured in $L^p(\mu)$ with $\operatorname{supp}\mu \subset M$, then neural networks can learn local charts and partitions of unity and achieve rates that depend on d' rather than the ambient d. This replaces ambient complexity by the intrinsic geometry of M, yielding polynomial (rather than exponential) scaling.

A second route is *spectral/Barron regularity* [39]: for targets with finite first Fourier moment (Barron class), two–layer networks attain a *dimension–independent* $O(n^{-1/2})$ L^2–approximation error with n hidden units. Deep and compositional Barron spaces extend this beyond smooth settings, providing broad families where the CoD provably disappears.

A third line exploits *PDE structure*. For many high–dimensional linear and semilinear parabolic/elliptic problems, stochastic representations (e.g., Feynman–Kac) and NN emulations of time–stepping yield approximation bounds whose complexity scales polynomially in d, again sidestepping the CoD under standard conditions.

Depth then clarifies *why* these escapes are efficient. Certain functions admit polynomial–size deep representations but require exponentially wide shallow ones (Eldan–Shamir separations), and deep ReLU networks generate exponentially more affine regions or oscillations per parameter than single–hidden–layer models. Thus, depth supplies the compositional mechanism that turns structure (manifolds, spectral regularity, PDE priors) into dimension–tolerant approximation.

2.5.4 Generalisation in overparameterized networks

Modern networks often have far more parameters than training samples, yet they fit the data *and* perform well on new inputs. This looks paradoxical if we judge models only by parameter count. A better lens is to ask which interpolating solution the training dynamics actually pick: test error is governed less by how large the hypothesis class is and more by the *kind* of function the algorithm converges to.

Two ideas make this concrete. First, stability: if small changes in the dataset produce only small changes in the learned predictor, the train–test gap remains small even at scale,

$$\left|\mathcal{R}(\hat{\theta}) - \hat{\mathcal{R}}(\hat{\theta})\right| \text{ is controlled.}$$

Second, implicit regularization: stochastic gradient methods (with finite batches, step sizes, and momentum) do not search the space uniformly; they are biased toward predictors with lower effective complexity—e.g., larger margins, smaller norms, or flatter basins in parameter space. In very wide nets, the dynamics are well-approximated by a linearized (kernel-like) model that interpolates while implicitly minimizing a function-space norm; at realistic widths, the same bias persists in practice. This also explains *double descent*: near the interpolation threshold test error can worsen, then improves again as capacity grows and the dynamics have more low-complexity interpolants to select from.

2.5.5 Optimisation in deep networks

Training is a large-scale, nonconvex problem, yet simple first-order methods reliably drive the loss to (near) zero. Two features help. Overparameterization and modern architectures (residual connections, normalisation) shape the landscape into broad, connected valleys of low loss and maintain usable descent directions through depth. As a result, poor isolated minima are less of a practical obstacle.

Stochastic gradient descent can be viewed as gradient flow with noise whose effective "temperature" is set mainly by the learning-rate–to–batch-size ratio. A higher temperature encourages escape from sharp minima and biases the search toward wider, flatter regions that tend to generalise better; annealing the temperature over time refines the final solution. Momentum accelerates along coherent directions and damps high-frequency noise; normalisation keeps activations well-scaled so larger, stable steps are possible. In short, the optimiser does more than find zero training error—it determines *which* zero-error solution you reach, and that choice largely sets generalisation quality.

2.5.6 Implications for physical modelling

In physical sciences, target functions such as wavefunctions are continuous, smooth (except at singularities like cusps), and often structured by symmetries. Neural networks thus provide a flexible ansatz space:

1. They can approximate smooth functions and their derivatives, which is crucial when differential operators appear in the loss (as in PINNs).
2. Depth allows efficient capture of multi-scale and compositional structure, relevant for many-body interactions.
3. Constraints such as permutation invariance or antisymmetry can be incorporated to encode physics explicitly.

These theoretical guarantees justify the use of neural networks as universal, symmetry-aware function approximators, laying the groundwork for our discussions of activation functions, initialisation strategies, and physics-informed extensions.

2.6 Physics-Informed Neural Networks

Traditional machine learning focuses on fitting models to labeled data. In scientific applications, however, governing equations such as differential equations encode a wealth of prior information about the system. *Physics-Informed Neural Networks (PINNs)* [40] integrate these constraints directly into the learning process, combining data-driven approximation with analytical structure. The idea of training a network against a differential operator predates the name by two decades [41]; what [40] added was automatic differentiation, which makes the residual cheap enough to evaluate that it can drive the optimisation directly. Figure 2.1 illustrates how the behaviour of neural networks changes when working with higher order derivatives, as is common in PINNs.

2.6.1 Formulation of PINNs

Let $u : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ denote the unknown solution of a partial differential equation (PDE) of the form

$$\mathcal{N}[u](x) = 0, \qquad x \in \Omega, \tag{2.31}$$

subject to boundary conditions $u(x) = g(x)$ for $x \in \partial\Omega$. Here $\mathcal{N}$ is a (possibly nonlinear) differential operator.

In a PINN, $u_\theta(x)$ is represented by a neural network, and the loss function enforces both data fitting and PDE residual minimization:

$$\mathcal{L}(\theta) = \lambda_{\text{data}} \sum_{i=1}^{n_d} |u_\theta(x_i) - y_i|^2 + \lambda_{\text{PDE}} \sum_{j=1}^{n_r} |\mathcal{N}[u_\theta](\xi_j)|^2 + \lambda_{\text{BC}} \sum_{k=1}^{n_b} |u_\theta(\zeta_k) - g(\zeta_k)|^2. \tag{2.32}$$

Here $\{x_i, y_i\}$ are observed data, $\{\xi_j\}$ are collocation points in the domain for enforcing the PDE, and $\{\zeta_k\}$ are boundary points. The weights $\lambda_{\text{data}}, \lambda_{\text{PDE}}, \lambda_{\text{BC}}$ balance the different contributions.

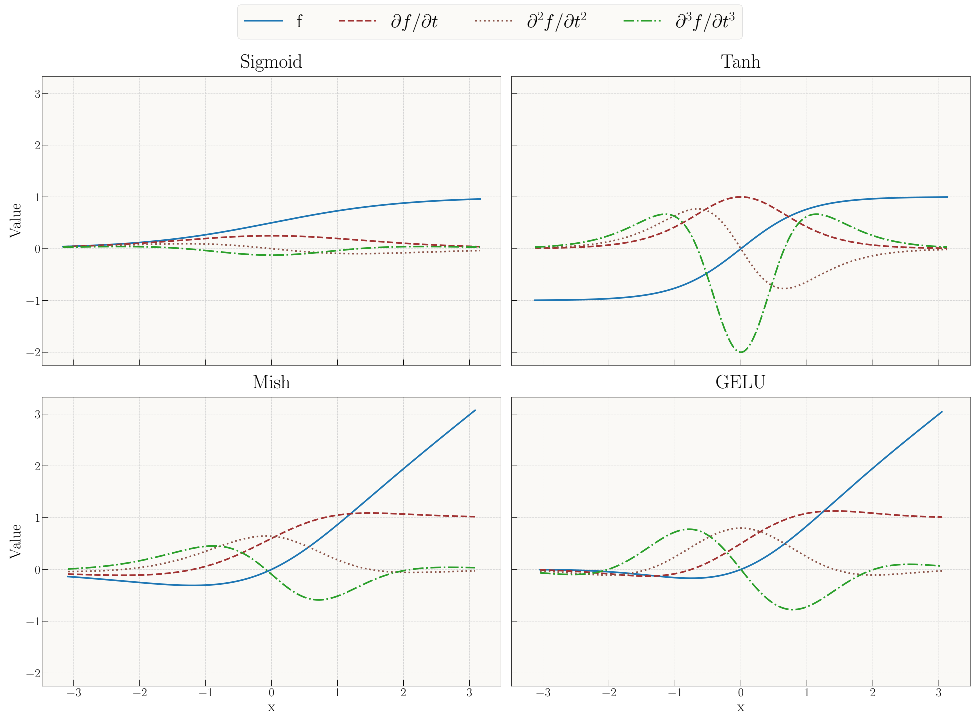


Figure 2.1: Comparison of common activation functions and their first, second, and third derivatives. Smooth activations (GELU, Mish) yield stable derivatives, while Tanh and Sigmoid have less stable higher derivatives.

2.6.2 Automatic differentiation

A key ingredient of PINNs is the use of *automatic differentiation* (AD) to evaluate ∇u_θ and Δu_θ. Unlike finite-difference approximations, AD computes derivatives exactly (up to floating-point error), making it feasible to incorporate higher-order differential operators directly into the loss. This enables end-to-end training of networks constrained by physical laws.

2.6.3 Theoretical difficulties

Although PINNs are conceptually straightforward, making them reliable and scalable raises several non-trivial challenges that go beyond classical universal approximation results.

Approximation in Sobolev spaces. Universal approximation theorems control function error in L^2/L^∞, but strong-form PINN losses depend on *derivatives* through $L[u_\theta]$. To make residuals small one must approximate in matching Sobolev norms $W^{\ell,p}$ (with ℓ the PDE order), which imposes smooth activations (e.g., tanh/SiLU for $\ell \geq 2$) and architecture choices that control higher derivatives. Low-regularity targets (discontinuities, cusps) break global $W^{\ell,p}$ control and favor weak/variational or entropy-based formulations.

Stability (residual-to-error). A small residual does not imply a small solution error unless the forward problem is (conditionally) stable. For well-posed elliptic/parabolic problems one can bound $\|u - u_\theta\|$ by $\|L[u_\theta] - f\|$, but inverse/ill-posed settings may require regularization, priors, or restricted hypothesis classes to prevent "residual overfitting" with large state error.

Sampling and generalisation. Collocation-based training minimizes a sampled surrogate of a continuum loss. Generalisation requires capacity control (e.g., bounded weights/Lipschitz loss) and sufficient sampling of interior vs. boundary/IC points; otherwise the empirical residual can be small while the population residual remains large.

Optimisation and conditioning. PINN losses couple multiple physics terms (interior, boundary, data) with disparate scales, producing ill-conditioned gradient dynamics. In the linearized/NTK view, convergence speed is governed by the spectrum of a problem-dependent kernel; large condition numbers lead to slow or stalled training.

Spectral bias. Gradient-based training tends to fit low-frequency components first, delaying convergence for solutions with sharp layers or oscillations unless the sampling, features, or architecture emphasize high-frequency content (e.g., Fourier features, windowed bases, curriculum).

Mitigating these challenges requires careful design of architectures, activations, initialisation, loss weighting, and sampling strategies. These topics are explained and discussed in detail in [42]

2.6.4 Relevance to this thesis

For scientific machine learning, PINNs provide a principled framework for embedding physical structure into neural networks. In this thesis, they motivate the choice of smooth activations (for stable derivatives), careful initialisation (to prevent exploding Hessians), and architectural constraints (to encode symmetry and physical laws). Thus, PINNs serve both as a methodological tool and as a theoretical lens for understanding the interplay between learning and physics.

2.6.5 Implications for scientific machine learning

For physics-informed models, generalisation takes on a slightly different meaning: the goal is not merely prediction on unseen data, but consistency with underlying physical laws. Inductive biases (symmetry, smoothness, conservation laws) are therefore not optional but essential. Implicit regularization from gradient-based optimisation complements these biases by favoring solutions that are both smooth and physically plausible. Generalisation in neural networks arises from an interplay of:

1. Algorithmic stability of gradient-based methods,
2. Implicit bias toward simple, smooth, low-complexity solutions, and
3. Architectural inductive biases encoding domain knowledge.

In scientific computing, these mechanisms explain why neural networks, despite their size, can approximate physically meaningful solutions. They also justify the careful design choices of activations, initialisation, and constraints emphasized throughout this chapter.

Mitigation in practice. The challenges discussed for PINNs do not resolve themselves; they must be engineered away. In brief: use weak/variational formulations (or energy principles) when higher-order derivatives make strong forms unstable; hard–enforce essential boundary conditions and adapt weights for the rest; non-dimensionalize and balance loss terms to improve conditioning; prefer smooth activations and architectures that control derivatives; exploit symmetry, equivariance, and locality to reduce effective complexity; precondition optimisation with second-order or natural-gradient methods (e.g., L-BFGS, Stochastic Reconfiguration); and design sampling (importance/stratified, curricula in time/space) to close the generalisation gap. Together, these measures turn inductive bias into concrete numerical stability and reliable, law-consistent generalisation.

2.7 Summary and Outlook

Three threads from this chapter carry into the rest of the thesis. Framing learning as empirical risk minimisation makes *generalisation*—not fitting—the real problem, and locates it in the inductive bias of the model rather than in raw capacity. The optimisation view adds that the training *trajectory* matters as much as the objective: gradient descent carries its own implicit bias, and the natural gradient re-expresses that trajectory in the geometry of the model's own outputs, the geometry we will later read directly through the tangent kernel. The physics-informed view supplies the binding constraint: a wavefunction ansatz must be differentiated twice, cleanly, to yield a local energy, so smooth activations, stable initialisation, and hard-wired symmetries are prerequisites rather than cosmetic choices. These are the principles that guide the architecture and diagnostics of the chapters that follow.

Part III

Methods

No great thing is created suddenly, any more than a bunch of grapes or a fig.

EPICTETUS
Discourses

Chapter 3

Trial Wavefunction

In this chapter we specify the concrete modelling and algorithmic choices used in our computations. We proceed from the physical setup and trial states to objectives, with an emphasis on stability, symmetry, and differentiability—all of which are critical for Laplacian-based local-energy evaluations and stochastic reconfiguration (SR). Unless otherwise stated, we work in atomic units.

3.1 Preliminaries

Hamiltonian and units. We study N spin-$\frac{1}{2}$ fermions in two spatial dimensions confined by a harmonic trap and interacting via Coulomb repulsion,

$$H \;=\; \sum_{i=1}^{N} \Big(-\tfrac{1}{2}\nabla^2_{\mathbf{r}_i} + \tfrac{1}{2}\,\omega^2 r_i^2 \Big) \;+\; \sum_{1\le i<j\le N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \qquad (\hbar = m_e = e = 1). \tag{3.1}$$

Spin does not appear in H; eigenstates factorize into a spatial function with good L_z and a spin function with good (S, S_z).

Coordinates and scaling. We collect positions as $R = (\mathbf{r}_1, \ldots, \mathbf{r}_N) \in \mathbb{R}^{2N}$ and use trap scaling $\tilde{\mathbf{r}}_i = \sqrt{\omega}\,\mathbf{r}_i$, $\tilde{R} = \sqrt{\omega}\,R$. This aligns typical length scales to $\mathcal{O}(1)$, improves conditioning of second derivatives, and makes the noninteracting ground state ω-independent. When useful, we also use pair distances $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ and their scaled versions $\tilde{r}_{ij} = |\tilde{\mathbf{r}}_i - \tilde{\mathbf{r}}_j|$.

Sectors and focus. We primarily report *closed-shell* cases (unique spin singlet at the noninteracting level). Open shells introduce a degenerate manifold and additional choices for the reference; we keep this to brief remarks only where needed.

Spin–orbitals and basis. One-particle spatial orbitals are the Cartesian two-dimensional harmonic-oscillator eigenstates $\{\phi_{n_x n_y}(\mathbf{r})\}$ with $E_{n_x n_y} = \omega(n_x + n_y + 1)$, $n_x, n_y \in \mathbb{N}_0$ (the basis used in the implementation; for the closed shells studied it spans the same filled shells as any complete basis of those shells). Spin–orbitals are $\varepsilon_\alpha(x) = \phi_{n_x n_y}(\mathbf{r})\chi_\sigma(s)$, $\sigma \in \{\uparrow, \downarrow\}$.

3.2 Overview of the ansatz

Our trial state combines exact antisymmetry, an analytic short-range cusp, and a compact neural residual, while optionally evaluating the Slater determinant on backflowed coordinates:

$$\Psi_{\theta,\beta}(R) = \mathrm{SD}\big(\tilde{R}'\big)\ \exp\Big(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(\tilde{R})\Big), \tag{3.2}$$

$$\tilde{R}' = \tilde{R} + \Delta_\beta(\tilde{R}). \tag{3.3}$$

Here SD enforces antisymmetry; u hard-wires the Kato cusp; W_θ (the NQS correlator) learns smooth medium/long-range structure; and Δ_β (backflow) adjusts the nodes by smoothly warping the coordinates used inside the determinants; its per-particle displacements $\Delta_{\beta,i}$ are written $\Delta\mathbf{x}_i$ in the results chapters. We deliberately feed the NQS with $\tilde{R}$ (not $\tilde{R}'$) to avoid self-reinforcing feedback and keep local-energy variance low.

Design principles. (i) *Separate guaranteed physics from learned details:* antisymmetry and the cusp are hard-wired; the network learns the rest. (ii) *Condition the geometry:* work in trap units and use soft-core pair features so that first/second derivatives are bounded. (iii) *Keep the learned object small and smooth:* once the cusp is explicit, a compact NQS suffices across (N, ω).

3.3 Slater reference

For closed shells we use a restricted product of spin-block determinants of the lowest $N/2$ harmonic oscillator orbitals (trap units):

$$\mathrm{SD}(\tilde{R}) = \det\big[\phi_a(\tilde{\mathbf{r}}_i)\big]_{i\in\uparrow,\ a\le N/2}\ \det\big[\phi_a(\tilde{\mathbf{r}}_j)\big]_{j\in\downarrow,\ a\le N/2}. \tag{3.4}$$

The correlator reweights amplitudes *within* nodal pockets; without backflow, the nodes are those of (3.4).

3.4 Neural correlator W_θ

As with the backflow, the correlator comes in two forms that we compare directly (Q1a, Chapter 7): a *separable* network that encodes particle- and pair-features independently and pools them, and a copresheaf *message-passing* (CTNN) network. We describe the separable network first—it is the baseline and also supplies the safe features and analytic cusp shared by both—and then the CTNN.

3.4.1 Separable correlator (DeepSets baseline)

The separable NQS aggregates *particlewise* and *pairwise* embeddings with permutation-invariant pooling and appends two global statistics, all in $\tilde{R}$. A core safety requirement is that pair channels satisfy $ds_k/d\tilde{r} \to 0$ as $\tilde{r} \to 0$ so that W_θ never injects spurious $1/\tilde{r}$ terms into $\nabla \log \Psi$ or $\Delta \log \Psi$.

Particle branch (DeepSets).

This part of the network is typically referred to as Deep Sets [34]. The permutation invariance of quantum many-body mechanics is naturally captured by this architecture. A neural network ϕ creates a latent embedding for each particle individually. All embeddings are pooled together to form a global representation, which is used later in the readout. Formally, we have

$$\mathbf{h}_i^\phi = \phi(\tilde{\mathbf{x}}_i) \in \mathbb{R}^{d_L}, \qquad \overline{\mathbf{h}}^\phi = N^{-1}\sum_i \mathbf{h}_i^\phi \in \mathbb{R}^{d_L}.$$

Pair branch (soft-core, even channels).

Although the particle branch could theoretically produce pair correlations, in practice it struggles to represent sharp near-field structure. We therefore add a dedicated pair branch that builds features from interparticle distances. This network processes all pairs $(i, j), i \neq j$ individually and pools the resulting latent embeddings. The problem is that raw distances $\tilde{r}_{ij}$ have unbounded derivatives at coalescence, which destabilizes Laplacian evaluations. We therefore build *safe* pair features with bounded first and second derivatives as $\tilde{r} \to 0$. With $\tilde{r}^{\text{soft}} = \sqrt{\tilde{r}^2 + \varepsilon^2}$ ($\varepsilon \in [0.15, 0.30]$ in trap units), we build six "safe" features

$$
\begin{aligned}
s_1 &= \log\bigl(1 + (\tilde{r}^{\text{soft}}/\varepsilon)^2\bigr), \quad s_2 = \tfrac{\tilde{r}^2}{\tilde{r}^2+\varepsilon^2}, \quad s_3 = (\tilde{r}^{\text{soft}}/\varepsilon)^2 \exp\bigl[-(\tilde{r}^{\text{soft}}/\varepsilon)^2\bigr], \\
\text{rbf}_g &= \exp(-g\, s_1), \quad g \in \{0.25, 1, 4\}, \qquad \mathbf{u}_{ij} = [s_1, s_2, s_3, \text{rbf}_{0.25}, \text{rbf}_1, \text{rbf}_4] \in \mathbb{R}^6.
\end{aligned} \tag{3.5}
$$

An MLP ψ maps $\mathbf{u}_{ij} \mapsto \mathbf{h}_{ij}^{\psi} \in \mathbb{R}^{d_L}$. A short-range gate

$$
\chi(\tilde{r}) = \frac{\tilde{r}^2}{\tilde{r}^2 + r_g^2}, \quad r_g \approx 0.3, \quad \chi(0) = \chi'(0) = 0,
$$

suppresses learned responses *at* coalescence; downstream we use $\chi\, \mathbf{h}_{ij}^{\psi}$.

3.4.2 Analytic short-range cusp (shared)

The analytic cusp below is shared by both correlator variants (and is independent of the backflow). Because the safe features are necessary in the pair branch, the NQS cannot represent the Kato cusp condition exactly. We therefore add an analytic cusp term to the exponent in (3.2), written in *physical* coordinates so that its slope at contact is manifestly correct:

$$
u_{\sigma\sigma'}(r) = \gamma_{\sigma\sigma'}\, r\, e^{-r/a_{\text{ho}}}, \qquad a_{\text{ho}} = 1/\sqrt{\omega}, \qquad \gamma_{\uparrow\downarrow} = \tfrac{1}{d-1}, \quad \gamma_{\uparrow\uparrow} = \tfrac{1}{d+1}. \tag{3.6}
$$

Because the linear prefactor is the physical separation r, the slope at contact is $\partial_r u|_0 = \gamma_{\sigma\sigma'}$, exactly the Kato condition in general d; for $d = 2$ this gives $\gamma_{\uparrow\downarrow} = 1$ and $\gamma_{\uparrow\uparrow} = \frac{1}{3}$, the standard two-dimensional quantum-dot values, derived—including the parallel-spin case, where antisymmetry already forces a linear node—in Appendix B.2. Only the exponential taper carries the trap length a_{ho}, suppressing long-range interference so the NQS need not unlearn cusp structure at medium and long range; in trap units ($\tilde{r} = \sqrt{\omega}\, r$) the same term reads $u = (\gamma_{\sigma\sigma'}/\sqrt{\omega})\, \tilde{r}\, e^{-\tilde{r}}$.

3.4.3 Separable readout

With contributions from the particle and pair branches, we form a global representation for the final readout. In addition to the pooled embeddings $\overline{\mathbf{h}}^{\phi}$ and $\overline{\mathbf{h}}^{\psi}$, we also include two global statistics that help the network gauge overall system size and density. We average pairs by default or apply degree-normalised radial attention

$$
w_{ij} = \exp[-(\tilde{r}_{ij}/r_c)^p], \quad \hat{w}_{ij} = w_{ij} \Big/ \sum_{k \neq i} w_{ik}, \quad (r_c, p) = (0.4, 6),
$$

then form $g_i = \sum_{j \neq i} \hat{w}_{ij}\, \mathbf{h}_{ij}^{\psi}$ and $\overline{\mathbf{h}}^{\psi} = N^{-1} \sum_i g_i$. Two system-level scalars in trap units,

$$
\mathbf{g}(\tilde{R}) = \Bigl[\tfrac{1}{Nd} \sum_i \|\tilde{\mathbf{x}}_i\|^2,\ \tfrac{2}{N(N-1)} \sum_{i<j} s_1(\tilde{r}_{ij})\Bigr] \in \mathbb{R}^2,
$$

are concatenated with $\overline{\mathbf{h}}^{\phi}$ and $\overline{\mathbf{h}}^{\psi}$ and mapped to a scalar by a tiny MLP ρ:

$$
W_\theta(\tilde{R}) = \rho\Bigl(\overline{\mathbf{h}}^{\phi} \parallel \overline{\mathbf{h}}^{\psi} \parallel \mathbf{g}(\tilde{R})\Bigr). \tag{3.7}
$$

3.4.4 Message-passing correlator (CTNN)

The message-passing correlator (code: `CTNNJastrowVCycle`) keeps the same node/edge embeddings, safe pair features, and analytic cusp, but replaces the independent pooling of the separable network with the copresheaf transport of §2.4.4: at each round the endpoint nodes are transported into an edge update and the updated edges are transported back and aggregated onto the nodes,

$$\mathbf{h}_{ij} \mathrel{+}= F_e\big(\mathbf{h}_{ij} \,\|\, \rho_{v\to e}\mathbf{h}_i \,\|\, \rho_{v\to e}\mathbf{h}_j\big), \qquad \mathbf{h}_i \mathrel{+}= F_v\Big(\mathbf{h}_i \,\|\, \operatorname*{Agg}_{j\neq i} w_{ij}^{\text{spin}}\, \rho_{e\to v}\mathbf{h}_{ij}\Big), \tag{3.8}$$

with residual updates and per-round transport maps. Unlike the single-round backflow, the correlator stacks these rounds in a multiscale "V-cycle": $n_\downarrow$ down-pass rounds, a linear coarsening to a bottleneck width, then $n_\uparrow$ up-pass rounds with skip connections back to the down-pass features (defaults $n_\downarrow = n_\uparrow = 2$). A permutation-invariant readout over the final node and edge statistics returns the scalar $W_\theta(\tilde{R})$, exactly as in (3.7). Holding the Slater reference, cusp, and training fixed and swapping only the separable readout for this V-cycle isolates the relational channel studied in Q1a (Chapter 7).

3.5 Backflow

Backflow deforms the coordinates *used only inside the Slater reference* so that the nodal surface can adapt to interparticle correlations:

$$\tilde{R}' \;=\; \tilde{R} + \Delta_\beta(\tilde{R}), \qquad \Psi_{\theta,\beta}(R) \;=\; \mathrm{SD}(\tilde{R}')\, \exp\Big(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(\tilde{R})\Big). \tag{3.9}$$

The correlator W_θ and cusp u are evaluated on $\tilde{R}$ to avoid feedback loops in the local energy; only the determinant "sees" $\tilde{R}'$. This choice cleanly separates (i) amplitude reweighting within nodal pockets (handled by $u + W_\theta$) from (ii) nodal *geometry* (handled by backflow).

Design goals. (i) *Stability near coalescence:* messages are built from mollified radii and short-range gates so that $\partial\Delta_\beta/\partial\tilde{r}$ stays bounded; (ii) *Permutation symmetry:* permutation equivariance holds by construction (shared maps and symmetric aggregation); (iii) *Close to identity at initialisation:* a zero-initialised last layer and a positive learnable scale keep $\Delta_\beta \approx 0$ at the start. Rotational symmetry, by contrast, is only *approximate*: the displacement head reads out a d-vector from node features that include the raw coordinates, so the map is not rotation-equivariant by construction, and it is encouraged instead by the geometry-based edge features and by rotationally-augmented sampling. A global translation is not a symmetry of the trap in any case; the zero-mean projection (3.19) is a regulariser that removes a spurious centre-of-mass drift, not an imposed physical symmetry.

We study *two* backflow architectures. Both map the configuration to a per-particle displacement field and share the output bounding, spin masking, and centre-of-mass projection below; they differ only in how a particle's displacement is made to depend on the others. The conventional network is the baseline of the architecture comparison in Chapter 7; the copresheaf CTNN is the production ansatz.

3.5.1 Conventional backflow (single-round messages)

The baseline (code: `BackflowNet`) uses a single round of pairwise messages. For each ordered pair (i, j) with $i \neq j$,

$$\mathbf{m}_{ij} = \phi_\beta\big(\tilde{\mathbf{x}}_i,\ \tilde{\mathbf{x}}_j,\ \mathbf{r}_{ij},\ \tilde{r}_{ij}^{\text{soft}},\ (\tilde{r}_{ij}^{\text{soft}})^2\big)\cdot w_{ij}^{\text{spin}}, \qquad \tilde{r}_{ij}^{\text{soft}} = \sqrt{\tilde{r}_{ij}^2 + \varepsilon^2}, \tag{3.10}$$

with ϕ_β a small MLP and $w_{ij}^{\text{spin}} \in \{0, 1\}$ a spin mask (same-spin only or all pairs). Messages are aggregated to a node state and a node network returns the displacement,

$$\mathbf{m}_i = \operatorname*{Agg}_{j\neq i} \mathbf{m}_{ij}, \qquad \delta\tilde{\mathbf{x}}_i = \psi_\beta\big(\tilde{\mathbf{x}}_i \,\|\, \mathbf{m}_i\big), \qquad \mathrm{Agg} \in \{\mathrm{sum}, \mathrm{mean}, \mathrm{max}\}. \tag{3.11}$$

There is no persistent edge state: each message is recomputed from the raw geometry and discarded after aggregation.

3.5.2 CTNN backflow (copresheaf)

The message-passing architecture (code: `CTNNBackflowNet`) instead maintains explicit *node and edge* features and transports information between them through learned linear maps—the copresheaf-style step of §2.4.4. Node features are embedded from the position and a spin scalar, edge features from the relative geometry,

$$\mathbf{h}_i^{(0)} = W_v\,[\tilde{\mathbf{x}}_i,\, s_i], \qquad \mathbf{h}_{ij}^{(0)} = \mathrm{MLP}_e\big(\mathbf{r}_{ij},\, |\mathbf{r}_{ij}|,\, |\mathbf{r}_{ij}|^2\big). \tag{3.12}$$

A node-to-edge transport $\rho_{v\to e}$ feeds the endpoint nodes into an edge update; an edge-to-node transport $\rho_{e\to v}$ returns the updated edges to the nodes, which are refined by a residual node update; a linear head reads out the displacement:

$$\mathbf{h}'_{ij} = \mathrm{MLP}'_e\big(\mathbf{h}_{ij}^{(0)} \,\|\, \rho_{v\to e}\mathbf{h}_i^{(0)} \,\|\, \rho_{v\to e}\mathbf{h}_j^{(0)}\big), \tag{3.13}$$

$$\mathbf{m}_i = \operatorname*{Agg}_{j\neq i} w_{ij}^{\mathrm{spin}}\, \rho_{e\to v}\big(\mathbf{h}'_{ij}\big), \tag{3.14}$$

$$\mathbf{h}_i = \mathbf{h}_i^{(0)} + \mathrm{MLP}_v\big(\mathbf{h}_i^{(0)} \,\|\, \mathbf{m}_i\big), \qquad \delta\tilde{\mathbf{x}}_i = W_\Delta\, \mathbf{h}_i. \tag{3.15}$$

The learned transport maps $\rho_{v\to e}, \rho_{e\to v}$ are the channel whose ablation collapses the backflow at the Wigner crystal (Chapter 7). The backflow uses a single copresheaf round; the message-passing *correlator* of the architecture comparison (`CTNNJastrowVCycle`) iterates the same style of update over a down/up "V-cycle".

3.5.3 Shared output, scaling, and initialisation

Both variants take inputs in trap units ($\tilde{\mathbf{x}} = \sqrt{\omega}\,\mathbf{x}$), bound the displacement, and start at the Slater nodes,

$$\Delta_\beta(\tilde{R}) \;=\; \tanh\big(\delta\tilde{X}\big)\cdot s_\beta, \qquad s_\beta = \mathrm{softplus}(s_\beta^{\mathrm{raw}}) > 0, \tag{3.16}$$

with the last linear layer (ψ_β or W_Δ) zero-initialised so that $\Delta_\beta \approx 0$ initially. Empirically this parameterisation stabilises the SR/energy tails and protects the Laplacian.

3.5.4 Short-range behaviour and gates

As $\tilde{r}_{ij}\to 0$, raw $1/\tilde{r}$ factors can make $\nabla\cdot\Delta_\beta$ spiky. We therefore (i) mollify $\tilde{r}_{ij}$ as in (3.10), and optionally (ii) gate the message with

$$\chi_{\mathrm{bf}}(\tilde{r}_{ij}) = \frac{\tilde{r}_{ij}^2}{\tilde{r}_{ij}^2 + r_g^2}, \qquad \chi_{\mathrm{bf}}(0) = \chi'_{\mathrm{bf}}(0) = 0, \quad r_g \approx 0.3, \tag{3.17}$$

or with degree-normalised radial attention

$$w_{ij} = \exp\big[-(\tilde{r}_{ij}/r_c)^p\big], \qquad \hat{w}_{ij} = \frac{w_{ij}}{\sum_{k\neq i} w_{ik}}. \tag{3.18}$$

Either choice keeps the learned response smooth at coalescence and limits the BF contribution to the local energy variance.

3.5.5 Symmetries and COM neutrality

Permutation symmetry is automatic (pairwise messages and symmetric aggregation). To avoid a spurious centre-of-mass (COM) drift, we project the flow to zero mean:

$$\Delta_\beta \;\leftarrow\; \Delta_\beta \;-\; \frac{1}{N}\sum_{i=1}^{N} \Delta_{\beta,i}. \tag{3.19}$$

The messages are built from the relative vectors $\mathbf{r}_{ij}$ and the zero-mean projection removes the spurious centre-of-mass drift; as noted under the design goals, the map is not translation-invariant or rotation-equivariant by construction (the displacement head reads a d-vector from the absolute coordinates).[1]

[1] Implementation note: the mean is taken over particles *within each configuration*, not across configurations in the batch, so the constraint removes centre-of-mass motion configuration by configuration. It is a one-liner and removes the small residual COM drift seen in ablations.

3.5.6 Effect on the nodes and variational safety

Because only SD is evaluated at $\tilde{R}'$, backflow changes the *nodes* but leaves the correlator exponent on $\tilde{R}$. No Jacobian term is introduced (we are not performing a change of variables of the full wavefunction). This "Feynman–Cohen style" [43] design targets fixed-node error directly while keeping the amplitude model simple and smooth; the same idea in its modern quantum-Monte-Carlo form is due to Holzmann and Ceperley [44].

3.5.7 Regularization, initialisation, and cost

We found the following defaults robust across (N, ω): (i) $\varepsilon \in [10^{-6}, 10^{-4}]$ (trap units) for the soft norm; (ii) bounded output tanh with a learnable scale s_β initialised in $[0.02, 0.10]$; (iii) optional gate χ_{bf} with $r_g \approx 0.3$ (trap units). Zero-initialising the last layer of ψ_β makes the ansatz start exactly at the Slater nodes and improves optimiser stability. The compute/memory cost scales as $O\big(BN^2(d+H)\big)$ for batch size B and hidden width H.

3.5.8 Mapping to implementation

In the conventional network, ϕ_β and ψ_β correspond to `BackflowNet.phi` (message MLP) and `BackflowNet.psi` (node/update MLP). In the copresheaf network, the embeddings, transport maps, and update MLPs of (3.13)–(3.15) correspond to `CTNNBackflowNet`'s `node_embed/edge_embed`, `rho_v_to_e/rho_e_to_v`, `edge_update/node_update`, and `dx_head`. For both, aggregation is selectable (`sum|mean|max`); spin handling (`use_spin`, `same_spin_only`) produces the mask w_{ij}^{spin}; output control matches (3.16) via `out_bound="tanh"` and s_β = softplus(`bf_scale_raw`); and (3.19) is enforced by subtracting the per-configuration mean over particles of the predicted Δ_β before forming $\tilde{R}'$.

Summary. Backflow provides a compact, symmetry-respecting way to move the Slater nodes toward the true correlated nodes. With mollified distances, optional short-range gates/attention, zero-mean projection, and bounded outputs, it integrates cleanly with the analytic cusp and smooth NQS correlator, designed to reduce nodal error while controlling local-energy variance.

3.6 Architecture sizes and hyperparameters

Table 3.1 collects the representative settings, so that the architecture claims of Chapter 7 can be read as matched-capacity comparisons rather than size effects. The controlled correlator/backflow study fixes each pair of variants at the same capacity tier; the reported ground-state energies use the wider production networks. A key point is that the message-passing and separable variants are held at comparable parameter counts, so a difference in outcome cannot be attributed to one simply being larger—and where Chapter 7 departs from matched capacity it does so in the *separable* network's favour, scaling the DeepSet from 20k to 164k parameters against a fixed 80k CTNN without closing the gap. All settings below are those of the accompanying code (`scripts/run_depth_analysis.py` for the comparison tiers; `scripts/source_state_check.py` for the production widths).

Table 3.1: Representative architecture and training hyperparameters. Widths are hidden dimensions; "rounds" are message-passing steps ($n_\downarrow + n_\uparrow$ for the V-cycle). The two correlator variants (and the two backflow variants) are matched in capacity within the comparison tier.

Component	Variant	Comparison tier	Production	Params
Correlator	CTNN (V-cycle)	node/edge 32, bottleneck 16, 2+2 rounds	width 128, 2 layers	79.8k
	DeepSet	pair-MLP 64 × 4, out 32	width 128, 2 layers	47.7k
Backflow	CTNN (copresheaf)	node/edge 64, 1 round	width 128, 3 layers	83k
	Conventional	msg 64, node-MLP 64 × 3	width 128, 3 layers	83k
Readout	(correlator)	hidden 64, 2–3 layers, SiLU	—	—

Optimiser: Adam, lr $= 5 \times 10^{-4}$ (backflow), 5×10^{-5} (Jastrow); CG-SR with Tikhonov damping λ, trust-region cap and CG restarts. Collocation batch $n_{\mathrm{coll}} \sim 3 \times 10^3$, oversampling $m = 8$–16.

Chapter 4

Optimisation

4.1 Primary objective and two–stage training

Given a parameterised wavefunction Ψ_θ, the variational ground–state energy is

$$E(\theta) = \frac{\langle \Psi_\theta | H | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} = \mathbb{E}_{\tilde{R} \sim \pi_\theta}\big[E_L(\tilde{R})\big], \qquad \pi_\theta(\tilde{R}) = \frac{|\Psi_\theta(\tilde{R})|^2}{\int |\Psi_\theta|^2}. \tag{4.1}$$

The local energy is evaluated in *physical* coordinates, consistently with the Hamiltonian (3.1),

$$E_L(R) = -\tfrac{1}{2} \sum_{i=1}^{N} \Big[\Delta_{\mathbf{r}_i} \ln \Psi_\theta + \|\nabla_{\mathbf{r}_i} \ln \Psi_\theta\|^2 \Big] + \tfrac{1}{2}\, \omega^2 \sum_{i=1}^{N} r_i^2 + \sum_{i<j} \frac{1}{r_{ij}}, \tag{4.2}$$

and the zero–variance principle holds: $\mathrm{Var}_{\pi_\theta}[E_L] = 0$ iff Ψ_θ is an eigenstate. The network *features* are computed in trap units $\tilde{\mathbf{r}} = \sqrt{\omega}\,\mathbf{r}$ for conditioning, but the local energy itself is formed in physical units, so no ω factors are absorbed into the kinetic, trap, or Coulomb terms. (In trap units the same operator reads $E_L = \omega\big[-\frac{1}{2}\sum_i (\Delta_{\tilde{\mathbf{r}}_i} \ln \Psi_\theta + \|\nabla_{\tilde{\mathbf{r}}_i} \ln \Psi_\theta\|^2) + \frac{1}{2}\sum_i \tilde{r}_i^2\big] + \sum_{i<j} \sqrt{\omega}/\tilde{r}_{ij}$; the kinetic and trap terms scale by ω and the Coulomb term by $\sqrt{\omega}$.)

Our optimisation follows two stages that mirror the implementation:

1. **Residual–based pretraining (stage I).** We fit the residual of E_L to a moving scalar target E_{eff} using the loss $\mathcal{L}_{\text{res}}(\theta) = \mathbb{E}\big[(E_L(\tilde{R}) - E_{\text{eff}})^2\big]$. We begin with a *self–target* (stabilizer) $E_{\text{eff}} = \mu$ where $\mu = \mathbb{E}[E_L]$ over the current batch. This choice is equivalent to variance minimization and prevents the network from chasing a potentially inaccurate external target before it can represent low–variance pockets. We then anneal toward a trusted reference E_{DMC} via

   $$E_{\text{eff}}(\alpha) \;=\; \alpha\, E_{\text{DMC}} + (1-\alpha)\, \mu, \qquad \alpha \in [0,1], \tag{4.3}$$

   with a smooth cosine schedule $\alpha = \alpha(t)$ that ramps from α_{start} to α_{end} over a prescribed fraction of epochs (code: `alpha_start`, `alpha_end`, `alpha_decay_frac`). Setting the `objective` to `"residual"` uses $E_{\text{eff}} = \mu$; `"energy"` uses $E_{\text{eff}} = E_{\text{DMC}}$; and `"energy_var"` uses the annealed blend above.

2. **Stochastic Reconfiguration (stage II).** After residual pretraining, we refine with SR (natural–gradient VMC). Writing $O(\tilde{R}) = \partial_\theta \log \Psi_\theta(\tilde{R})$,

   $$S \;=\; \mathbb{E}_\pi\big[O^\top O\big], \qquad g \;=\; 2\,\mathbb{E}_\pi\big[(E_L - \mathbb{E}_\pi[E_L])\, O\big], \tag{4.4}$$

   we solve the damped normal equations

   $$(S + \lambda I)\, \Delta\theta \;=\; -\,g, \tag{4.5}$$

   using (preconditioned) conjugate gradients with restarts, a trust–region scale on the final step, and batchwise centering/whitening of O (code: `damping`, `max_param_step`, CG options).

This pipeline exploits the zero–variance principle for stable shaping of W_θ (and backflow) before applying the geometry–aware SR update.

4.2 Configuration–space sampling

Guiding principle (permutation invariance). In a many–body fermionic system, *which* particle indices are close is irrelevant; what matters is *how many* close pairs and at which length scales. Consequently, we can design samplers that (i) emphasize *configurational motifs* (clusters, shells, dimers, long tails) rather than specific labels, and (ii) deliberately permute particles and randomly rotate configurations to explore the space efficiently without biasing towards any ordering.

4.2.1 Stratified capped–simplex mixture

We draw collocation batches $X \in \mathbb{R}^{B\times N\times d}$ from a five–component mixture,

$$\text{(centre, tails, mixed, shells, dimers)},$$

with nonnegative weights $w \in \Delta^4$ projected to a *capped simplex* $\{\, w : \sum_k w_k = 1,\ 0 \le w_k \le 0.3 \,\}$ to enforce coverage (code: `_project_simplex_with_caps`). Each component proposes x as follows (all in physical units; the code internally scales by $a_{\text{ho}} = 1/\sqrt{\omega}$):

- **Centre:** narrow isotropic Gaussian around the trap centre (samples near the density core).
- **Tails:** wide Gaussian (probes low–density outskirts and large–r behaviour).
- **Mixed:** hybrid batches mixing narrow and moderate spreads per configuration (tests robustness to inhomogeneous spreads).
- **Shells:** points placed on K hyperspherical shells with trap–scaled radii $\{r_k\}$ and occupancy $\{q_k\}$, with small radial/tangential jitter (captures ring/shell order at weak traps).
- **Dimers:** we induce a few close pairs per configuration by sampling short interparticle offsets with random directions and log–uniform distances (enforces near–coalescence coverage without singularities).

We then apply a random in–plane rotation (for $d = 2$), and a random permutation of particle indices; both preserve the target expectations by symmetry.

Adaptive mixture and shell occupancy. Per epoch we compute simple difficulty scores and update w and q by exponentiated–gradient (EG) steps with momentum, temperature, and an exploration term; w is then projected back to the capped simplex (code: `eg_eta`, `eg_temp`, `eg_momentum`, `explore_gamma`, `prob_floor`). Mixture difficulty uses the residual $(E_L - E_{\text{eff}})^2$ per component; shell difficulty uses a proxy

$$\underbrace{V_i}_{\text{harm.+Coulomb per particle}} + \underbrace{\gamma_{g^2}\,\|\nabla_{\mathbf{x}_i}\log\Psi\|^2}_{\text{stiff gradients}} + \underbrace{\gamma_{\text{cusp}}\, r_{\min}^{-1}}_{\text{coalescence pressure}},$$

aggregated by nearest shell index (code: `g2_weight`, `cusp_gamma`, safe clip `rmin_clip`). This shifts probability mass toward underfit regions *without* collapsing coverage thanks to the caps.

Hard–example injection. We maintain a small buffer of previously challenging configurations (based on residuals) and stochastically inject them, with mild multiplicative and additive jitter, into the next batch (code: `sampler_hard_*`). This provides a curriculum–like pressure while avoiding overfitting to outliers.

Robustness tweaks. We trim extreme local–energy outliers by quantiles (default 3%) *before* forming losses/updates; we also micro–batch the evaluation to keep memory bounded, and apply gradient clipping to f_θ (and backflow) parameters.

4.3 Residual–based training (stage I)

For a batch $\{\tilde{R}_b\}_{b=1}^{B}$ we compute $E_L(\tilde{R}_b)$ and the batch mean $\mu = \frac{1}{B}\sum_b E_L(\tilde{R}_b)$. The residual loss is

$$\mathcal{L}_{\mathrm{res}}(\theta;\alpha) = \frac{1}{B}\sum_{b=1}^{B}\big(E_L(\tilde{R}_b) - E_{\mathrm{eff}}(\alpha)\big)^2, \qquad E_{\mathrm{eff}}(\alpha) = \alpha\, E_{\mathrm{DMC}} + (1-\alpha)\,\mu. \tag{4.6}$$

- **Self–target warmup ($\alpha = 0$).** Minimizing $\mathbb{E}[(E_L - \mu)^2]$ is exactly variance minimization; it teaches the network to produce smooth, low–variance amplitudes inside nodal pockets and reduces sensitivity to early sampler noise.
- **Anneal to reference ($\alpha \uparrow$).** As capacity grows, we blend in the external target, gently steering the mean energy while retaining the variance pressure. The cosine ramp (code: `alpha_start`, `alpha_end`, `alpha_decay_frac`) avoids abrupt objective switches.
- **Pure energy fit ($\alpha = 1$).** If desired, we can end stage I with $E_{\mathrm{eff}} = E_{\mathrm{DMC}}$.

In practice we compute $\nabla_\theta \mathcal{L}_{\mathrm{res}}$ by autodiff, use micro–batches of size `micro_batch`, apply optional gradient clipping (code: `grad_clip`), and log per–component usage and shell radii in physical units.

Local–energy derivatives in practice. We support three Laplacian backends:

1. **Exact:** chunked second partials (reference–accurate, heavier).
2. **HVP–Hutchinson:** $\Delta \log\Psi = \mathbb{E}_v[v^\top H_{\log\Psi} v]$ with Rademacher v (fast and stable; fallback to FD if a probe is non–finite).
3. **FD–Hutchinson:** centred finite differences of directional gradients (robust in corner cases; slightly noisier).

We average a small number of probes (2–16 in ablations), and trim quantiles before forming losses.

4.4 Stochastic Reconfiguration (stage II)

Stochastic reconfiguration [31, 45] preconditions the energy gradient with the (inverse) quantum geometric tensor, and is the imaginary-time / natural-gradient step for a variational state [29, 32]. It uses the covariance form of the energy gradient

$$\partial_{\theta_k} E = 2\,\mathrm{Cov}_\pi\big(E_L, O_k\big) = 2\Big(\langle E_L O_k\rangle - \langle E_L\rangle\,\langle O_k\rangle\Big), \tag{4.7}$$

and approximates a natural–gradient step by solving $(S + \lambda I)\Delta\theta = -g$ with the quantum geometric tensor S of §2.2.5. Implementation details:

- **Centering/whitening.** We centre O per batch and whiten by the diagonal of S before CG to improve conditioning.
- **Trust region.** The final step is scaled to satisfy a maximum parameter norm (code: `max_param_step`).
- **Damping.** A small λ (code: `damping`) stabilizes low–variance directions.

Because our backflow only alters the Slater coordinates (and W_θ stays on safe features of $\tilde{R}$), the SR statistics remain well–behaved even when nodes shift.

Optimiser variants compared. Three first-order updates appear in the results, and the comparison between them is itself a finding (Chapter 7, §7.2): (i) **Adam** [27], a diagonal adaptive step used as the robust default, with separate learning rates for the Jastrow and backflow blocks; (ii) **plain SR**, the natural-gradient step above with a fixed diagonal damping λ; and (iii) **CG-SR**, the same step solved matrix-free by (preconditioned) conjugate gradients with Tikhonov damping and a trust-region cap, which converges faster where S is well conditioned but becomes unstable in the weak-confinement regime where S is nearly singular. Solving the SR step without forming S is what makes it affordable at $\mathcal{O}(10^5)$ parameters; the same obstacle is addressed from the other side by the minSR and kernel-formulation reductions, which exploit the fact that the number of samples, not the number of parameters, sets the true rank of the problem [46, 47]. Under $|\Psi|^2$ sampling the Adam and SR *fixed points* coincide; they differ in conditioning and in the collocation regime, which is the subject of §7.2.

4.5 Sampling paradigms and the weak-form collocation objective

The pipeline above draws its batches from an analytic proposal rather than a Markov chain, so the choice of *sampling measure* must be stated explicitly: the results treat it as a variable in its own right. Two paradigms appear in this thesis. In *variational Monte Carlo* (VMC) the batch is drawn from $|\Psi_\theta|^2$ itself, usually by a Metropolis chain, so the estimators of the QGT and of the gradient share the very metric they are meant to precondition. In *collocation* the batch is drawn from a fixed analytic proposal $q(\mathbf{R})$ and reweighted to $|\Psi_\theta|^2$ by importance weights $w_i = |\Psi_\theta(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$, so that the gradient training requires no Markov chain. A Metropolis probe is used only outside the gradient loop—for periodic checkpoint selection and for the final energy certification.

For the collocation objective we use a *weak form*: the local energy E_L enters only as a reward in a score-function (REINFORCE) gradient [48],

$$\nabla_\theta \langle E\rangle = 2\,\mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}\big[(E_L(\mathbf{R}) - b)\,\nabla_\theta \log|\Psi_\theta(\mathbf{R})|\big], \tag{4.8}$$

with a variance-reducing baseline b (a running mean of E_L), and the Laplacian inside E_L is never differentiated through. This is deliberate. Differentiating the Laplacian through a backflow-shifted determinant is the source of the nodal instability analysed in Appendix C; the weak form keeps the full local-energy signal as a reward while sidestepping that pathology. Two robustness controls stabilise the reward: local energies are clipped to a robust median/MAD location–scale band ($\sim 5\sigma$), and the most extreme rewards are quantile-trimmed before the gradient is formed.

4.5.1 Analytic proposal and importance resampling

The collocation points are drawn from an analytic mixture of isotropic Gaussians centred at the trap minimum and reweighted to $|\Psi_\theta|^2$ [49],

$$q(\mathbf{R}) = \frac{1}{K}\sum_{k=1}^{K}\mathcal{N}\big(\mathbf{R};\,0,\,\sigma_{f,k}^2/\omega\; I_{Nd}\big), \quad \sigma_f \in \{0.8, 1.3, 2.0\}, \qquad w_i = \frac{|\Psi_\theta(\mathbf{R}_i)|^2}{q(\mathbf{R}_i)}, \tag{4.9}$$

where the $1/\omega$ scaling ties the proposal width to the oscillator length $\ell = 1/\sqrt{\omega}$ and the three components cover core, bulk, and tails. A batch of n_{coll} points is obtained by multinomial resampling on $p_i = w_i/\sum_j w_j$ from $m\,n_{\text{coll}}$ candidates ($m \approx 8$–16). Reweighting quality is tracked by the effective sample size $\text{ESS} = (\sum_i w_i)^2/\sum_i w_i^2$ and, as a heavier-tailed diagnostic, the Pareto $\hat{k}$ of the weights [50]; both degrade when q ceases to overlap $|\Psi_\theta|^2$, the failure mode that bounds the paradigm at large N and weak ω. Three controls manage the tails: weight tempering ($w \to w^\alpha$, $\alpha < 1$), log-weight clipping at an upper quantile, and ESS-adaptive oversampling that raises m when the achieved ESS falls below a floor. An instability-rollback reverts to the last good checkpoint and decays the learning rate if the energy jumps beyond a threshold.

4.5.2 Physics-informed proposal for the Wigner regime

The origin-centred proposal (4.9) places its mass where a Wigner molecule has almost none: at strong correlation the density collapses onto a shell at the classical ring radius r_0, and $|\Psi_\theta|^2$ has essentially no support near the origin, so the ESS collapses. We therefore replace q by a proposal built from the Wigner shell itself—a radial Gaussian centred at r_0 (set by balancing confinement against Coulomb repulsion) times an angular distribution over the ring, so that proposed configurations resemble electrons spread around the ring rather than clustered at the centre. This restores a usable ESS and carries MCMC-free training to $\omega \approx 0.0035$ (Chapter 7, §7.4), far past where the naive proposal collapses; the remaining wall is *angular* crystallisation, analysed there. Difficult confinements are reached by *cascade warm-starting*: parameters converged at one ω initialise the next, following $\omega : 1.0 \to 0.5 \to 0.1 \to 0.01 \to 0.001$. Whether the two paradigms reach the same *state*, and where they diverge, is one of the questions of Chapter 7.

4.6 Units, scaling, and batch construction

All features are computed in trap units $\tilde{\mathbf{x}} = \sqrt{\omega}\,\mathbf{x}$ so that length scales and gates (e.g. the short–range gate in the NQS and backflow) behave consistently across ω. The sampler internally widens/narrows Gaussians and shell radii using $a_{\text{ho}} = 1/\sqrt{\omega}$ in a fixed, explicit way, so that the five components cover comparable *dimensionless* regimes for $\omega \in [0.01, 1]$. Batches are i.i.d. by design; we rely on the stratified mixture (with caps, EG–adaptation, and hard–injection) rather than long MCMC chains to traverse configurations, which markedly reduces correlation and simplifies accounting.

4.7 Workflow summary

1. **Sample collocation set** X from the stratified capped–simplex mixture (centre, tails, mixed, shells, dimers), then apply rotation and permutation; optionally inject hard examples.
2. **Compute** E_L using the selected Laplacian backend; trim outliers.
3. **Stage I (residual).** Minimize $\frac{1}{B}\sum_b (E_L(\tilde{R}_b) - E_{\text{eff}}(\alpha))^2$ with α ramped from 0 to α_{end}; update mixture/shell weights via EG.
4. **Stage II (SR).** Form S and g, solve $(S + \lambda I)\Delta\theta = -g$, apply a trust–region step.

This procedure keeps the early dynamics variance–dominated and sampler–aware, then transitions to a geometry–aware natural–gradient refinement. Empirically it yields stable convergence across (N, ω), including weak traps where shell/dimer coverage and permutation–respecting stratification are essential.

4.8 Statistical uncertainties

Every energy and observable we report carries a statistical error, and because our samples are correlated—within an MCMC chain, and across the score-function updates that shape the final ensemble—a naive standard error over B samples understates it. We therefore estimate the error on a scalar mean $\bar{A} = \frac{1}{B}\sum_i A(\mathbf{R}_i)$ by *blocking*: successive samples are averaged into blocks of growing size and the plateau of the block standard error as the block size exceeds the autocorrelation time is taken as the uncertainty [51, 52]. For the final ground-state energy this is the dominant reported error; a parenthetical digit, e.g. 155.8738(7), denotes one standard error on the last quoted figure.

For quantities that are *nonlinear* functions of sample statistics—participation ratios, effective ranks, PCA cosines, and the coupling diagnostics of Chapter 6—a blocked standard error is not available in closed form, and we use a *block bootstrap* instead: blocks (of the plateau size above) are resampled with replacement, the statistic is recomputed on each resample, and its spread across resamples is the reported error (§5.2.9). Where a number is quoted as a seed average, we additionally report the spread across independent random seeds, which captures optimisation—not just sampling—variability; the two are distinguished in the captions. Reference energies drawn from the literature carry the uncertainties quoted by their sources.

Chapter 5

Analysis

This chapter describes *what* we measure on fresh $|\Psi_\theta|^2$ samples, *how* we compute the diagnostics, and *why* each quantity is informative. Results and numbers are presented in Chapter 6.

5.1 Structural diagnostics and Wigner markers

Unless stated otherwise, curves are accumulated in *trap units* $\mathbf{y} = \sqrt{\omega}\,\mathbf{x}$ to compare shapes across ω, while all headline distances are reported in *Bohr* (physical units $\mathbf{x}$). Production chains are thinned; we split samples into halves for stability checks (acceptance window, JSD between halves, and reproducibility under restarts). Using trap units to compare shapes across confinements is standard in parabolic dots; see the overview in [10].

5.1.1 Pair distribution and radial statistics

We estimate the pair distribution function $g(r)$ via a shell-area corrected histogram in trap units,

$$g(r_\alpha) \;=\; \frac{A_{\text{eff}}}{N(N-1)}\,\frac{\big\langle \sum_{i<j} \mathbf{1}\{r_{ij} \in [r_\alpha^-, r_\alpha^+)\}\big\rangle}{2\pi r_\alpha\,\Delta r_\alpha}, \qquad r_{ij} = \|\mathbf{y}_i - \mathbf{y}_j\|_2, \tag{5.1}$$

with $A_{\text{eff}} = \pi R_{0.95}^2$ and $R_{0.95}$ the 95% quantile of single-particle radii in trap units. This $g(r)$-based structural analysis follows established practice in mesoscopic 2D electron systems and quantum dots [7, 10, 12]. We also form the normalised *radial probability* $P(r) \propto r^{d-1} g(r)$ and report the mode r_{mode}, mean $\bar{r}$, standard deviation σ_r, and FWHM (all in Bohr). A compact width–to–scale marker is

$$\gamma \;=\; \sigma_r / r_{\text{mode}}, \tag{5.2}$$

which decreases as correlations strengthen (narrower distribution around a larger typical radius), consistent with the liquid→Wigner crossover picture [7, 12].

5.1.2 Automatic shell detection and topologies

For each frame we sort the single-particle radii $s_1 \le \cdots \le s_N$ (trap units), compute gaps $g_k = s_{k+1} - s_k$, and flag the frame as *two-shell* if the largest gap exceeds a robust threshold:

$$\max_k g_k \;\ge\; \tau\,\text{median}(g_k), \qquad \tau \in [2.5, 3.5] \text{ (default 3.0)}. \tag{5.3}$$

The cut radius is the midpoint of the maximal gap; the inner multiplicity n_{in} defines the topology $(n_{\text{in}}, N - n_{\text{in}})$. Shelling and discrete ring occupancies are well known in parabolic Coulomb clusters and quantum dots [8–10]; our detector provides a reproducible, data-driven assignment in that spirit. We report: (i) the fraction of two-shell frames, (ii) the histogram over n_{in} among two-shell frames, and (iii) for selected topologies (e.g. $(1,5)$ or $(3,9)$) the fraction among *all* frames.

Shell-resolved reconstruction of $g(r)$. On two-shell frames we decompose pairs into II, IO, and OO and accumulate histograms $g_{\mathrm{II}}, g_{\mathrm{IO}}, g_{\mathrm{OO}}$. With empirical weights $w_{\mathrm{II}}, w_{\mathrm{IO}}, w_{\mathrm{OO}}$ (combinatorial counts) we reconstruct $\tilde{g}(r) = w_{\mathrm{II}}g_{\mathrm{II}} + w_{\mathrm{IO}}g_{\mathrm{IO}} + w_{\mathrm{OO}}g_{\mathrm{OO}}$ and report $\cos\angle(g, \tilde{g})$. Values ≈ 1 indicate that the observed $g(r)$ shape is explained by shell geometry rather than sampling artifacts, in line with shell-based descriptions in [8–10].

5.1.3 Orientational order and "ring stiffness"

On each detected ring (inner or outer) we centre the mean angle to remove global rotations and define angles $\{\phi_k\}_{k=1}^{n}$. Bond-orientational order is

$$\Phi_m \;=\; \frac{1}{n}\sum_{k=1}^{n} e^{\,\mathrm{i}m\phi_k}, \qquad m = n, \tag{5.4}$$

so that $|\Phi_m| \in [0, 1]$ measures n-fold orientational order (e.g. m=5 for $(1, 5)$, m=9 for the outer ring of $(3, 9)$), analogous to standard bond–orientational order parameters used in 2D Coulomb/Yukawa crystals [11]. As a complementary, dimensionless *angular Lindemann* number we use

$$\lambda_\phi \;=\; \frac{\mathrm{std}(\Delta\phi)}{\mathbb{E}[\Delta\phi]}, \qquad \Delta\phi \;=\; \text{nearest-neighbour angular spacing on the ring.} \tag{5.5}$$

Smaller λ_ϕ indicates a stiffer (more Wigner-like) ring. This is a Lindemann–type orientational metric tailored to finite rings, motivated by the use of Lindemann criteria to diagnose intershell rotational (angular) melting in mesoscopic Coulomb clusters and dots [9, 12].

5.1.4 Scaling check for $N = 2$

To connect with the classical strong-coupling limit we fit

$$r_{\mathrm{mode}}(\omega) \;=\; C\,\omega^{\alpha} \tag{5.6}$$

by linear regression in $\big(\log\omega, \log r_{\mathrm{mode}}\big)$ and compare the slope with the classical reference $\alpha_{\mathrm{cl}} = -2/3$ (two charges in a harmonic trap at strong coupling; see, e.g., the reviews in [10]). Approach toward α_{cl} as $\omega\downarrow$ is one of our Wigner crossover indicators, consistent with the crossover phenomenology reported in [7, 12].

5.1.5 Reporting and stability checks

For all scalars and histograms we report standard errors calculated over thinned blocks or frames. We verify: (i) acceptance in the target window, (ii) Jensen–Shannon divergence and ℓ_2 distance between halves of production (stationarity), (iii) robustness of topology fractions and $|\Phi_m|$ under $\tau \in [2.5, 3.5]$ and mild pre-smoothing of radii, and (iv) reproducibility of $g(r)$ and $P(r)$ under independent sampler restarts. All diagnostics operate directly on saved bundles $\{\mathbf{X}_{\mathrm{Bohr}},\, r,\, g(r)\}$.

5.2 Representation analysis

We probe what the networks learn by evaluating fixed, trained models on fresh $|\Psi_\theta|^2$ samples. Unless stated otherwise, features are computed in *trap units* $\mathbf{y} = \sqrt{\omega}\,\mathbf{x}$ to match the correlator's internal scaling. All estimates use thinned MCMC frames, split into two equal blocks A and B; PCA fits on A and reports on B (and vice versa) to avoid circularity. We use `float64` throughout.

Feature taps. Let $z_\theta \in \mathbb{R}^D$ denote the concatenated correlator representation

$$z_\theta = \big[\overline{\phi} \mid \overline{\psi} \mid \mathbf{g}\big],$$

where $\overline{\phi}$ are per-particle summaries, $\overline{\psi}$ are pairwise summaries, and $\mathbf{g}$ are global/extras. The residual head is linear:

$$f_{\mathrm{base}}(z) = \rho_\theta(z) = w^{\top}z + b, \qquad f = f_{\mathrm{base}} + f_{\mathrm{cusp}}. \tag{5.7}$$

Backflow returns a displacement field $\Delta x \in \mathbb{R}^{N\times d}$ that enters only the Slater determinant; the correlator and cusp are evaluated on the unshifted $\tilde{R}$ (§3.5). The coupling diagnostics below additionally probe the correlator on the shifted coordinates as a counterfactual sensitivity.

5.2.1 Preprocessing and PCA

Given a sample matrix $Z \in \mathbb{R}^{K\times D}$ (rows are frames), we standardize each column using means and standard deviations estimated on block A:

$$\tilde{Z} = (Z - \mu_A) \odot \sigma_A^{-1}.$$

We perform PCA on $\tilde{Z}_A$ (SVD of the covariance) to obtain orthonormal loadings $U = [u_1, \ldots, u_D]$. Scores on B are $S_B = \tilde{Z}_B U$. Unless noted, all quantities reported below are computed on the held-out block.

5.2.2 Entropy effective rank

Let $\{\lambda_i\}_{i=1}^{D}$ be the eigenvalues of $\mathrm{Cov}(\tilde{Z})$ (on the held-out block). With $p_i = \lambda_i / \sum_j \lambda_j$, the *entropy effective rank* is

$$r_{\mathrm{eff}}(Z) \;=\; \exp\Big(-\sum_{i=1}^{D} p_i \log p_i\Big),$$

which equals 1 for perfectly one-dimensional representations and increases with spectral spread.

5.2.3 PC ablations of the head

To test how many latent axes the head truly uses, we project $\tilde{Z}$ onto the top-k PCs:

$$\Pi_k = \sum_{i=1}^{k} u_i u_i^{\top}, \qquad z^{(k)} = \Pi_k \tilde{z},$$

and re-evaluate the trained linear head *without refitting*: $\hat{f}^{(k)} = \rho_\theta(z^{(k)})$. We report a *normalised* error on block B,

$$\text{rel-MAE}(k) \;=\; \frac{\mathbb{E}_B \,|\, \hat{f}^{(k)} - f_{\text{base}} \,|}{\mathrm{std}_B(f_{\text{base}})},$$

together with the learning curve in k.

5.2.4 PC1 block power decomposition

To attribute the leading axis to branches, we partition u_1 conformably with $z = [\overline{\phi} \,|\, \overline{\psi} \,|\, \mathbf{g}]$ and report

$$\mathrm{power}(\overline{\phi}) = \frac{\|u_{1,\overline{\phi}}\|_2^2}{\|u_1\|_2^2}, \quad \mathrm{power}(\overline{\psi}) = \frac{\|u_{1,\overline{\psi}}\|_2^2}{\|u_1\|_2^2}, \quad \mathrm{power}(\mathbf{g}) = \frac{\|u_{1,\mathbf{g}}\|_2^2}{\|u_1\|_2^2}.$$

Because PCA is performed on standardized features, these shares are comparable across blocks.

5.2.5 Linear probes from PCs to coarse physics

We regress interpretable summaries per configuration on the first k PC scores $S^{(k)}$ using ridge regression (tiny penalty $\alpha = 10^{-6}$):

$$y \approx S^{(k)}\beta, \qquad y \in \big\{\bar{r},\ \mathrm{Var}(r),\ \Pr(r < r_0),\ \text{shell contrast}\big\}.$$

We report R^2 on the held-out block; r_0 is a fixed near-contact cutoff in trap units.

5.2.6 Near-field gradient concentration (residual head)

For each frame we compute the minimum pair distance $r_{\min}$ (trap units) and the gradient of the residual head with respect to particle coordinates, $\nabla f_{\text{base}}(\mathbf{y})$. The concentration at quantile q is

$$\text{share}(q) \;=\; \frac{\mathbb{E}\big[\|\nabla f_{\text{base}}\|_2^2\, \mathbf{1}\{r_{\min} \le Q_q\}\big]}{\mathbb{E}\big[\|\nabla f_{\text{base}}\|_2^2\big]},$$

where Q_q is the q-quantile of $r_{\min}$. Values $\gg 1$ indicate that residual stiffness is concentrated near contact (the analytic cusp is insufficient); values ≈ 1 indicate that the cusp handles most short-range structure.

5.2.7 Backflow geometry and energy-locality shares

Field spectrum. We vectorize the backflow displacement per frame, $\delta = \text{vec}(\Delta x) \in \mathbb{R}^{Nd}$, standardize across frames, and compute $r_{\text{eff}}(\Delta x)$ via Sec. 5.2.2.

Where backflow acts. On the same thinned frames we evaluate local energies with and without backflow, $E_{\text{loc}}^{\text{BF}}$ and $E_{\text{loc}}^{\text{noBF}}$, using the *same* coordinates. We form the signed correction $\Delta E_{\text{loc}} = E_{\text{loc}}^{\text{BF}} - E_{\text{loc}}^{\text{noBF}}$ and report near-field shares at quantile q of $r_{\min}$:

$$\text{NF-}\Delta E(q) \;=\; \frac{\mathbb{E}\big[|\Delta E_{\text{loc}}|\, \mathbf{1}\{r_{\min} \le Q_q\}\big]}{\mathbb{E}\big[|\Delta E_{\text{loc}}|\big]}.$$

(Using absolute values isolates magnitude; signed averages are also available.)

5.2.8 Alignment of correlator manifolds (noBF vs BF)

We collect two standardized feature matrices $\tilde{Z}^{\text{noBF}}$ and $\tilde{Z}^{\text{BF}}$ on identical frames (each standardized by its own μ, σ), compute their PC1 loadings u_1^{noBF} and u_1^{BF}, and report the cosine similarity

$$\cos\big(u_1^{\text{noBF}}, u_1^{\text{BF}}\big) \;=\; \frac{\langle u_1^{\text{noBF}},\, u_1^{\text{BF}}\rangle}{\|u_1^{\text{noBF}}\|_2\, \|u_1^{\text{BF}}\|_2}.$$

Cosines near 1 indicate that backflow acts as a structured reweighting along the same dominant axis, rather than rotating the correlator's manifold.

5.2.9 Uncertainty and stability

For scalars we report standard errors over blocks. For PCA-based quantities and cosines we use a block bootstrap (resampling blocks with replacement) to obtain 68%/95% CIs. Stability checks include: (i) swapping train/held-out blocks for PCA, (ii) varying the near-field quantiles $q \in \{1, 5, 10\}\%$, (iii) re-running analyses on independent sampler restarts.

Reproducibility package. All diagnostics consume the saved bundles $\{\mathbf{X}_{\text{Bohr}},\, r,\, g(r)\}$ and the model snapshot, and emit JSON/CSV tables with r_{eff}, rel-MAE(k) curves, block-power shares, R^2 probes, near-field shares, $r_{\text{eff}}(\Delta x)$, and noBF/BF cosines. As in Sec. 5.1, we log JSD/ℓ_2 splits to confirm stationarity.

5.3 Tangent-kernel diagnostics

The representation diagnostics above describe the ansatz's *features*; a second family, developed with the geometry of §2.2.5, describes its *tangent space*, and it is these that carry the architecture, optimiser, and paradigm comparisons of Chapter 7. From a batch we assemble the centred score matrix $O_{ki} = \partial_{\theta_i} \log|\Psi_\theta(\mathbf{x}_k)|$ and form the quantum geometric tensor $S = O^\top O/B$ (or, when $B < P$, the tangent kernel $K = OO^\top$, whose nonzero spectrum equals that of S up to the overall factor B). From the spectrum $\{\lambda_a\}$ we report the effective dimension $d_{\text{eff}} = (\sum_a \lambda_a)^2 / \sum_a \lambda_a^2$ and the condition number $\kappa(S) = \lambda_{\max}/\lambda_{\min}$. Cross-architecture comparisons of d_{eff} are made on a *common probe set*—all models scored on the same configurations, not on each model's own $|\Psi|^2$—so that the comparison reflects the ansatz and not its sampling measure, and we report only participation ratios that have converged in the batch size.

To ask whether two trained models are the *same* state rather than merely equal in energy, we use a symmetric, importance-sampled squared overlap,

$$\mathcal{S}^2(A, B) = \big\langle \Psi_B/\Psi_A \big\rangle_{|\Psi_A|^2} \, \big\langle \Psi_A/\Psi_B \big\rangle_{|\Psi_B|^2},$$

each factor evaluated by a log-sum-exp mean over samples of the corresponding state. It equals 1 for identical states, and on the analytic two-electron ground state the trained ansatz reproduces $\mathcal{S}^2 = 1.000000$, which anchors the tooling.

One convention must be stated, because the ratio Ψ_B/Ψ_A is signed for a fermionic state. We evaluate the ratios from $\log|\Psi|$, so the estimator is strictly a *modulus* overlap, $\langle|\Psi_B/\Psi_A|\rangle\langle|\Psi_A/\Psi_B|\rangle$. For the comparisons made here this is an accurate proxy for the signed overlap: both states share the same Slater reference, and the correlator and cusp enter as a strictly positive exponential, so the sign of Ψ is fixed by the determinant and the two states agree in sign except on the measure-suppressed set where their backflows carry a configuration across a node differently. Where the two states genuinely diverge the modulus overlap is an upper bound on the signed one, so a reported $\mathcal{S}^2 \to 0$ is if anything conservative. These quantities—together with the energetic price of ablating a component—are invariant to how correlation is divided between the Jastrow and the backflow, and it is on them, rather than on the gauge-dependent component ranks of the previous section, that the architectural conclusions of Chapter 7 rest.

Part IV

Results

The least initial deviation from the truth is multiplied later a thousandfold.

Aristotle
On the Heavens

Chapter 6

The state, and the physics inside it

A variational wavefunction is two things at once. It is an answer, and it is an instrument. This chapter uses it as both, in that order.

The quantum dot is an unforgiving place to try. The Coulomb interaction is singular wherever two electrons meet. The kinetic energy carries second derivatives, which magnify any error in the wavefunction's curvature. And the confinement ω sweeps the physics across three decades of correlation strength—from a compact, kinetic-energy-dominated droplet at $\omega = 1$ to a dilute Wigner molecule at $\omega = 10^{-3}$, where the electrons settle several oscillator lengths apart and the repulsion between them, not the trap, decides the shape of the state. We work at $N \in \{2, 6, 12, 20\}$ electrons and $\omega \in \{1.0,\ 0.5,\ 0.1,\ 0.01,\ 0.001\}$.

Three questions follow one another. *Can the state be trusted?* Section 6.1 puts the energies against every published reference that exists—diffusion Monte Carlo for $N \leq 12$, and the tabulated N=20 energies of Haas [53], which are not DMC and are treated as the weaker benchmark they are. Those references reach down to $\omega = 0.1$ for every N, and to $\omega = 0.01$ for N=2. Below that, for $N \geq 6$, nothing published exists, and the boundary is marked wherever it occurs, because it is the single most important feature of the tables.

What does the state show us? Section 6.2 turns the trained wavefunction on the physics and follows the Wigner crossover from the inside—shell topologies, angular order, and melting—from two electrons up to twenty. The structure we are looking for is invisible in the observable one would naturally plot, which is why it has to be looked for this way.

And how does the network hold all of that? Section 6.3 dissects the trained wavefunction itself. The answer is smaller than the physics that produced it: a correlator living on one to three effective directions, whose *content* reorganises across regimes while its dimensionality does not, alongside a backflow that divides the labour with it in a way that shifts as the trap weakens. One caveat governs how those internal numbers may be read, and it is stated where they are: several of them are gauge-like, fixed by the training route rather than by the state. Separating those from what is invariant is the business of the next chapter.

6.1 Training, and the energies it reaches

Unless otherwise stated, we first apply residual-based pretraining to obtain a well-conditioned initialiser, and then run a Stochastic Reconfiguration (SR) tail to refine the last digits of the energy. In the residual phase, the network initially chooses its own target energy as a stabilizer, before we switch to the DMC reference energy (and optionally add a variance penalty). The SR phase then uses on the order of 10^3–3×10^3 iterations with batches of $\sim 3{\times}10^3$ samples per iteration. Throughout this section—and wherever else the thesis says "SR tail" without further qualification—SR is applied to *variational Monte Carlo with $|\Psi|^2$ sampling.* This matters because the same optimiser reappears in Chapter 7 on the collocation objective, where the sampling measure differs and SR behaves quite differently: here it is a refinement that buys the last digits; there it is what makes the optimisation viable at all.

We compare two backflow architectures throughout. Both condition each particle's displacement on the interparticle coordinates, but they do so differently: the *conventional* BackflowNet applies a single round of pairwise messages aggregated per particle, whereas the copresheaf *message-passing* network (CTNN), realised as the CTNNBackflowNet, maintains explicit node and edge features and transports information between them through learned maps (a single copresheaf round) before the displacement is read out; the contrast between them is the subject of the backflow analysis in Chapter 7. The architecture size is identical for all N: the base PINN correlator has ≈ 54k parameters, the conventional BackflowNet adds ≈ 83k (total PINN+BF ≈ 137k), and the CTNNBackflowNet adds ≈ 182k (total PINN+CTNN $\approx$236k). All components are fully trainable; the parameter count is independent of N because particle interactions enter through permutation-equivariant features rather than per-particle weights.

Three parameter budgets appear in this thesis and should not be confused. These *production* networks carry the SR–VMC energies above. The *collocation* runs of Chapter 7 use deliberately smaller models ($\sim$ 25k Jastrow-only, $\sim$ 49k with backflow), because that programme is limited by gradient noise and memory rather than by capacity. And the *controlled comparison* of Chapter 7 uses its own matched ladder (Table 3.1), so that architecture can be varied with size held fixed. Numbers from one budget are never compared with numbers from another.

6.1.1 Energies

The tables in this section are the thesis's only contact with an external ground truth. Two things matter in them, and neither is the raw energy. The first is *where the agreement holds*: a method that matches DMC at $\omega = 1$ and drifts at $\omega = 0.1$ is a different object from one that holds across the whole benchmarked range, because the physics changes character between those points while the architecture does not. The second is *where the tables stop*. Below $\omega = 0.1$ there is no DMC reference for $N \geq 6$, and the numbers reported there are variational predictions rather than validated ones. That boundary is the single most important feature of what follows, and we have marked it everywhere it occurs rather than letting a uniformly formatted table imply a uniform level of evidence.

Table 6.1 begins with the easiest case, and its purpose is diagnostic rather than competitive: two electrons are few enough that the architecture can be tested nearly in isolation. It reports two-electron energies from *residual-only* training (PINN and PINN+BF). The result to notice is a negative one—for N=2, backflow is *not* required to reach DMC accuracy; it slightly reduces the variance and smooths convergence, and nothing more. The Slater–Jastrow correlator already places the nodes well enough at this size. This matters because it establishes a baseline against which the backflow's later contribution, at larger N and weaker traps, can be read as a genuine effect rather than as an artefact of having added parameters.

Table 6.1: Ground-state energies (Hartree) for two electrons in a harmonic trap using *residual-only* training—that is, *without* the SR tail. To be read against Table 6.2, which reports the same systems after SR refinement: the comparison isolates what pretraining alone achieves, and the residual bias at $\omega = 0.01$ visible here is the one the SR tail removes. DMC references from [16]. Parentheses denote 1σ on the last digits.

ω	PINN+BF	PINN	DMC (Ref.)
1.00	2.99998(3)	2.999940(5)	3.00000(1)
0.50	1.65975(2)	1.65979(3)	1.65977(1)
0.10	0.440795(3)	0.440833(1)	0.44079(1)
0.01	0.0740724(6)	0.0741679(6)	0.073839(2)

Table 6.2 compares diffusion Monte Carlo (DMC) references with our *final* PINN+BF energies after residual pretraining and an SR tail. From the ultra–shallow two-electron case (ω=0.01) to the many-electron systems, absolute deviations are small; for N=2 they lie within DMC uncertainties, and for $N \in \{6, 12\}$ the relative deviations span 0.008–0.05% (see also Fig. 6.1).

The same benchmark, without a Markov chain. Everything above was trained with a Markov chain: residual pretraining, then an SR tail that draws its samples from $|\Psi|^2$ by Metropolis. A second route through this thesis removes the chain entirely, replacing it with an analytic proposal and importance weights. Why that works, where it stops, and what the stopping teaches are the subject of Chapter 7; the energies it reaches belong here, beside the others, because the question they answer is the same one.

Table 6.2: Residual-based pretraining + SR tail. Reference DMC and our PINN+BF and PINN+CTNN ground-state energies (Hartree) for selected (N, ω). The reference column collects the best available benchmark for each size: DMC from [16] (whose study spans $N \leq 13$) for $N \leq 12$, and the reference energies tabulated in the master's thesis of Daniel Haas [53] for $N = 20$, which lies beyond the range of [16]. Parentheses denote 1σ on the last digit(s); the % err columns are relative deviations from the reference. A dash in the reference column marks (N, ω) for which no published DMC value exists (all $\omega = 0.001$, and $\omega = 0.01$ for $N \geq 6$); the relative-error columns are correspondingly blank there, as no external ground truth is available to score against.

N	ω	Reference (DMC)	PINN+BF	% err	PINN+CTNN	% err
2	0.001	—	0.0137948(8)	—	**0.013778(1)**	—
	0.01	0.073839(2)	0.073838(1)	−0.0014	0.0738382(5)	−0.0014
	0.10	0.44079(1)	0.44079(1)	+0.0000	0.440796(4)	−0.0000
	0.50	1.65977(1)	1.65976(2)	−0.0006	1.65976(1)	−0.0006
	1.00	3.00000	2.99999(1)	−0.0003	3.00000(2)	−0.0000
6	0.001	—	0.140913(1)	—	**0.140832(1)**	—
	0.01	—	0.69125(2)	—	**0.69036(1)**	—
	0.10	3.55385(5)	3.5549(1)	+0.0295	**3.55388(5)**	+0.0008
	0.50	11.78484(6)	11.7895(4)	+0.0395	**11.7847(2)**	−0.0000
	1.00	20.15932(8)	20.1610(6)	+0.0083	**20.1585(3)**	−0.0041
12	0.001	—	0.515823(3)	—	**0.515365(4)**	—
	0.01	—	2.48620(5)	—	**2.47363(4)**	—
	0.10	12.26984(8)	12.2731(2)	+0.0266	**12.2743(2)**	+0.0364
	0.50	39.1596(1)	39.1786(8)	+0.0485	**39.1604(3)**	+0.0018
	1.00	65.7001(1)	65.717(1)	+0.0257	**65.69556(5)**	−0.0069
20	0.001	—	—	—	**1.293033(6)**	—
	0.01	—	—	—	**6.14645(5)**	—
	0.10	$29.9779(1)^{\mathrm{H}}$	—	—	29.9888(2)	+0.0364
	0.50	$93.8752(1)^{\mathrm{H}}$	—	—	**93.8789(5)**	+0.0040
	1.00	$155.8822(1)^{\mathrm{H}}$	—	—	**155.8738(7)**	−0.0024

Reference-energy provenance: values marked $^{\mathrm{H}}$ are from the master's thesis of Daniel Haas [53] ($N = 20$); all other reference energies are diffusion Monte Carlo from Pederiva *et al.* [16] ($N \leq 12$, whose study spans $N \leq 13$). Dashes mark (N, ω) with no published value.

Table 6.3: Best ground-state energies from collocation-assisted training. "Multi-stage" denotes the best result from earlier campaigns using hand-tuned continuation chains; "Campaign (best)" denotes the best single result from the 30-run reliability campaign for N=6 (Table 7.5), and the best result across all independent training runs for N=12 (no dedicated reliability campaign was conducted for N=12). The N=2 rows report a single collocation run each (the zero-variance-extrapolated energy under Multi-stage; the campaign column does not apply), reaching DMC accuracy for $\omega \geq 0.01$ and degrading only in the deep-Wigner limit (+0.15% at $\omega = 0.001$). All values are VMC-evaluated ($\geq$20k samples) at the best checkpoint. The reference column is the DMC energy where one exists and, for (N, ω) with no published DMC value (N=6 at $\omega \leq 0.01$; see Table 6.2), our own converged variational (PINN+CTNN) energy; the corresponding % err is then a measure of internal consistency, not of absolute accuracy. The tabulated N=6 low-ω errors are the *reliability-campaign* bests and are deliberately conservative: the best single runs we observed reached +0.06–0.13% at $\omega \leq 0.1$ (Chapter 7, §7.3.4). All tabulated rows use the backflow+Jastrow (BF) architecture; the memory-limited N=20 case is discussed separately in Appendix D.2. The % err column scores the "Campaign (best)" energy where that column exists (N=6, 12) and the "Multi-stage" energy for N=2, where it does not. Parentheses denote 1σ statistical uncertainty on the last digit(s).

N	ω	Reference	Multi-stage	Campaign (best)	% err	Arch
2	1.0	3.00000	3.00035	—	+0.012	BF
	0.5	1.65977(1)	1.66014	—	+0.023	BF
	0.1	0.44079(1)	0.44077	—	−0.004	BF
	0.01	0.073839(2)	0.073840	—	+0.002	BF
	0.001	0.013778	0.013799	—	+0.154	BF
6	1.0	20.15932(8)	20.1613(23)	20.1620(56)	+0.013	BF
	0.5	11.78484(6)	11.7847(20)	11.7850(48)	+0.002	BF
	0.1	3.55385(5)	3.5571(4)	3.5632(7)	+0.263	BF
	0.01	0.69036	0.69141(8)	0.69200(1)	+0.237	BF
	0.001	0.140832	0.14118(1)	0.14118(1)	+0.250	BF
12	1.0	65.7001(1)	65.712(4)	65.706(4)	+0.009	BF
	0.5	39.1596(1)	39.171(4)	39.169(4)	+0.024	BF
	0.1	12.2698(1)	12.285(1)	12.2824(6)	+0.102	BF

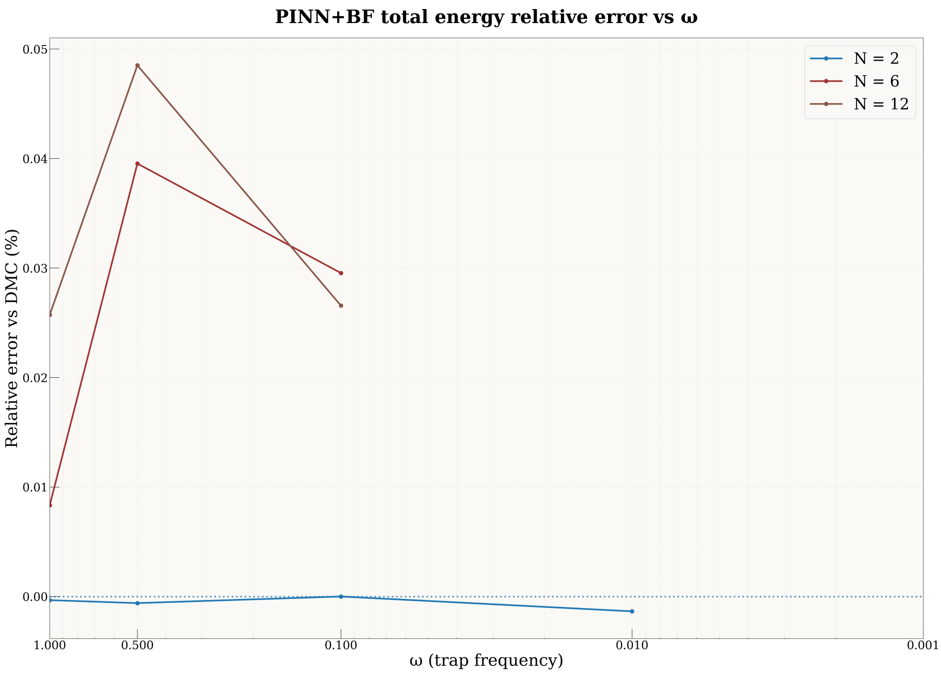


Figure 6.1: Relative deviation of PINN+BF ground-state energies from DMC references as a function of trap frequency ω for the cases where DMC data are available. For N=2 the deviations are at or below the DMC error bars ($\sim 10^{-3}\,\%$), while for N=6 and N=12 the errors span 0.008–0.05 % across ω, with no sign of deterioration in the shallow- or tight-trap limits.

Read against Table 6.2, the comparison is closer than one might expect. At strong confinement the Markov-chain-free route is inside DMC uncertainty for $N \in \{6, 12\}$, and it stays within a few tenths of a percent of our own converged variational energies well into the correlated regime. It loses ground where the importance sampling loses the target—at $\omega \leq 0.1$, and at N=20, where it does not reach the reference at all. The pattern is not a defect of the objective but of the proposal, which is a distinction Chapter 7 makes precise and then acts on.

Where the tables stop. One subtlety matters here, because the message-passing (CTNN) ansatz occasionally sits *below* the quoted DMC reference (for example N=12, ω=1: 65.6956 vs. 65.7001 Ha). This does not violate the variational principle, which bounds the energy only from below by the *exact* ground state, $E_{\mathrm{VMC}} \geq E_0$. The DMC benchmarks are *fixed-node* calculations and therefore carry their own nodal (systematic) bias; a neural ansatz with a better nodal surface can legitimately lie below a fixed-node DMC value obtained with poorer nodes. We therefore do not treat sitting above the published DMC number as a requirement, and where the CTNN falls below it we read this as a plausibly improved nodal structure rather than an error — though establishing that rigorously would require the DMC trial nodes and a matched uncertainty analysis, which we have not undertaken.

The pattern is clean wherever we can check it. For N=2 the ansatz reaches DMC accuracy at every ω for which a DMC number exists, and for $N \in \{6, 12\}$ the relative deviation stays within 0.008–0.05% across the benchmarked range ($\omega \geq 0.1$; Fig. 6.1). Below $\omega = 0.1$ there is no DMC reference for $N \geq 6$, and we make no claim of absolute accuracy there; what we can say is that the energies join smoothly onto the benchmarked regime and, in the deep-Wigner limit, follow the power-law scaling fitted in §6.2.3, whose exponent sits close to the classical Wigner value 2/3. The efficiency is as notable as the accuracy: the SR tail converges in 10^3–3×10^3 iterations of $\sim 3{\times}10^3$ samples each, a modest optimisation budget for a fully many-body wavefunction. We attribute this stability to a small set of conditioning choices—trap-unit scaling, soft-core pair features with $ds/dr \to 0$ at coalescence, and an explicit analytic cusp—each of which keeps the gradients and the Laplacian numerically tame from the shallowest trap to the tightest.

Wherever an external number exists, the state matches it. We can stop asking whether it is right, and start asking what it knows.

6.2 Wigner–molecule crossover

Up to this point the wavefunction has been the thing under examination. Here it becomes the instrument.

What we are looking for cannot be seen in the observable one would naturally plot. In a rotationally symmetric trap, the electrons of a Wigner molecule localise in *relative* coordinates—concentric rings, polygon-like angular correlations—while the whole pattern is free to rotate. Average over that rotation, as any lab-frame density does, and the rings smear into something close to uniform. The order is there; the density hides it. Every diagnostic below exists to look past that averaging.

We combine one-body densities, label-invariant shell topology, rotation-registered angular order, shell radii and gaps, pair statistics, and Lindemann-type disorder, across $N = 2, 6, 12, 20$ and $\omega \in \{1, 0.5, 0.1, 0.01, 0.001\}$ [7, 10]. No one of them is decisive on its own. They fail in different ways, which is why they are reported together.

6.2.1 Observables and diagnostics

For each (N, ω) we draw $\mathbf{X} \sim |\Psi|^2$ and report (i) the pair distribution $g(r)$, (ii) the single-particle radial density $P(r) \propto r\,\rho(r)$, and summary statistics: r_{mode}, $\langle r\rangle$, σ_r, FWHM, quantiles q_{10}, q_{50}, q_{90}, and a localization ratio $\gamma = \sigma_r / r_{\text{mode}}$ (smaller $\gamma \Rightarrow$ stiffer localization). For multi-electron systems we additionally monitor shell-resolved bond-orientational order $|\Phi_m|$ and a dimensionless Lindemann-type ratio for positional fluctuations on each ring [11].

Density parameter r_s. Following Egger *et al.* [7], we estimate the Wigner–Seitz parameter from the first maximum r^* of the spin-summed pair correlation, $r_s \equiv r^*/a_B^*$. The Fermi–liquid $\to$ Wigner–molecule crossover occurs near $r_s \simeq 4$ in parabolic dots; our weak-trap cases lie well beyond this threshold [7, 12].

6.2.2 One-body densities: visual crossover

Figures 6.2 and 6.3 compare one-body densities at strong confinement ($\omega = 1$) and weak confinement ($\omega = 10^{-3}$) for $N = 2, 6, 12, 20$. At $\omega = 1$ the density remains compact and the dominant change with N is progressive shell filling under the harmonic trap. At $\omega = 10^{-3}$ the density broadens dramatically and develops pronounced radial structure for all N, consistent with Coulomb-dominated localization and the onset of Wigner-molecule ordering.

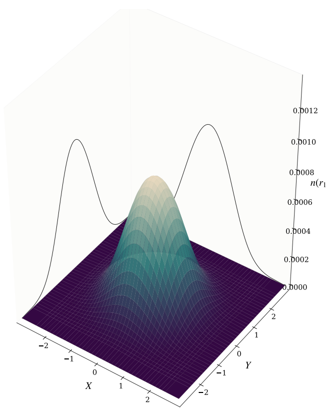
6.2.3 Energy-based crossover diagnostics

Log–log fits $E(\omega) \propto \omega^\alpha$ on the grid $\omega \in \{10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1\}$ give $\alpha_T \simeq 1.00$, $\alpha_{V_{\text{int}}} \simeq 0.66$, and $\alpha_{V_{\text{trap}}} \simeq 0.81,\ 0.73,\ 0.72$ for $N = 2, 6, 12$ respectively. The kinetic energy thus scales essentially linearly with ω, while both potential terms grow sublinearly with nearly N-independent exponents. Across $\omega \in [1, 10^{-3}]$ the interaction-to-kinetic ratio $\Gamma = \langle V_{\text{int}}\rangle / \langle T\rangle$ increases from $\Gamma \simeq 0.9 \to 8.7$ (N=2), $2.4 \to 25.4$ (N=6), and $3.6 \to 39.9$ (N=12), while T/V_{int} falls correspondingly (Fig. 6.4a).

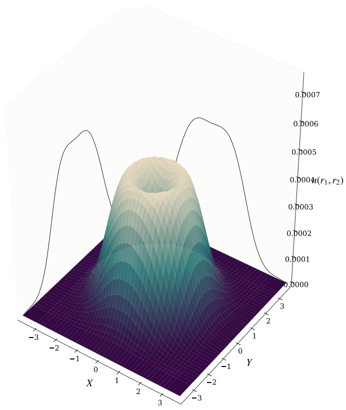
The partitioning between trap and interaction energies becomes quasi-classical at weak confinement. At $\omega = 10^{-3}$ we find $2\langle V_{\text{trap}}\rangle / \langle V_{\text{int}}\rangle \simeq 1.25,\ 1.12,\ 1.02$ for $N = 2, 6, 12$, so that N=12 already satisfies the classical virial condition $V_{\text{int}} \approx 2V_{\text{trap}}$ to within 2% (Fig. 6.4b). Using $V_{\text{trap}} = \frac{1}{2}\omega^2 \sum_i \langle r_i^2\rangle$ we extract an rms radius $r_{\text{rms}} = \sqrt{2V_{\text{trap}}/(N\omega^2)}$, which expands from ≈ 1.1–$2.0\,a_B^*$ at $\omega = 1$ to ≈ 70–$169\,a_B^*$ at $\omega = 10^{-3}$. Fitting $r_{\text{rms}} \propto \omega^{\alpha_r}$ yields $\alpha_r \simeq -0.60$ to -0.64, i.e. $\langle r^2\rangle \propto \omega^{-\beta}$ with $\beta \simeq 1.19$–1.28: the cloud expands *faster* than the non-interacting ω^{-1} scaling, consistent with interaction-driven swelling and shell formation.

6.2.4 Radial localization and pair correlations

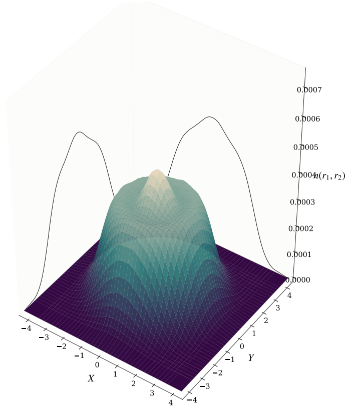
Two numbers carry this subsection. The typical radius r_{mode} grows as a power law, $r_{\text{mode}} \sim \omega^\alpha$ with $\alpha \simeq -0.62$ at N=2—close to the classical two-body value $-2/3$, which nothing in the ansatz knows about. And the width-to-scale ratio $\gamma = \sigma_r / r_{\text{mode}}$ falls, which is the shell stiffening rather than merely dilating (Table 6.4).



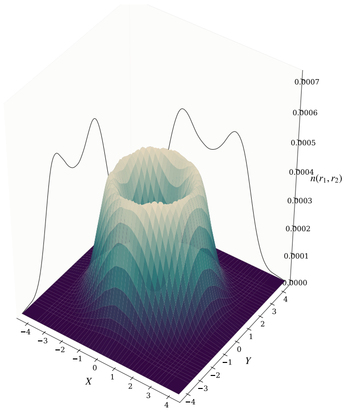
(a) N=2, $\omega = 1.0$



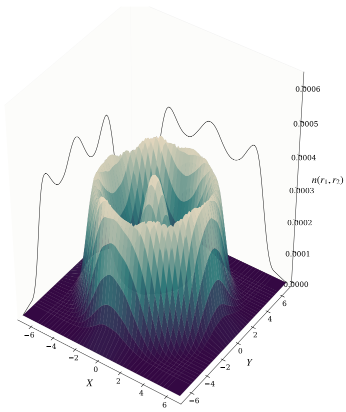
(b) N=6, $\omega = 1.0$



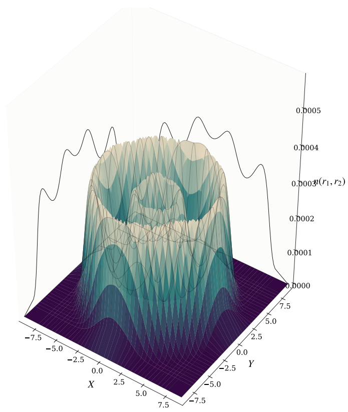
(c) N=12, $\omega = 1.0$



(d) N=2, $\omega = 0.001$

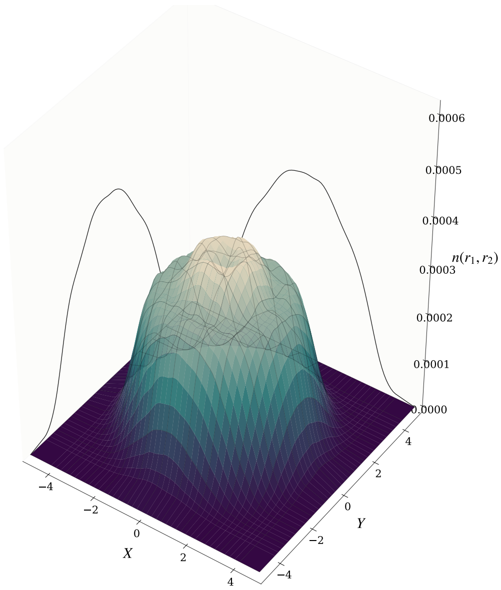


(e) N=6, $\omega = 0.001$

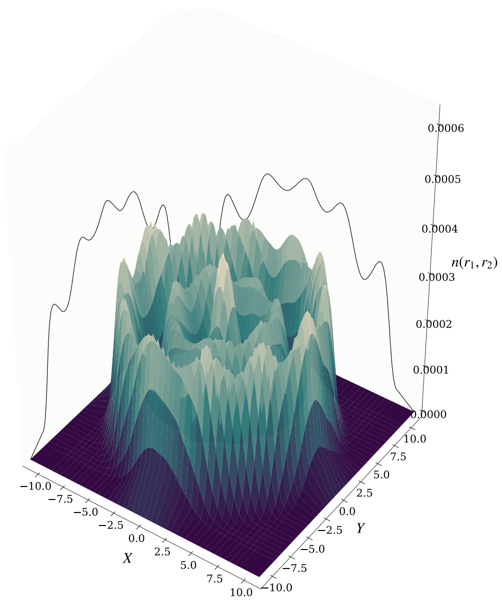


(f) N=12, $\omega = 0.001$

Figure 6.2: One-body densities for $N = 2, 6, 12$ at $\omega = 1.0$ (top) and $\omega = 0.001$ (bottom). As confinement weakens, the densities broaden and develop pronounced shell structures characteristic of Wigner molecules.

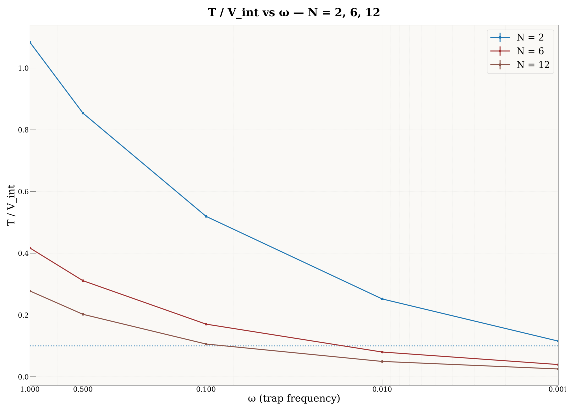


(a) N=20, $\omega = 1.0$

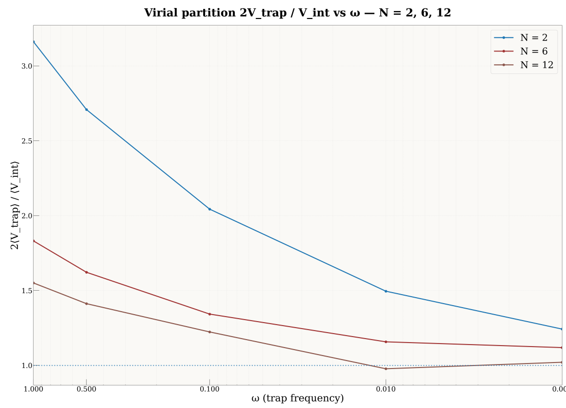


(b) N=20, $\omega = 0.001$

Figure 6.3: One-body densities for N=20 at strong and weak confinement. The weak-trap case shows strong radial broadening and clear multi-shell organization, consistent with deep Wigner-molecule ordering.



(a) T/V_{int} vs. ω for N=2, 6, 12. The horizontal line marks T/V_{int}=0.1.



(b) $2V_{\text{trap}}/V_{\text{int}}$ vs. ω for N=2, 6, 12. The horizontal line marks the classical virial ratio $2V_{\text{trap}} = V_{\text{int}}$.

Figure 6.4: Energy-based Wigner diagnostics as functions of trap frequency. As ω decreases the dots become interaction dominated ($T/V_{\text{int}} \ll 1$) and the trap–interaction partition approaches the classical virial form, most clearly for N=12.

Table 6.4: Radial localisation across the confinement range. r_{mode} is the mode of the single-particle radial density in Bohr; $\gamma = \sigma_r/r_{\text{mode}}$ is the width-to-scale ratio, which falls as the shell stiffens. Full diagnostics—mean, width, FWHM, quantiles, ring-angle spread and pair Lindemann ratio—are in Appendix E.

	N=2		N=6	
ω	r_{mode}	γ	r_{mode}	γ
1.00	1.44	0.491	2.09	0.442
0.50	2.30	0.437	3.07	0.445
0.10	3.60	0.480	8.17	0.429
0.01	14.19	0.396	33.15	0.420
0.001	64.29	0.287	113.55	0.421

The two sizes stiffen differently, and the difference is physical. At $\omega = 10^{-3}$ the N=2 pair sits at separation $r_{12} \approx 122$ with Lindemann ratio $\gamma_{r_{12}} \approx 0.20$ and a relative angle sharply peaked at π: a dilute, anti-aligned dimer. At N=6 the ring's angular fluctuations and pair Lindemann ratio fall monotonically while γ plateaus near 0.42. The shell is stiffening internally even as its radial profile broadens, because the equilibrium radius is growing faster than the width.

6.2.5 Shell topology, angular order, and the rotating Wigner molecule

This subsection resolves an apparent contradiction: the densities plotted above look like smooth rings, yet we are about to claim the electrons form a crystal. Both are true, and the reason is symmetry. Because the Hamiltonian is rotationally invariant, so is its ground state, and any lab-frame average necessarily smears an angularly ordered configuration into a featureless ring. The order is real; it simply lives in relative coordinates, and looking for it in the lab frame is looking in the one place it is guaranteed not to be. Everything below follows from taking that seriously.

Concretely, a Wigner molecule's internal angular order can only be observed after registering frames to a co-rotating reference. For a shell s with n_s particles we compute a rotation-registered bond-orientational parameter $|\Phi_{n_s}|$ and a corresponding Lindemann-type ratio quantifying angular stiffness. Lab-frame densities remain ring-like (the "rotating Wigner molecule" scenario), while the co-rotating frame reveals crystal-like polygon order [7, 10].

N=6: single ring with sector mixing. At $\omega = 0.01$ the system fluctuates between a $(1,5)$ sector (fraction ≈ 0.69) and a $(0,6)$ single-ring sector (fraction ≈ 0.31), with bond order $|\Phi_5| = 0.453 \pm 0.214$ and $|\Phi_6| = 0.466 \pm 0.206$ respectively. At $\omega = 10^{-3}$ the ring stiffens further: $(1,5)$ fraction ≈ 0.80, $(0,6) \approx 0.20$, with $|\Phi_5| = 0.597 \pm 0.224$ (Lind $\approx$0.225) and $|\Phi_6| = 0.663 \pm 0.183$ (Lind $\approx$0.205). Classically

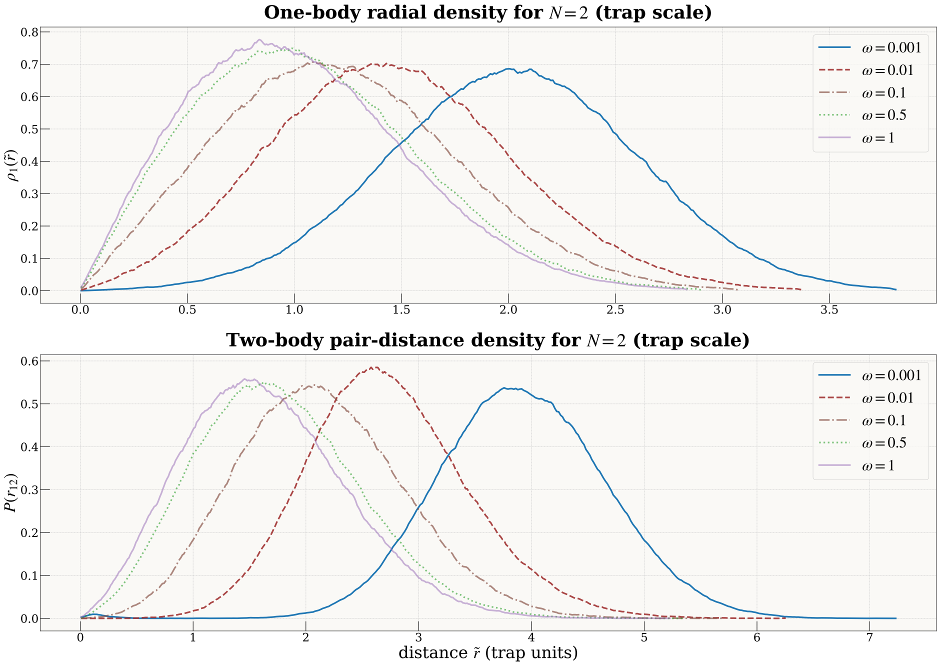


Figure 6.5: Two-electron radial densities $P(r)$ across ω.

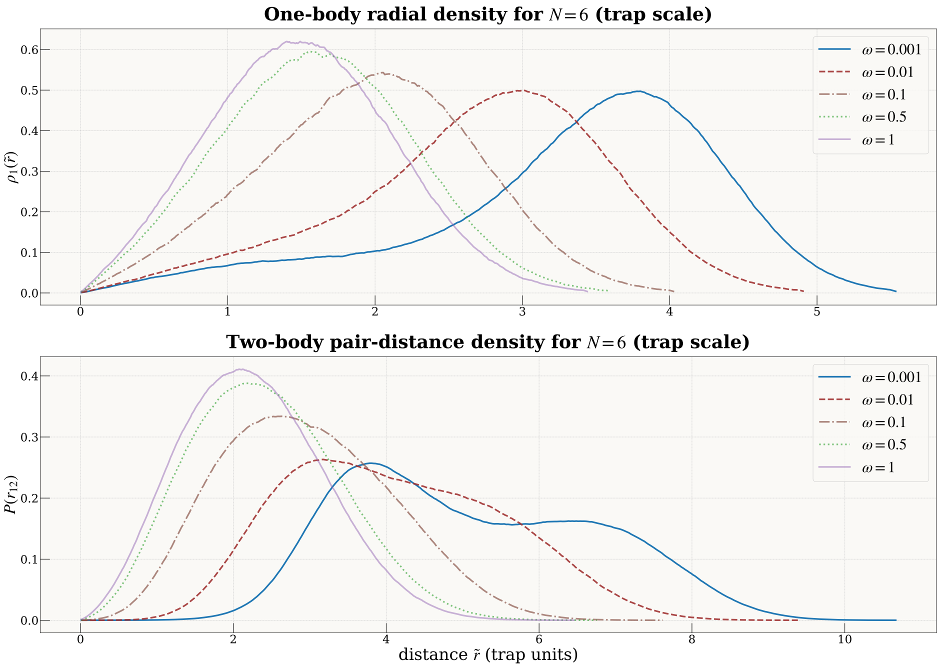


Figure 6.6: Six-electron radial densities $P(r)$ across ω.

$(1,5)$ is the ground-state topology and $(0,6)$ a proximate competitor; the observed quantum mixture mirrors this hierarchy [8, 9]. In both weak-trap cases the global $g(r)$ is reproduced to numerical precision by the appropriate inner–inner / inner–outer / outer–outer (II/IO/OO) mixtures.

N=12**: two shells and commensurability.** At $\omega = 0.01$ two-shell formation is already frequent (fraction 0.761 at radial-gap threshold τ=3.0), distributed across sectors $(1,11) = 32.6\%$, $(2,10) = 25.7\%$, $(3,9) = 13.7\%$. At $\omega = 10^{-3}$ shelling becomes ubiquitous ($f_{2\text{shell}} \approx 0.99$–$0.92$ for $\tau = 2.0$–3.0) and the *commensurate* $(3,9)$ split emerges as modal (43.3% of quality-cut frames), with strong bond order on both rings: $|\Phi_3| = 0.687 \pm 0.229$ ($\text{Lind}_{\text{in}} = 0.223$), $|\Phi_9| = 0.526 \pm 0.194$ ($\text{Lind}_{\text{out}} = 0.224$). A shell-resolved $g(r)$ reconstruction (II/IO/OO mixture with the combinatorial weights 0.045:0.409:0.545 of the ideal $(3,9)$ topology) reproduces the global $g(r)$ with cosine similarity 0.9982 (Fig. 6.7). Since the sector histograms and the total are built from the same samples, this is a test of *decomposability* rather than of sampling correctness: it shows the pair structure is cleanly explained by the identified shell sectors and, because the weights are the idealised $(3,9)$ combinatorics and not empirical sector fractions, that this topology accounts for it. Sampling correctness itself rests on the independent restarts, threshold robustness, and chain-stationarity diagnostics reported above.

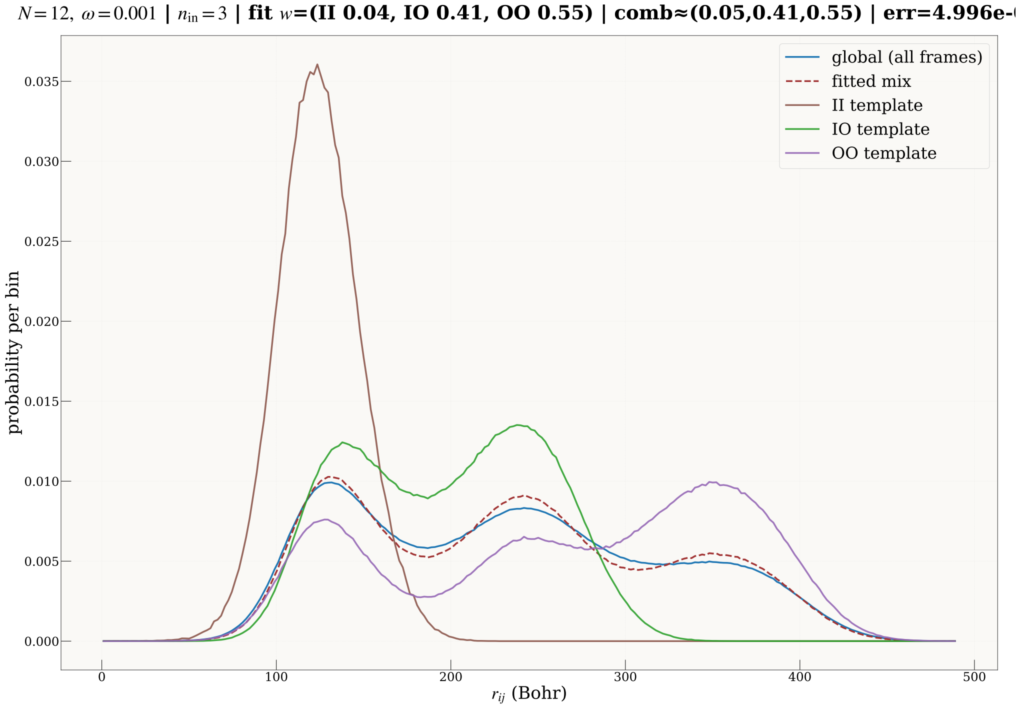


Figure 6.7: Shell-resolved reconstruction of $g(r)$ at $\omega = 10^{-3}$ for N=12. The global pair distribution is reproduced by a convex mixture of inner–inner (II), inner–outer (IO) and outer–outer (OO) histograms with weights set by the observed $(n_{\text{in}}, n_{\text{out}})$ statistics.

N=20**: three-shell molecule and rapid topology collapse.** At $\omega = 10^{-3}$ the dominant shell topology is $(1,7,12)$—a clear three-shell Wigner molecule with widely separated rings at radii $r_0 \approx 38$, $r_1 \approx 132$, $r_2 \approx 247$ (gaps ≈ 94 and ≈ 115). Rotation-registered angular order is strong ($\tilde{C}_{\text{global}} \approx 0.79$), while the lab-frame pinning metric remains small ($C_{\text{global}} \approx 0.06$), as expected for a freely rotating molecule. Lindemann disorder on the two occupied outer shells is low: $L_{7,\text{reg}} \approx 0.35$ and $L_{12,\text{reg}} \approx 0.27$.

Already at $\omega = 10^{-2}$, the three-shell pattern collapses to $(1,19)$: the shell radii contract by an order of magnitude ($r_0 \approx 10$, $r_1 \approx 45$), the registered angular order drops to $\tilde{C}_{\text{global}} \approx 0.20$, and the outer-shell Lindemann rises to ≈ 0.57. At $\omega = 1$ the same $(1,19)$ topology persists with compact radii ($r_0 \approx 0.55$, $r_1 \approx 2.3$), negligible internal order ($\tilde{C}_{\text{global}} \approx 0.20$), and high disorder ($L_{\text{out}} \approx 0.69$). The sharpness of this transition—a single decade in ω separates a stiff three-shell crystal from a disordered single-shell state—is the most dramatic structural collapse observed in our system-size series.

Two implications follow, and the first needs its scope stated precisely. Because the sector histograms and the global $g(r)$ are built from the *same* samples, this is a test of *decomposability*, not of sampling

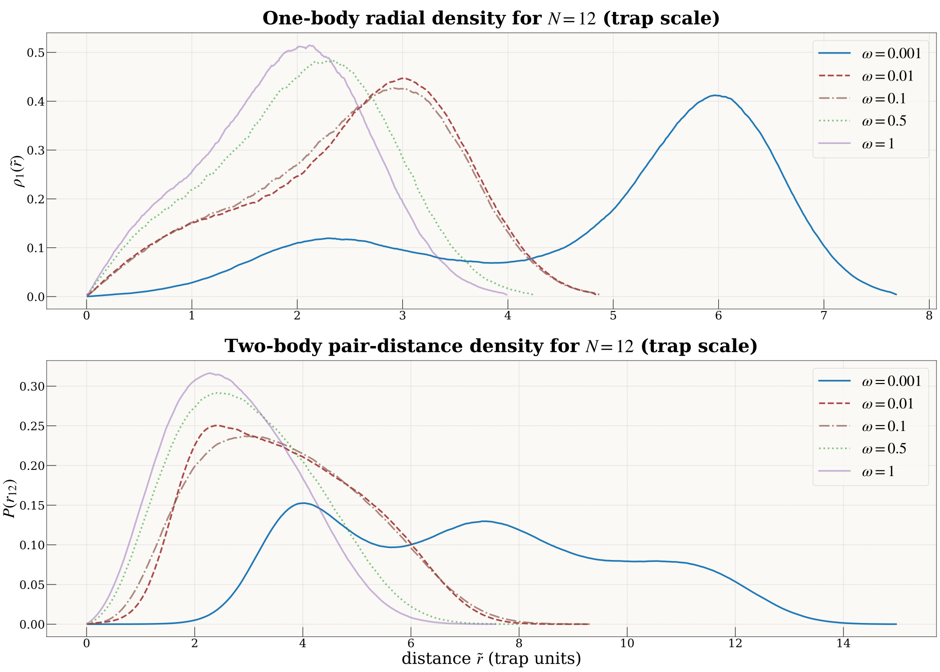


Figure 6.8: Twelve-electron radial densities $P(r)$ across ω.

correctness: it shows that once the shell occupancies and topology statistics are fixed, the global $g(r)$ is determined by shell geometry and combinatorics alone, leaving no residual structure unaccounted for. It cannot, and does not, rule out a sampling artefact common to both—that case rests independently on the restart, threshold and chain diagnostics. Second, subject to that scope, the ansatz is not only reproducing the energy but also the detailed two-body correlation structure expected from classical and quantum studies of parabolic Coulomb clusters.

A caveat on total spin. Our reference states are the closed-shell singlets of the non-interacting problem, with $N_\uparrow = N_\downarrow$ held fixed throughout. This is safe at weak coupling, but in the deep-Wigner regime the ground-state *spin* can change: for parabolic dots the interaction can drive spin transitions as it strengthens (for instance an $S = 0 \to S = 1$ change for $N = 6$ at strong coupling) [7]. Fixing $N_\uparrow = N_\downarrow$ does not by itself pin the total spin, since an $S = 1$ state also has an $S_z = 0$ component. We do not scan total spin here, so the weak-trap energies and structures below should be read as the lowest states we find *within the fixed spin-projection sector*, not as proven global-spin ground states. Determining the true spin ordering across the crossover, by comparing sectors of different S, is left to future work.

6.2.6 What the crossover looks like, taken together

Six diagnostics, four particle numbers, five confinements—and they agree. As ω falls from 1 to 10^{-3} the densities expand and develop rings; the shell topology resolves into the discrete finite-N patterns that classical Coulomb clusters predict—$(1,5)$ at N=6, $(3,9)$ at N=12, $(1,7,12)$ at N=20 [8, 9]; the shells separate and stiffen; rotation-registered angular order rises while lab-frame order stays near zero, which is what a freely rotating molecule looks like; the Lindemann disorder falls; and the energy partitioning approaches the classical virial form, with Γ reaching $\sim$40 for N=12 at the weakest trap. None of these was built into the ansatz. The commensurate $(3,9)$ sector at N=12 carries real weight in $(1,11)$ and $(2,10)$ besides, which is the quantum rotational mixing the rotating-Wigner-molecule picture predicts [10, 12]. At N=20 the three-shell structure and its collapse toward a single shell within one decade in ω is the sharpest transition in the series, and the sector-resolved $|\Phi_m|$ measured conditionally on shell occupancy have not, to our knowledge, been reported before for $N \geq 12$ parabolic dots.

Two qualifications bound all of it. The states are the lowest we find within a fixed spin-projection sector, and we have not scanned total spin, so these are not proven global-spin ground states. And the shell-resolved reconstruction of $g(r)$ establishes decomposability rather than sampling correctness, which rests on the independent restart and chain diagnostics.

From physics back to method. These structural facts are also what make part of the method possible. The shell radius that grows as $\omega^{-2/3}$ and the angular order that sharpens as the trap weakens are exactly the quantities a sampler must respect to reach the Wigner regime without MCMC. Chapter 7 uses them directly: an origin-centred proposal aims where the wavefunction no longer has any mass, whereas a proposal built from the measured shell radius and angular order tracks it, and carries MCMC-free training far into the correlated regime (§7.4). What the trained model reveals about the physics is what later lets us train it more cheaply.

So much for the state seen from outside. How much of that did the network need to hold internally? Less than one would think.

6.3 What the network holds

The physics of the last section is elaborate: rings, occupancies, angular stiffness, a topology that shifts with the trap. The network that produced it is not. Across $N \in \{2, 6, 12, 20\}$ and $\omega \in \{10^{-3}, 10^{-2}, 0.1, 0.5, 1.0\}$, the correlator W_θ turns out to live on a manifold of one to three effective dimensions, and it stays there. What changes across the crossover is not how many directions it uses but which quantities they are built from—a $\phi \leftrightarrow \psi$ recomposition in the leading principal axis. Beside it, the backflow displacement field is the opposite kind of object: high-rank, close to its full $2N$ dimensions, and yet it leaves the correlator's manifold almost untouched. The two work in complementary subspaces, and the division of labour between them shifts sharply near $\omega \approx 0.1$—the same place the physics changes character, where $\Gamma = V_{\text{int}}/T$ crosses over (§6.2.3).

Three things are probed below: the correlator (§6.3.1), the displacement field (§6.3.2), and the coupling between them (§6.3.3). All diagnostics are evaluated on samples drawn from $|\Psi|^2$.

One caveat frames the whole section, and it is not a small one. The Jastrow and the backflow are partial substitutes: how the labour is divided between them is a nearly flat direction of the loss, settled by the route the training took rather than by the state it arrived at. Quantities living on a single component—the backflow's rank and displacement magnitude, the per-branch loadings, the coupling itself—are therefore *gauge-like*. What follows is a faithful dissection of one high-quality solution, and no less interesting for that; the regularities it shows are the more telling for being ones the network was free *not* to adopt. But which of them are properties of the *architecture* rather than of this particular training run is a question this section cannot answer. Chapter 7 answers it by controlled comparison, and establishes there which quantities survive a change of gauge and which do not.

6.3.1 Correlator geometry: a compact manifold that reorganises

We characterise the correlator feature space $Z = \overline{\phi} \| \overline{\psi} \| \mathbf{g}$ by its entropy-based effective rank $r_{\text{eff}}(Z)$, the correlation $|\rho(\text{head}, \text{PC1})|$ between the head output and its first principal component, and the power of PC1 decomposed across architectural branches. Table 6.5 reports these quantities for all CTNN configurations.

The manifold is low-dimensional everywhere. The effective rank stays below 3.1 across the entire grid, and the head output is almost perfectly reconstructed from PC1 alone ($|\rho| > 0.94$ for all $N \geq 6$ and all ω). This holds even at strong correlation where the real-space density develops pronounced shell structure—the correlator does not respond by expanding its latent dimensionality. The modest peak at N=12, ω=0.1 ($r_{\text{eff}} = 3.08$) coincides with the intermediate-correlation regime where neither the shell picture nor the crystalline picture provides a clean single-mode description.

A $\phi \leftrightarrow \psi$ transition in the leading axis. The more interesting variation is in *which branch dominates PC1.* At moderate-to-weak correlation ($\omega \geq 0.1$, $N \geq 6$), PC1 is predominantly ψ-loaded: the pair branch carries 60–73% of the variance, meaning the correlator's primary degree of freedom tracks pair correlations. At strong correlation ($\omega \leq 0.01$, $N \geq 12$), the loading shifts toward ϕ (orbital) or $\mathbf{g}$ (auxiliary) branches, with ψ dropping below 15%. The transition is sharpest at N=12, where ψ goes from 0.09 at ω=0.01 to 0.63 at ω=0.1.

The physical interpretation is direct. When interactions are weak relative to the trap ($\omega \gtrsim 0.1$), the non-interacting shell structure provides a good zeroth-order description, and the correlator's primary role is refining pair correlations on top of that. When interactions dominate ($\omega \lesssim 0.01$), the leading correlator

Table 6.5: Correlator geometry (CTNN). r_{eff}: entropy-based effective rank of Z. $|\rho|$: absolute correlation between head output and PC1 reconstruction. $\phi/\psi/\text{g}$: fraction of PC1 variance from the orbital, pair, and auxiliary branches. The ψ-dominated entries ($\psi > 0.5$) are shown in bold.

N	ω	r_{eff}	$\|\rho\|$	ϕ	ψ	g
2	0.001	1.00	1.00	0.00	**1.00**	0.00
2	0.01	1.04	0.66	0.00	**1.00**	0.00
2	0.1	1.31	0.99	0.00	**0.53**	0.46
2	0.5	1.25	1.00	0.00	**0.58**	0.42
2	1.0	1.53	0.98	0.00	0.35	**0.65**
6	0.001	1.52	0.97	0.01	0.12	**0.87**
6	0.01	1.38	0.99	0.05	0.44	**0.50**
6	0.1	2.73	0.94	**0.66**	0.21	0.14
6	0.5	2.69	0.98	0.15	**0.63**	0.22
6	1.0	2.69	0.99	0.03	**0.62**	0.35
12	0.001	1.66	0.96	**0.54**	0.13	0.33
12	0.01	1.54	0.97	0.43	0.09	**0.48**
12	0.1	3.08	0.99	0.12	**0.63**	0.26
12	0.5	2.32	0.99	0.12	**0.69**	0.19
12	1.0	1.76	0.99	0.08	**0.73**	0.19
20	0.001	2.59	0.99	0.18	**0.74**	0.08
20	0.01	1.09	0.79	0.04	**0.94**	0.01
20	0.1	2.20	0.97	0.36	0.44	0.20
20	0.5	2.30	1.00	0.17	**0.70**	0.13
20	1.0	2.32	0.99	0.18	**0.71**	0.11

coordinate shifts away from pair features toward the single-particle and auxiliary branches. The network reorganises the features through which it represents the state, and that reorganisation—not any growth in dimension—is how it adapts. The network discovers this distinction autonomously: regime changes appear as a rotation of a low-dimensional axis across branches, not as a growth of latent dimensionality.

6.3.2 Backflow: a high-rank field with a low-rank footprint

Table 6.6 characterises the CTNN displacement field. We report the normalised displacement magnitude $\|\Delta\mathbf{x}\|/\ell$, the effective rank of $\Delta\mathbf{x}$ (viewed as a $B \times 2N$ matrix of flattened displacements), and the CKA similarity between the correlator features with and without backflow.

Displacement magnitude scales with ω but saturates near $\|\Delta\mathbf{x}\|/\ell \sim 0.1$. At strong correlation ($\omega \leq 0.01$), displacements are small relative to the harmonic length, $\|\Delta\mathbf{x}\|/\ell \sim 0.02$–$0.05$. As the trap tightens and the system enters the moderate-to-weak correlation regime, the magnitude grows and saturates at $\sim$8–10% of ℓ for $\omega \gtrsim 0.1$. The CTNN never produces large displacements—its tanh output bound and copresheaf message-passing structure act as implicit regularisers, in contrast to the conventional BackflowNet which can exhibit displacements comparable to the system scale in poorly converged regimes (e.g. $\|\Delta\mathbf{x}\|/\ell \sim 0.95$ for N=6, ω=0.001 with BackflowNet).

Full-rank displacements, near-unity CKA. The effective rank of the displacement field is consistently $\approx 2N - 2$, saturating the available spatial degrees of freedom modulo centre-of-mass conservation (confirmed by COM shifts of order 10^{-14}–10^{-17}). The backflow transformation is not a low-rank collective mode such as a breathing or rigid rotation; the displacement field spans essentially all COM-free spatial directions, so each particle receives a correction conditioned on the full configuration. The field stays near that ceiling across the whole grid, including the deepest Wigner points, where one might have expected the correction to become collective. Whether that is a property of *this* architecture or of trained backflows generally is not something the present measurement can decide; Chapter 7 settles it by holding everything else fixed (§7.1.2).

Yet the CKA between correlator features with and without backflow exceeds 0.99 in nearly all cases (and exceeds 0.979 everywhere). The correlator manifold is almost unperturbed by the displacement field. This establishes a clean separation of concerns: the CTNN's action leaves the correlator's feature

Table 6.6: CTNN backflow geometry. $\|\Delta\mathbf{x}\|/\ell$: mean displacement magnitude normalised by harmonic length. Rank: effective rank of the displacement covariance. $2N-2$: maximum possible rank after centre-of-mass projection. CKA: linear centred kernel alignment between correlator features with and without backflow (1.0 = identical feature manifolds).

N	ω	$\|\Delta\mathbf{x}\|/\ell$	Rank	$2N-2$	CKA
2	0.001	0.003	1.9	2	1.000
2	1.0	0.029	2.0	2	1.000
6	0.001	0.017	10.0	10	0.999
6	0.01	0.035	9.9	10	0.998
6	0.1	0.096	10.0	10	0.995
6	0.5	0.093	10.0	10	0.991
6	1.0	0.089	10.0	10	0.979
12	0.001	0.017	21.9	22	1.000
12	0.01	0.030	21.9	22	0.998
12	0.1	0.096	21.9	22	0.993
12	0.5	0.096	21.9	22	0.993
12	1.0	0.091	21.9	22	0.995
20	0.001	0.023	37.7	38	0.999
20	0.01	0.046	37.4	38	1.000
20	0.1	0.075	37.8	38	0.997
20	0.5	0.099	37.6	38	0.990
20	1.0	0.099	37.6	38	0.991

geometry almost unchanged, providing configuration-dependent corrections that the globally organised correlator does not and need not represent.

6.3.3 PINN–CTNN coupling: from independence to cooperation

The previous subsection established that backflow leaves the correlator manifold intact. But are the PINN and CTNN *functionally* coupled—does the CTNN's correction depend on what the PINN computes? We measure this through three diagnostics: (i) the mean cosine similarity $\langle\cos(\nabla_{\mathbf{x}}f,\, \Delta\mathbf{x})\rangle$ between the PINN's gradient in configuration space and the CTNN displacement (gradient alignment implies the CTNN pushes along directions the PINN is sensitive to); (ii) the fraction of configurations where $\Delta\mathbf{x}$ is aligned with ∇f; and (iii) $|\delta f|/\sigma_f$, where $\delta f = f(\tilde{R}+\Delta\mathbf{x}) - f(\tilde{R})$ is the change in the correlator output when it is evaluated on the backflow-shifted coordinates, in units of the correlator's own standard deviation σ_f. Two cautions attach to this last quantity. It is a *counterfactual sensitivity*: in the production ansatz the correlator sees the unshifted $\tilde{R}$ (only the determinant sees $\tilde{R}+\Delta\mathbf{x}$; §3.5), so δf measures how much the correlator *would* respond to the displacement, not an actual term in the wavefunction. And it is a typical relative change, not a variance decomposition: $|\delta f|/\sigma_f = 0.7$ means the counterfactual change is 0.7 standard deviations, not that 70% of the variance is attributable to backflow. Table 6.7 reports these alongside the fraction of particles pushed toward their nearest neighbour.

A transition at $\omega \approx 0.1$. The data reveal a sharp regime change. At $\omega \le 0.01$, the PINN and CTNN are weakly coupled: $\langle\cos\rangle \sim 0.1$–$0.2$, the alignment fraction is well below unity, and backflow barely affects the PINN output ($|\delta f|/\sigma_f < 0.07$). In this regime the CTNN operates in a subspace largely orthogonal to the PINN's gradient—it modifies the wavefunction in directions that W_θ is insensitive to, consistent with its role being to adjust the Slater determinant's orbital positions rather than the correlator.

At $\omega \ge 0.1$, the picture changes qualitatively. The gradient cosine jumps to 0.5–0.8, the alignment fraction reaches 1.0 for $N \ge 6$, and the counterfactual output change reaches $|\delta f|/\sigma_f = 0.25$–$0.69$ (i.e. up to $\sim$0.7 standard deviations of the correlator). Here the CTNN acts as a cooperative refinement: it pushes configurations along directions the PINN is maximally sensitive to, systematically correcting the correlator's residual errors. The transition is sharpest between $\omega = 0.01$ and $\omega = 0.1$, coinciding with the crossover in $\Gamma = V_{\mathrm{int}}/T$ from $\Gamma > 10$ (interaction-dominated) to $\Gamma \sim 3$–6 (moderate correlation).

Directional structure: correlation-hole deepening. The nearest-neighbour direction fraction provides a complementary signature. At $\omega \ge 0.1$ for $N \ge 6$, only 12–25% of particles are displaced toward

Table 6.7: PINN–CTNN coupling diagnostics (CTNN ansatz). $\langle\cos\rangle$: mean cosine between ∇f and $\Delta\mathbf{x}$. Aligned: fraction of configs with $\nabla f \cdot \Delta\mathbf{x} > 0$. $|\delta f|/\sigma_f$: BF-induced change in PINN output relative to its standard deviation. $\to$NN: fraction of particles displaced toward their nearest neighbour (0.5 = isotropic). Only $N \geq 6$ shown; N=2 displacements are near-zero for most ω.

N	ω	$\langle\cos\rangle$	Aligned	$\lvert\delta f\rvert/\sigma_f$	$\to$NN
6	0.001	0.14	0.63	0.03	0.43
6	0.01	0.18	0.77	0.05	0.28
6	0.1	0.49	0.99	0.25	0.16
6	0.5	0.62	1.00	0.33	0.19
6	1.0	0.77	1.00	0.41	0.18
12	0.001	−0.48	0.01	0.03	0.61
12	0.01	0.22	0.82	0.05	0.29
12	0.1	0.56	1.00	0.39	0.12
12	0.5	0.65	1.00	0.49	0.20
12	1.0	0.75	1.00	0.55	0.21
20	0.001	0.32	0.97	0.06	0.44
20	0.01	0.22	0.89	0.07	0.45
20	0.1	0.39	1.00	0.25	0.17
20	0.5	0.65	1.00	0.58	0.22
20	1.0	0.78	1.00	0.69	0.25

their nearest neighbour. The CTNN preferentially pushes particles *apart*, actively deepening the correlation hole. This directional bias is strongest at N=12, ω=0.1 ($\to$NN = 0.12), exactly where the correlator effective rank peaks and the $\phi \leftrightarrow \psi$ transition occurs. At strong correlation ($\omega \leq 0.01$, large N), the fraction rises toward 0.5 (isotropic), because the large equilibrium inter-particle separations leave little room for further correlation-hole deepening.

6.3.4 What this is, and what it is not

Put together, the picture is a division of labour. The correlator supplies a small set of learned collective coordinates—one to three directions, whose leading axis tracks the dominant global coordinate of the cloud and changes identity, rather than growing in number, as the physics changes. The backflow supplies a high-rank, configuration-dependent displacement that leaves those coordinates almost untouched (CKA > 0.98) and works instead in the space they do not span. At strong correlation the two operate independently. At moderate coupling they lock together, the displacements aligning with the correlator's gradient. The switch is sharp, and it happens where the physics switches.

That is an appealing account, and every number in it is real. It is also, at this stage, an account of *one* trained solution. Nothing above rules out that a different training route would have divided the same labour differently and reached the same energy—which, for two components that are partial substitutes, is exactly what one should expect it to do. The anatomy cannot tell us which of its features belong to the architecture. For that we need to hold everything fixed except the architecture, and measure what changes.

Chapter 7

Why the method works: geometry, optimisation, and sampling

The previous chapter showed that the ansatz works. It reaches reference quality wherever a reference exists, the states it produces are genuine Wigner molecules, and it holds that structure on a handful of internal directions. Why it works is a separate question, and a harder one.

Two networks of the same size, trained the same way, landing on the same energy, need not be the same object. A natural-gradient step that is decisive in one training regime does nothing in another. An importance-sampled objective reproduces a Markov chain's answer in one corner of the phase diagram and parts company with it in the next. Three unrelated puzzles, on the face of it. They have one answer, and it is geometric.

Everything turns on the per-sample log-derivative matrix

$$O_{ki} \;=\; \frac{\partial \log|\Psi_{\boldsymbol{\theta}}(\boldsymbol{x}_k)|}{\partial\theta_i}, \qquad k=1,\ldots,B,\;\; i=1,\ldots,P, \tag{7.1}$$

and on its two Gram matrices, the quantum geometric tensor $S = O^\top O/B$ and the neural tangent kernel $K = OO^\top$. Architecture sets the shape of that geometry. The optimiser inverts it. The sampling measure decides whether it can be estimated at all. The chapter follows them in that order.

The instrument throughout is the *effective dimension* of the tangent space—the participation ratio of the QGT spectrum $\{\lambda_a\}$,

$$d_{\text{eff}}(S) \;=\; \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2}, \tag{7.2}$$

which counts how many directions actually carry Fisher weight, whatever the parameter count says. Beside it sit the local-energy variance $\mathrm{Var}(E_L)$, zero only for an exact eigenstate; the effective sample size; and the condition number $\kappa(S)$.

Three constraints hold across the chapter. Architectures are compared on a *common probe set*—the same configurations for every model, never each model's own $|\Psi|^2$. Backflow comparisons are seeded and held at reference quality. And below $\omega = 0.1$ every relative error is measured against our own converged variational energies, which makes it a statement about internal consistency and not about accuracy. The tooling is anchored on the analytic two-electron ground state, where the ansatz returns $E = 3.000127$ Ha ($+0.004\%$) at squared overlap 1.000000.

The road. We begin with what message passing changes in the tangent geometry, in the amplitude and in the nodes; then ask what that geometry implies for the optimiser and for the measure that builds it; then, taking the measure as something we may choose, remove the Markov chain from training altogether and find where the choice stops working. One caveat runs underneath all three and is itself a result: quantities living on a single component—backflow rank, displacement magnitude—are gauge-like, fixed by the training route rather than by the state. What the conclusions rest on instead is set out in §7.5, once there are conclusions to rest.

7.1 Message passing reorganises the variational geometry

What does a message-passing network learn that a separable one cannot? The answer is the same in both learnable objects, and it is a statement about the tangent kernel. In the correlator, message passing *compresses*: the same ground state is reached along fewer effective directions. In the backflow it prevents a *collapse*: the displacement field keeps its rank where a separable one loses it. We take them in turn.

7.1.1 The correlator: compression onto physical directions

At $N = 6$, $\omega = 1$ the two Jastrow families are *energetically indistinguishable*: trained to DMC quality across a 20k–164k parameter ladder, every model lands within ±0.1% of the reference, and the best raw energy belongs to a 164k DeepSet—twice the CTNN parameter count. Total energy is a blunt discriminator; both have saturated the variational floor.

The difference is geometric. On a common probe set the CTNN reaches the ground state in $d_{\text{eff}} \approx 1.2$–$1.7$ (three seeds), while every DeepSet needs $d_{\text{eff}} \approx 2.9$–$3.6$—a factor of ~ 2. The gap is robust to the probe measure ($\lesssim 0.2$ whether scored on CTNN, DeepSet, or pooled points) and does not shrink as the DeepSet is given more capacity: the largest DeepSet in the ladder sits at 2.9, no lower than the smallest. The compression is architectural, not a matter of size or sampling mass. It is also a well-posed comparison despite d_{eff} not being a reparameterization-invariant absolute dimension (§2.2.5): both families are compared under matched normalisation conventions and a common probe measure, so the factor-of-two gap reflects how each architecture spends its tangent directions, not a change of coordinates.

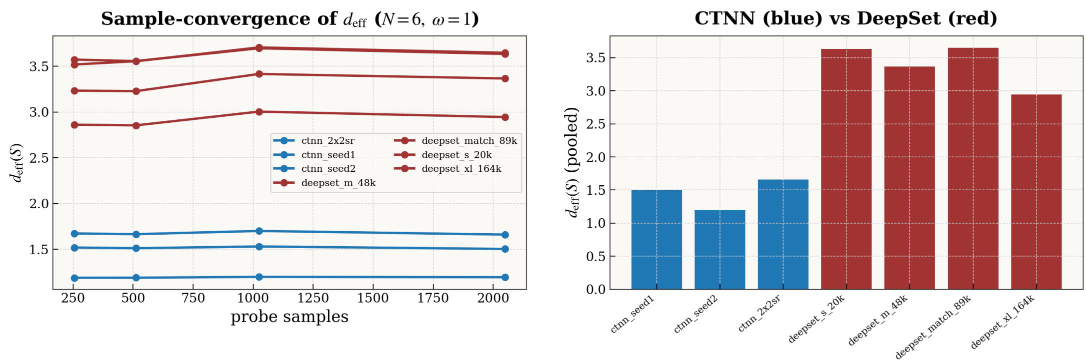


Figure 7.1: Message passing compresses the correlator (N=6, ω=1). *Left:* the effective rank of the quantum geometric tensor on a common pooled probe, converged in the number of probe samples. *Right:* the same pooled measure per model. The CTNN (blue) reaches the ground state in $d_{\text{eff}} \approx 1.2$–$1.7$, while every DeepSet (red) needs ≈ 2.9–3.6—and the gap does not close as the DeepSet is scaled from 20k to 164k parameters. The compression is architectural, not a matter of capacity.

One subtlety should be dispatched here, because the synthesis (§7.5) will caution that quantities split between the Jastrow and the backflow are gauge-like. That caution does not reach this comparison: it is between Jastrow families with *no backflow present*, so there is no correlation to divide between two substitutable channels. What remains is a comparison of how two architectures spend their tangent directions on the same ground state, and it is stable across seeds and across the probe measure—a property of the correlator, not an accounting artefact.

Across seeds at fixed capacity the separation is $d_{\text{eff}} = 1.40 \pm 0.17$ (CTNN) vs 3.25 ± 0.03 (DeepSet) at $\omega = 1$, $\sim 11\sigma$; the wider 2.9–3.6 quoted above is the spread across the *capacity ladder*, not across seeds. The gap is a *strong-confinement* effect: it closes toward the Wigner crystal, where both families reach ~ 3.7 but occupy *different* tangent subspaces. A single-seed dimension "inversion" at the crystal, reported in an earlier draft, did *not* survive seeding and is retracted here as a worked cautionary result (see §7.5).

The discriminator that survives where dimension does not. d_{eff} stops separating the families at the crystal, and rank will turn out to be gauge-like (§7.5). The local-energy variance does neither: it is defined on the state, vanishes only for an exact eigenstate, and is directly comparable between any two ansätze evaluated on their own $|\Psi|^2$. Measured at matched training protocol and production capacity, $\mathrm{Var}(E_L)$ is lower for the message-passing correlator at every confinement (Table 7.1), by factors of 2 to 7, while the two are indistinguishable in energy. At the crystal the separation is clean across seeds: three CTNN runs give $(1.95, 1.63, 1.96) \times 10^{-5}$ against three DeepSet runs at $(3.40, 3.23, 3.41) \times 10^{-5}$, with no overlap. Intermediate confinements ($\omega = 0.28, 0.05, 0.03$) reproduce the same ordering at ratios 1.2–1.8. The one place the ordering is not resolved is the smallest capacity tier used for the depth analysis (12k CTNN vs 9k DeepSet), where the two sit within 8% of each other at $\omega = 1$ and 0.5—so this is a statement about trained production models, not about the architectures at any size.

Table 7.1: Local-energy variance at N=6, matched training protocol and production capacity (79.8k CTNN vs 47.7k DeepSet), both at reference-quality energy ($|\mathrm{err}| \leq 0.13\%$). Lower is better; $\mathrm{Var}(E_L) \to 0$ only for an exact eigenstate.

ω	CTNN	DeepSet	ratio
1.0	1.09×10^{-2}	2.51×10^{-2}	2.3
0.5	4.62×10^{-3}	1.01×10^{-2}	2.2
0.1	5.63×10^{-4}	1.14×10^{-3}	2.0
0.01	4.11×10^{-6}	3.04×10^{-5}	7.4

Naming the manifold, at every size. The leading tangent directions are not abstract. At the exact $N = 2$ anchor an operator decomposition identifies them, to $R^2 = 1.00$, as the physical collective modes of H—a breathing mode (∂_ω), a correlation-hole mode, and the first relative excitation. This is not special to two bodies. Projecting the leading tangent mode of trained correlators onto a dictionary of physical operators (breathing/monopole, quadrupole, correlation-hole, pair-linear), seed-averaged over well-trained checkpoints (energy within 1.5% of reference), the CTNN reaches $R^2 \gtrsim 0.93$ at $N = 6$ and $R^2 \approx 1.00$ at $N = 12$ across the whole confinement range $\omega \in [0.01, 1]$, and both architectures reproduce the $N = 2$ naming at strong confinement (Fig. 7.2). The message-passing correlator finds the ground state by moving along the handful of coordinates the Hamiltonian actually couples.

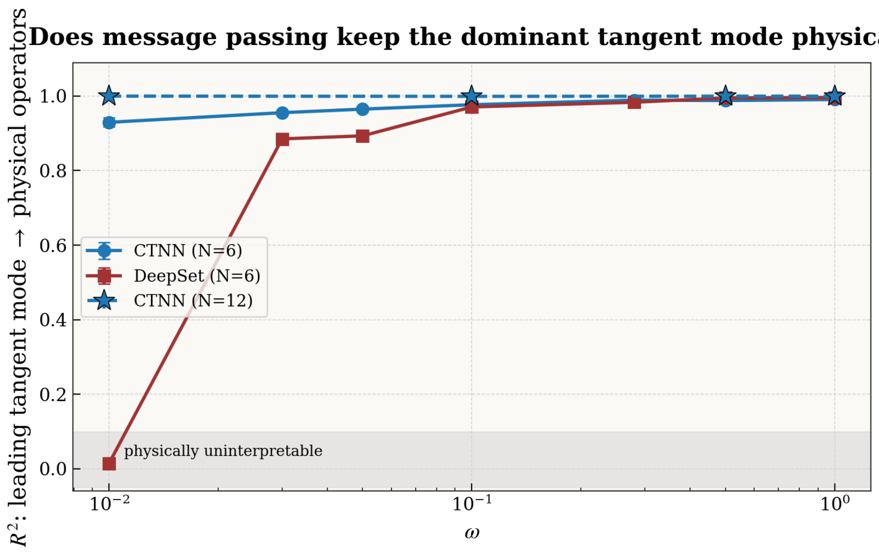


Figure 7.2: The leading tangent mode is a nameable physical mode. R^2 of the correlator's leading tangent-kernel eigenmode onto the physical-operator dictionary, seed-averaged over well-trained checkpoints. The CTNN stays physical at every ω and both sizes; the DeepSet matches it down to $\omega \approx 0.1$ and then loses the leading mode entirely in the deep-Wigner regime. Three seeds at $\omega = 0.01$.

The missing correlation channel is visible at *high* ω too. The leading-mode score is generous to the DeepSet: at $\omega = 1$ it is nominally the more physical of the two ($R^2 \approx 0.996$ vs. 0.991), with energies identical to four figures. Yet a correlation-specific probe exposes a deficit the leading mode hides. Write the eigenvalue-weighted fraction of the top tangent modes lying *outside* the physical-operator span as the *dictionary-orthogonal tangent mass*. The DeepSet carries a large excess at *every* confinement—0.44 against the CTNN's 0.06 at $\omega = 1$, a gap of $\sim$0.3–0.5 that persists from the strongly-confined limit down to the crystal (Fig. 7.3). The reading is that a separable network reaches the same energy—correlation being a small fraction of the total at strong confinement, so the energy looks converged—while leaving nearly half its tangent budget along directions a compact physical dictionary does not resolve. A direction outside the dictionary need not be unphysical; the robust statement is that the CTNN's tangent space is far more completely accounted for by a small set of physical operators than the DeepSet's. The deficiency is intrinsic and present throughout; weak confinement merely removes the disguise, because there correlation *is* the energy. At $N = 12$ the contrast is starker still: the CTNN's dictionary-orthogonal mass stays $\lesssim 0.02$ across ω, while a matched DeepSet (trained to +0.2%) sits at ≈ 0.77 at $\omega = 1$—the gap *widens* with system size rather than closing.

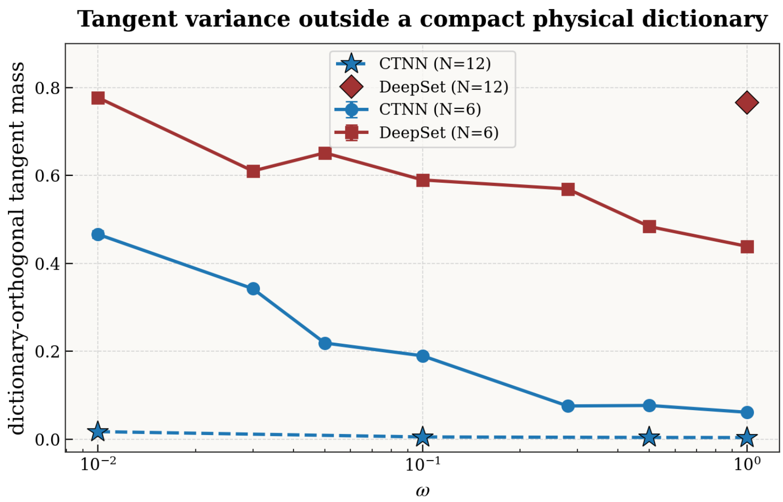


Figure 7.3: A limitation visible where the energy cannot see it. Dictionary-orthogonal tangent mass—the eigenvalue-weighted tangent variance outside the span of a compact physical-operator dictionary—versus ω. At $\omega = 1$ the two correlators are indistinguishable in energy (-0.001%) and in leading-mode physicality, yet the DeepSet already spends 44% of its tangent variance on directions not resolved by the chosen physical-operator dictionary, against the CTNN's 6%. The excess is roughly constant in ω: the correlation deficit is intrinsic, disguised at high ω only because correlation is a small slice of the energy there.

7.1.2 The backflow: a collapse averted

The same relational channel is decisive, in a sharper form, in the backflow. Here we fix everything but the backflow architecture (conventional single-round messages vs. the copresheaf CTNN) and train both to a Woodbury-SR endpoint. At $\omega = 1$ both reach reference quality, but they diverge as the trap weakens (Table 7.2): the CTNN backflow holds $\sim 0.02\%$ error across all N, ω, while the conventional backflow degrades to +0.5% at $N = 6, \omega = 0.01$ and $\sim 6\times$ the CTNN error at $N = 20, \omega = 0.01$ (+0.69% vs. +0.12%).

Table 7.2: Relative energy error (%) of the two backflow architectures, seed-averaged.

	$N = 6$		$N = 12$		$N = 20$	
ω	CTNN	conv	CTNN	conv	CTNN	conv
1.0	−0.03	+0.06	−0.01	+0.04	+0.06	+0.07
0.1	−0.05	+0.16	+0.01	+0.16	+0.02	+0.18
0.01	−0.04	+0.58	−0.01	+0.54	+0.12	+0.69

The energy gap has a sharp structural signature. Let the *backflow rank* be the participation ratio of the displacement field $\Delta\mathbf{x}$ (how many independent directions the backflow moves electrons along). The field lives in $2N$ dimensions, and we project out its centre of mass (Eq. 3.19), so the attainable ceiling is $2N-2$: 10, 22, 38 at $N = 6, 12, 20$. Both architectures sit close to that ceiling at large ω. Toward Wigner the *conventional* backflow collapses to a single collective mode (rank $\to 1$) while the CTNN backflow stays near the ceiling (Table 7.3); the collapse sharpens with size ($10 \to 1$ at $N = 6$, $22 \to 1$ at $N = 12$, $35 \to 1$ at $N = 20$). The tangent dimension tells the same story from the other side: toward Wigner d_{eff} grows for CTNN ($\sim 4 \to 10$) and collapses for the conventional backflow ($\sim 6 \to 1$). The collapse to *exactly* rank 1 is seed-robust at every size. Rank is not the discriminator above the crystal: at $\omega \geq 0.1$ the conventional field is if anything the marginally higher-rank of the two (Table 7.3), and the two architectures are then also within 0.2% of each other in energy. It is the collapse, not the level, that carries the finding.

Table 7.3: Backflow displacement rank (participation ratio), seed-averaged.

	$N = 6$		$N = 12$		$N = 20$	
ω	CTNN	conv	CTNN	conv	CTNN	conv
1.0	8.8	10.4	20.8	22.2	36.9	35.4
0.1	7.7	10.4	18.2	21.7	33.8	37.6
0.01	9.8	**1.0**	20.8	**1.0**	35.2	**1.0**

Pricing each component by deletion. We price each component by ablation on a *common probe*: configurations are drawn once from the full model's $|\Psi|^2$ and every arm is re-evaluated on them, so the trap and Coulomb terms are held fixed by construction and each figure below is the change in the kinetic term at fixed measure—the instantaneous cost of removing a component, not the energy of a re-relaxed state. Two facts are seed-stable. (i) The **relational work migrates from the nodes to the amplitude** as the trap weakens. Deleting the backflow costs +5.6% at $\omega = 1$ and +1.4% at $\omega = 0.1$, but only +0.013–0.017% at $\omega = 0.01$ (three seeds), while deleting the Jastrow messages runs the other way: +3.9% at $\omega = 1$ rising to +9–15% at $\omega = 0.01$. At the crystal the backflow is very nearly inert—with the Jastrow messages already removed, deleting the displacement as well changes the energy by under 0.01%. "Message passing buys kinetic energy through the nodes" is a high-ω statement only. (ii) **Within the backflow, the transport maps *are* the displacement**: zeroing the inter-particle transport maps $(\rho_{v\to e}, \rho_{e\to v})$ perturbs $\Delta\mathbf{x}$ by as much as deleting it outright—the mean $|\Delta\mathbf{x}_{\text{abl}} - \Delta\mathbf{x}|$ is comparable to the mean $|\Delta\mathbf{x}|$ at every (N, ω)—so what displacement the network produces comes from the messages rather than from the per-particle path. Both architectures condition each particle on the relative coordinates $\mathbf{r}_{ij}$, so neither is blind to inter-particle structure; the difference is that the conventional backflow uses a single round of pairwise messages aggregated per particle, whereas the CTNN maintains and transports explicit edge state across the graph. Empirically the single-round backflow does not *sustain* a full-rank relative displacement at the crystal—it collapses to a single collective displacement mode (rank 1)—while the CTNN's copresheaf transport holds the full-rank response, and the ablation shows that this transport carries essentially the entire effect.

7.1.3 One bottleneck, two channels

The two are one statement about how relational information is *routed*. The baselines are not blind to inter-particle structure—the DeepSet pools independently-encoded pairs, and the conventional backflow aggregates a single round of pairwise messages—but neither maintains a persistent, transported edge representation: each pairwise term is computed and discarded. The copresheaf CTNN does maintain one, and moves it back and forth between nodes and edges through learned transport maps $(\rho_{v\to e}, \rho_{e\to v})$; the message-passing correlator iterates this exchange over a V-cycle, while the backflow applies a single such copresheaf round. Where a compact representation exists this *compresses* the correlation onto fewer tangent directions (Jastrow: smaller d_{eff}); where the physics demands a genuinely collective, full-rank response the single-round baseline instead collapses to a single collective displacement mode (backflow: rank 1), which the copresheaf transport avoids. Both effects are visible only in the tangent-kernel geometry, not in the total energy, until weak confinement makes the geometric deficit an energetic one.

The two channels fail together. The correlator deficit and the backflow deficit set in at the *same* confinement. Holding the backflow architecture fixed (the conventional network, in both cases) and varying only the correlator, we track the correlator's correlation-hole capture and the backflow's displacement

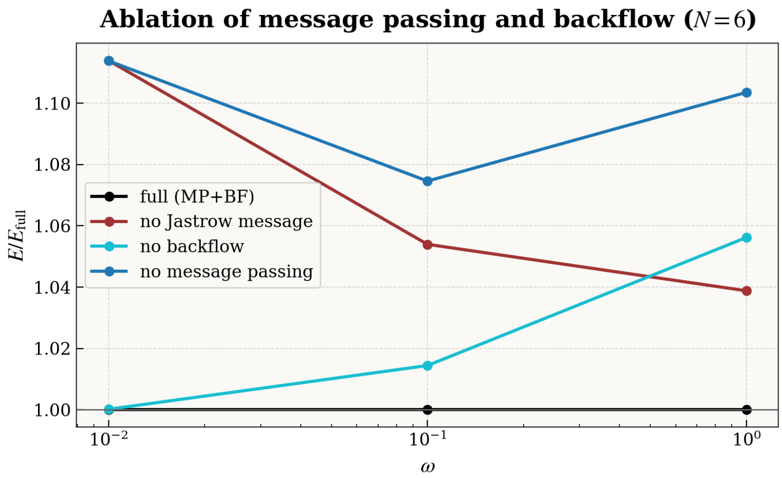


Figure 7.4: What message passing buys, by ablation (trained networks, N=6; energy relative to the full model, all arms evaluated on a common probe drawn from the full model's $|\Psi|^2$). Removing the backflow (orange) costs $\sim$6% at ω=1 but under 0.02% at ω=0.01, while removing the Jastrow messages (red) or all message passing (blue) grows more expensive as the trap weakens. The relational work migrates from the nodes to the amplitude: at the crystal it is the correlator's messages, not the backflow, that carry it.

rank down the ω-sweep (Fig. 7.5). With a DeepSet correlator both give way together at $\omega \approx 0.05$—the hole capture falls from 0.68 to 0.07 as the backflow rank drops from $\sim$10 to 1. With a CTNN correlator both hold until $\omega \approx 0.01$. The correlator's grip on the physical correlation modes and the backflow's ability to sustain a full-rank displacement are the same capability seen from two sides: when the amplitude model can no longer resolve the correlation hole, the nodal deformation can no longer stay high-rank, and a better correlator postpones *both*. The evidence—coincident transitions that shift together under a correlator swap—supports a common relational bottleneck beneath both rather than proving them literally identical.

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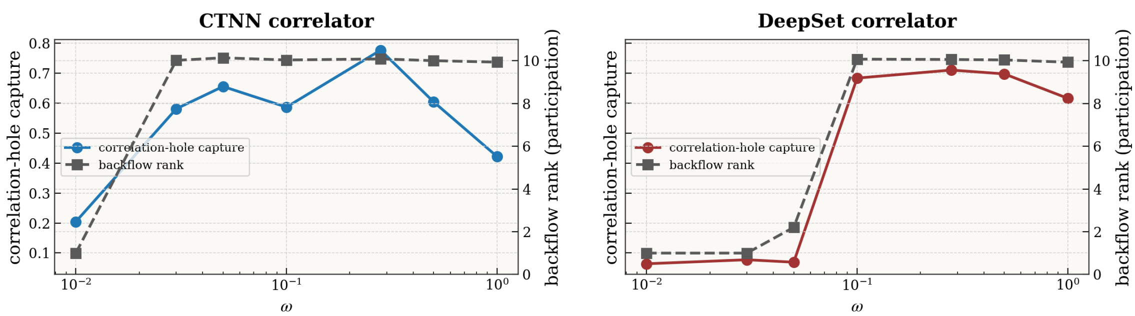

Figure 7.5: Correlator physicality and backflow rank collapse together ($N = 6$; the backflow architecture is held fixed, only the correlator changes). For each correlator, the correlation-hole capture (left axis) and the backflow displacement rank (right axis) give way at the same ω: $\omega \approx 0.05$ for the DeepSet, $\omega \approx 0.01$ for the CTNN. A correlator that keeps its physical structure lets the backflow keep its rank.

7.2 Conditioning: what the optimiser can do depends on the

measure

Two design choices remain, and the tangent geometry settles both at once. The optimiser question—what does a natural-gradient step buy over a diagonal one—and the paradigm question—what changes when the loss is defined by an analytic proposal instead of by $|\Psi|^2$—turn out to be the same question asked twice. Stochastic reconfiguration inverts the metric S. The sampling measure is what builds S in the first place. Neither can be understood without the other, so we take them together.

7.2.1 Whitening, and where it pays

Stochastic reconfiguration preconditions the gradient by the inverse QGT, $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, S^{-1}\nabla_{\boldsymbol{\theta}} E$; geometrically it *whitens* the tangent kernel. At $N = 2$ this is exact: the SR step aligns with the imaginary-time direction to numerical precision. The question is where whitening changes the outcome.

Fixing the ansatz and varying only the optimiser from a common stage-3 base: **for $|\Psi|^2$-VMC, SR and Adam are indistinguishable**—exact at $N = 2$ (var $\sim 10^{-9}$), and at $N = 6$ the energy gap $E_{\mathrm{Adam}} - E_{\mathrm{SR}}$ is $\lesssim 0.002$ across ω and both seeds. The VMC metric is well enough conditioned ($\kappa(S)$ moderate, and $|\Psi|^2$ sampling is self-preconditioning) that Adam already moves along the right directions. **For weak-form collocation, SR is essential**: there $E_{\mathrm{Adam}} - E_{\mathrm{SR}}$ is large and positive (e.g. +2.9 at $N = 6, \omega = 1$), because the importance weights make the objective badly conditioned and plain Adam fails to converge it.

So the verdict is neither "SR wins" nor "SR ties". **SR matters when the tangent metric is ill-conditioned and still estimable.** Its value is a function of $\kappa(S)$, set by the ansatz and the sampling measure, not an intrinsic property of the optimiser—which is exactly why an earlier low-ω VMC sweep saw a modest SR advantage that never grew into a decisive win.

The claim needs one sharpening, because "more ill-conditioned $\Rightarrow$ more useful SR" cannot hold without bound: inverting S requires S itself to be *reliably estimated*. SR helps when the tangent metric is anisotropic but its empirical estimate is trustworthy; once the sampling that builds S collapses—as the importance weights do deep in the Wigner regime—the empirical S becomes noisy and rank-deficient, and preconditioning by its inverse amplifies that noise rather than the signal. This is exactly the behaviour seen at ultra-weak confinement, where CG-SR on the collocation objective becomes unstable (Chapter 6, §7.3.3): not a counterexample to the conditioning story but its completion—SR pays off in the window where the metric is both ill-conditioned and estimable, and fails once estimation itself breaks down.

One further boundary deserves to be drawn explicitly. The VMC equivalence of Adam and SR is established here at $N = 2$ and $N = 6$; we have not verified that it survives to $N = 12$ and $N = 20$, where the conditioning of the VMC metric could in principle worsen with dimension. The mechanism we propose—that $|\Psi|^2$ sampling keeps $\kappa(S)$ moderate and so lets Adam's diagonal preconditioning suffice—is a prediction there, not yet a measurement, and testing it is the obvious extension.

7.2.2 The measure sets the conditioning, the variance, and the reach

Between $|\Psi|^2$-sampled VMC and importance-sampled collocation the wavefunction family is identical; only the measure that defines the loss changes. Three consequences follow, and they are one statement.

$|\Psi|^2$ is self-preconditioning; the covering proposal is not. Sampling from $|\Psi|^2$ makes the estimator of S and of the gradient share the same measure, which conditions the update. A fixed covering proposal instead drives a monotone *ESS collapse* as N and $1/\omega$ grow: at $N = 6, \omega = 0.01$ the effective sample size falls to 0.11% of the batch (about 5 of the 4096 retained points—the normalisation used for every ESS figure in this thesis). When ESS collapses the gradient estimator is degenerate—a hard limit, not undertraining.

This needs one reconciliation, because §7.3.4 below reports usable collocation energies at the same (N, ω) where the estimator is called degenerate here (+0.24–0.29% across three seeds, Table 7.5). Two things differ. First, the sampler: the collapse measured above is for the *static* origin-centred mixture, while the campaign's robust recipe refits its proposal every 30 epochs and oversamples 16×. The campaign's *baseline* rows, which keep the static proposal, are the ones that blow up (+3.9 to +7.2% at $\omega = 0.001$). Second, and more interesting, the two statements are not about the same quantity. The energy stays within a third of a percent while the state overlap goes to zero—which is the sharpest illustration in this thesis of why energy alone does not certify a state. The honest qualification is that $\mathcal{S}^2$ is itself a ratio estimator built from the same degenerate weights, so "$\to 0$" should be read as *unresolvable at this ESS* rather than as a measured orthogonality.

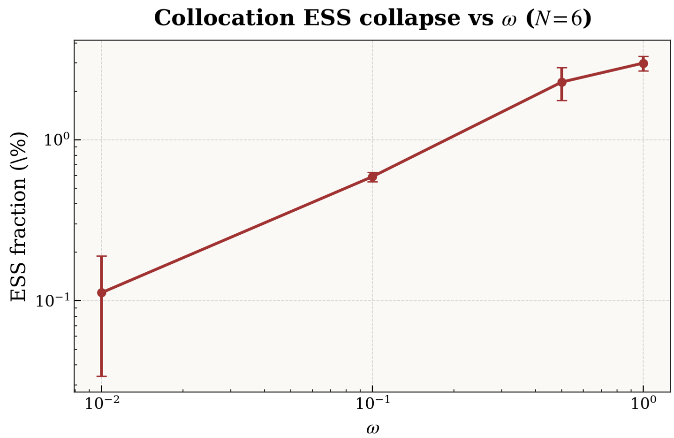

Figure 7.6: The naive collocation proposal collapses in the Wigner regime (N=6). The effective sample size (as a percentage of the batch) falls by more than an order of magnitude from ω=1 to ω=0.01, where it reaches ∼0.1%—about five usable points in a batch of 4096. This is a property of the sampling measure, not of optimisation, and it is what the physics-informed ring proposal (§7.4) is designed to repair.

Same state, until the sampling loses the target. We test whether the two paradigms reach the same *state* (not merely the same energy) via the symmetric overlap $\mathcal{S}^2 = \langle \Psi_B/\Psi_A \rangle_{|\Psi_A|^2} \langle \Psi_A/\Psi_B \rangle_{|\Psi_B|^2}$. At $N = 2$ they are identical ($\mathcal{S}^2 > 0.997$ for every ω, both optimisers, both seeds): the paradigm is efficiency-only, collocation paying $\sim 5\times$ the variance for the same wavefunction. At $N = 6$ they *diverge*: under Adam $\mathcal{S}^2$ falls $0.85 \to 0.77 \to\approx 0$ from $\omega = 1$ to 0.01 (both seeds), tracking the ESS collapse; SR partially rescues it ($\mathcal{S}^2 \approx 0.95$) but does not close the gap. Weak-form collocation thus has a **domain of validity**—equivalent to VMC when the importance sampling is healthy, degrading as correlation strengthens. The two halves of this section meet here: the paradigm that needs SR is the one that breaks down at scale. An adaptive proposal that tracks $|\Psi|^2$ is the natural remedy, pursued in §7.4.

Why the collocation energy readout is biased. For an exact eigenstate the local energy $E_L = \hat{H}\Psi/\Psi$ is constant over configuration space, so $\mathrm{Var}(E_L) \to 0$ as the ansatz approaches the ground state—the zero-variance principle, and the reason $\mathrm{Var}(E_L)$ is a sharp quality measure. That cancellation is a property of the *full* local energy, in which the kinetic (Laplacian) and potential terms combine to a constant; it is not a property of the Laplacian term alone. The collocation objective does *not* discard the Laplacian: it is a score-function (REINFORCE) estimator that *forms* the full E_L as a detached reward and differentiates only $\log|\Psi|$ (§7.3.1). What it gives up is the *measure*: the reward is averaged over the analytic proposal with importance weights (and, in the hardest regimes, tempered weights), so the importance-sampled energy *readout* is a biased estimate of $\langle E_L \rangle_{|\Psi|^2}$ and must not be used as the convergence criterion. The reliable criteria are $\mathrm{Var}(E_L)$ and the state overlap, both evaluated on $|\Psi|^2$ samples.

7.3 Training without a Markov chain

The conditioning story has a practical edge. If the sampling measure is what makes the metric estimable, then choosing that measure is a design decision, not a fact of nature—and the obvious thing to try is a measure with no Markov chain in it at all.

MCMC is what makes neural variational Monte Carlo sequential. Chains must equilibrate, successive samples are correlated, and mixing times grow with particle number and with correlation strength—which is to say, they grow fastest exactly where the physics gets interesting. Draw the configurations instead from a fixed analytic proposal and reweight them, and the whole gradient step becomes embarrassingly parallel. The question is what that costs.

Two ways of posing the problem behave very differently, and the difference decides everything downstream. The *strong form* asks the wavefunction to satisfy the local Schrödinger equation pointwise, minimising the variance of E_L and differentiating through the Laplacian to do it. The *weak form* uses a score-function (REINFORCE) gradient in which the Laplacian enters only as a reward and is never differentiated. Appendix C shows that the strong form meets a structural obstruction the moment it is asked to train a backflow: the learning signal lives at the nodal surface, where E_L diverges for any imperfect node, and no combination of sampling, loss or preconditioning breaks an accuracy floor near 0.3%—a floor set by the Jastrow alone, with the backflow contributing almost nothing. That is a property of the objective, not of the idea. Everything below uses the weak form.

So posed, the result is clean. On the same Slater–Jastrow–backflow ansatz, and with no Markov chain anywhere in the optimisation, the weak form recovers the variational energy to within DMC uncertainty at strong confinement and to a few tenths of a percent of our gold standard well into the correlated regime for $N \in \{6, 12\}$ (Table 6.3). Why it works, where it stops, and what the stopping is made of, follow below.

7.3.1 The objective, and why it is stable

The collocation training used here is exactly the MCMC-free pipeline specified in the Methods: a weak-form (REINFORCE) objective in which the local energy enters as a detached reward (§4.5, Eq. (4.8)), an analytic Gaussian-mixture proposal reweighted to $|\Psi_\theta|^2$ with ESS / Pareto-$\hat{k}$ monitoring and tail controls (§4.5.1), and—in the Wigner regime—the physics-informed ring proposal with cascade warm-starting (§4.5.2). We do not repeat those definitions; this section reports what the pipeline achieves and where it fails. Two methodological findings shape the results below. First, among the optimisers (§4.4), **Adam** was the robust default across all regimes, whereas **CG-SR** converged faster at $\omega \geq 0.1$ but diverged for $\omega \leq 0.01$, where the geometric tensor becomes ill-conditioned. Second, **cascade warm-starting**—initialising each confinement from the converged checkpoint of a stronger one—was decisive for reaching the deep-Wigner regime, exploiting the transfer of correlation structure across ω.

Three ansatz families were tried within this framework, all built on the same Slater–Jastrow core $\Psi(\mathbf{R}) = \det[\phi_i(\mathbf{r}_j + \Delta\mathbf{r}_j)]\, e^{J(\mathbf{R})}$ and differing only in what dresses it.

Jastrow-only (no backflow). The Jastrow factor is parameterised by a graph neural network (`CTNNJastrowVCycle`) with $\sim$25k parameters. It operates on node features (single-particle orbital values) and edge features (pairwise distances with soft-core regularisation). The V-cycle variant uses a coarse-to-fine message-passing scheme. This family was the fastest to train and the most robust, serving both as a standalone ansatz and as an initialiser for backflow runs.

Coordinate backflow + Jastrow (BF). The `CTNNBackflowNet` adds a copresheaf-style message-passing network that predicts electron displacement vectors $\Delta\mathbf{r}_j$ fed into the Slater determinant. Edge features in the backflow network scale as $\mathcal{O}(N^2 \times h_{\mathrm{bf}})$ where h_{bf} is the hidden dimension, creating a memory bottleneck at large N. For N=6 and N=12, the default h_{bf}=128 was used (total $\sim$49k parameters); for N=20, reduced variants with $h_{\mathrm{bf}} \in \{48, 64\}$ were necessary. The BF+Jastrow family produced the best results at N=6 and N=12.

Neural Pfaffian. The Pfaffian ansatz replaces the Slater determinant with a more general antisymmetric form $\mathrm{Pf}[A_{ij}]$, which in principle offers richer nodal topology. Representative results: E=20.261 ± 0.004 and E=20.245 ± 0.005 at N=6, ω=1.0 (+0.50% and +0.43% above DMC), compared to +0.013% for BF+Jastrow. The Pfaffian's greater expressivity was not exploitable within the current training pipeline; the simpler BF+Jastrow family consistently outperformed it.

Why the weak form is stable where the strong form is not. The catch-22 analysis (Appendix C) identified a fundamental obstacle for direct PDE-residual training of the many-body Schrödinger equation: the backflow learning signal is concentrated near nodes and coalescence points where the local energy E_L diverges, but these are precisely the configurations where second-derivative pathways are ill-conditioned. Backpropagating through the Laplacian in such regions produces pathological gradients that destabilise training.

The REINFORCE formulation resolves this by construction. The Laplacian enters only through the *reward* (the local energy in Eq. (4.2)), and gradients act exclusively on the score function $\nabla_\theta \log|\Psi|$. The stability is subtler than the score simply being bounded. For a *parameter-dependent* (backflow) node the score itself can diverge— $\nabla_\theta \log|\Psi| \supset (\nabla_\theta D)/D$ grows like $1/d_\theta$ as the learned determinant $D \to 0$ near its node. What matters is that this divergence is *integrable* under the $|\Psi|^2$ sampling measure, which weights it by $|\Psi|^2 \sim d_\theta^2$ (and the near-node points are further suppressed by the importance screening). REINFORCE avoids the *stronger*, non-integrable divergence of the strong-form residual, in which the Laplacian term $\nabla^2 D/D$ enters the gradient directly (Appendix C). Keeping the Laplacian in the *reward* but out of the *gradient flow* is what lets the collocation pipeline train backflow and Jastrow parameters stably in regimes where direct residual minimisation fails.

The superiority of pure REINFORCE (β=0) over the hybrid objective (β>0) has a related explanation. The direct weak-form term $\tilde{E} = \frac{1}{2}\|\nabla \log \Psi\|^2 + V$ is cheaper to evaluate (no Laplacian needed) but minimises a different functional: it drives the wavefunction toward low kinetic energy rather than low total energy. At nonzero β, the resulting gradient conflict between the two objectives manifests as oscillatory behaviour in the energy, with the REINFORCE term pulling toward the variational minimum and the direct term pulling toward a kinetic-energy minimum that is generally distinct.

7.3.2 What it achieves

N=6**: near-reference accuracy across all** ω**.** For six electrons, collocation training reaches DMC-level accuracy where DMC exists—at $\omega \geq 0.5$ the errors are $\lesssim 0.01\%$, inside the DMC statistical uncertainty. At the most extreme confinements, $\omega \in \{0.01, 0.001\}$, where the electrons are separated by $\sim$30–120 a_0^* (a few oscillator lengths) and no DMC reference is available, the energies stay within 0.1–0.25% of our converged variational gold standard; this is agreement between two independent training paradigms on the same ansatz, not a certificate of absolute accuracy. The best campaign result at ω=1.0 is E=20.1620 ± 0.0056 (+0.013%), and the best multi-stage run E=20.1613 ± 0.0023 (+0.010%); both are indistinguishable from DMC.

These results are achieved without any Markov chain in the training gradients. MCMC enters only outside the gradient loop: a small Metropolis probe periodically selects the best checkpoint, and the final heavy evaluation certifies the energy. The training itself uses 4096 importance-resampled collocation points per epoch with an oversampling factor of 8, requiring $\sim$32k proposal evaluations per gradient step. That is ten times the $\sim 3\times10^3$ samples per step used by the SR route, but each is a single independent amplitude evaluation rather than a step of a correlated chain that must first equilibrate—so the two are comparable in wall-clock cost, and the collocation batch is embarrassingly parallel where the chain is not.

The best N=6 result at ω=1.0 was obtained through a three-stage continuation chain (see §4.5.2): an initial BF+Jastrow joint REINFORCE run (900 epochs, lr=5×10^{-4}), followed by a low-LR continuation (750 epochs, lr=3×10^{-4}), and a final hard-focus stage (500 epochs, lr=2×10^{-4}). Each stage inherited the full checkpoint from the previous one. The monotonic improvement across stages—+0.08%, +0.05%, +0.010%—demonstrates that patient multi-stage polishing is effective within the collocation framework.

N=12**: sub-**0.03% **at strong confinement.** For twelve electrons, the BF+Jastrow collocation results track DMC closely at $\omega \geq 0.5$, with best-run errors of +0.009% and +0.024% at ω=1.0 and 0.5 respectively (Table 6.3; the hand-tuned multi-stage chains reach +0.018% and +0.029%). At ω=0.1 the error increases to +0.102%, reflecting the growing difficulty of importance sampling in 24 dimensions at weak confinement. The ESS at ω=0.1 is typically 5–15 per epoch (out of 4096 kept points), indicating that the Gaussian mixture proposal still maintains usable overlap with $|\Psi|^2$.

Attempts to extend to $\omega \leq 0.01$ at N=12 were unsuccessful: both direct starts and cascade warm-starts from ω=0.1 failed to converge, with the ESS collapsing to 1 and the rollback mechanism rejecting all gradient updates. This marks the boundary of the current collocation approach for N=12.

N=20 **exposes the memory–sampling trade-off.** Twenty electrons in two dimensions push the collocation pipeline against a hardware wall rather than a conceptual one. The backflow's edge-feature tensor scales as $\mathcal{O}(N^2 \times h_{\mathrm{bf}})$, which at N=20 forces the hidden dimension, the collocation budget, or the oversampling factor down far enough that the gradient quality suffers—so much so that a Jastrow-only ansatz, freed of that cost, trains better than the backflow ansatz it should in principle beat. Because the limitation is one of memory and gradient noise rather than of the method, we report the N=20 collocation results, and the reversal that exposes the bottleneck, in Appendix D.2 rather than here.

7.3.3 What worked and what did not

Most of the method space we explored did not work. The failures are reported here beside the successes, because a working configuration tells you only that some set of choices is sufficient, whereas a failure chased far enough to name its mechanism tells you which choice was load-bearing—and that transfers to systems other than this one. Each attempt below was, at the time, an intuitively reasonable idea. Each turned out to violate an assumption the method had never been forced to state.

Table 7.4: Collocation variants tried, and what each one settled. Errors are relative to the reference of Table 6.3. The three marked † are taken up in prose below; the rest are recorded for completeness.

Attempt	Outcome	Mechanism
Pure REINFORCE (β=0) over the hybrid objective	adopted	The direct weak-form term is cheaper but minimises a different functional; at $\beta > 0$ it pulls toward a kinetic-energy minimum distinct from the variational one, and the two gradients oscillate against each other.
Cascade warm-start from higher ω	essential	Cold starts never converged at ω=0.01 for $N \geq 6$. Cascading from ω=0.1 cut the initial error from ∼30% to ∼5% and made +0.15% reachable in ∼1000 epochs.
ESS-adaptive oversampling with instability rollback	essential	Raises the proposal budget when overlap degrades and reverts on energy spikes. Without both, ω=0.001 diverged within a few hundred epochs.
CG-SR below $\omega \simeq 0.01$ †	abandoned	+4–36% against +0.15–0.24% from Adam. Not ill-conditioning but unestimability; see below.
Langevin proposal refinement †	abandoned	Worse in every configuration tested (+153% against +5.7% at N=20, ω=0.1). The weights are computed against a proposal the samples have stopped following; see below.
Finite-difference residual loss	abandoned	+0.22% at N=6, ω=1 against +0.010% from REINFORCE. Sensitive to the step size and to noise in the Laplacian estimate.
Neural Pfaffian ansatz	abandoned	+0.43% at N=6, ω=1, forty times the BF+Jastrow error. The larger parameter space amplifies importance-sampling gradient noise faster than the richer nodal topology repays.
Backflow at N=20 †	abandoned	+18% against +1.45% for Jastrow-only. A memory limit rather than a conceptual one; see §7.3.5.

Why natural gradients stop helping. CG-SR gave the fastest convergence and the best final energies at $\omega \geq 0.1$, and then failed outright below it. The obvious explanation is ill-conditioning: at weak confinement the Fisher matrix $F_{ij} = \mathbb{E}[\partial_i \log |\Psi|^2 \, \partial_j \log |\Psi|^2]$ has eigenvalues spanning many orders of magnitude, because parameters controlling short-range correlation and parameters controlling the orbital envelope act on scales separated by $1/\sqrt{\omega}$.

The obvious explanation is wrong, or at least incomplete. Ill-conditioning is what makes natural gradients *useful*; a well-conditioned metric is one Adam can handle unaided. What disqualifies them is the failure to *estimate* the metric. F is built from importance-resampled points, and when the ESS is low a handful of high-weight samples dominate it, leaving it noisy and rank-deficient. Tikhonov damping stabilises the inversion at the cost of washing out the curvature information that motivated it.

So there are three regimes, not two. Where the metric is well conditioned, SR is sound and unnecessary. Where it is anisotropic and still estimable, SR is at its most valuable—which is the collocation case of §7.2. And where importance-sampling collapse makes the empirical metric unreliable, inverting it amplifies the estimation noise and SR does active harm. The failures above are the third regime, not the second, and the practical implication is about the estimator rather than the curvature: any second-order method applied to a weakly confined many-body system must certify that the metric it inverts is statistically resolved at the sample sizes in use.

Why pushing the samples uphill makes things worse. Langevin refinement was motivated by an intuition that is hard to argue with. If the Gaussian proposal overlaps $|\Psi|^2$ poorly, evolve the samples up the score $\nabla_{\mathbf{R}} \log |\Psi|^2$ for K steps before reweighting, and the overlap should improve. It failed in every configuration tested, and the reason is standard in the MCMC literature and easy to miss in an importance-sampling setting.

The weights $w_i = |\Psi(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$ are valid only if the $\mathbf{R}_i$ came from q. After K steps from an out-of-equilibrium start the samples follow some intermediate $\tilde{q}$ that depends on K and ε, while the weights still divide by q. The resulting bias is not noise. It pushes the wavefunction toward the regions the Langevin dynamics favours, which moves $\tilde{q}$ further the same way, and the loop closes on itself—which is what the +153% in the table is. Ten to twenty steps in Nd=40 dimensions do not equilibrate. Fixing it means running to equilibrium, adding a Metropolis–Hastings step, or computing annealed weights along the non-equilibrium path. The first two are MCMC under another name; the third costs enough to forfeit the advantage that made the route worth taking.

What the failures have in common. Read together these are not separate disappointments. Four of them—CG-SR, Langevin, the Pfaffian's parameter count, and the backflow at N=20—fail through one channel: a quantity that must be *estimated* from a finite sample degrades faster than the quantity it was meant to improve. Natural gradients need an estimate of the metric. Langevin needs samples that have actually equilibrated. A larger parameter space needs a correspondingly better-resolved gradient. None of these methods is wrong in principle, and each would be sound given samples we cannot afford. What fails is the estimate, and it fails precisely in the regime where the correction was supposed to help. That thread is what §7.3.4 follows deliberately, from the proposal mismatch at one end to the evaluation bottleneck at the other.

7.3.4 Reliability analysis: the gradient-quality chain

The results in Table 6.3 report the *best* energies obtained by collocation training, selected from multiple runs on different seeds. A natural question is whether these results are *typical* or whether they reflect rare fortunate outcomes from a high-variance process. This subsection reports a systematic 30-run reliability campaign designed to answer that question, and reveals a simple causal chain that explains both the successes and failures of the MCMC-free collocation approach.

Experimental design

All runs use the BF+Jastrow ansatz (∼49k parameters) initialised from the same pretrained checkpoint. The grid is N=6 across $\omega \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$ with three random seeds per cell (42, 137, 314), giving $5 \times 2 \times 3 = 30$ runs. Two training recipes are compared head-to-head:

- **Config A (baseline):** Adam optimiser with lr=5×10^{-4}, $\mathrm{lr}_{\mathrm{jas}}$=$5\times10^{-5}$, 4096 collocation points, 8× oversampling, local-energy clipping at 5σ, no reward normalisation, no rollback, fixed Gaussian mixture proposal (Eq. (4.9)).
- **Config B (robust):** Identical to Config A plus four stabilisation mechanisms: 16× oversampling (doubling the proposal budget), reward normalisation (standardising the REINFORCE reward to zero mean and unit variance before the gradient step), instability rollback (reverting to the previous checkpoint when the energy jumps exceed 3σ, with learning-rate decay of 0.95), and an adaptive Gaussian mixture proposal (8 GMM components refitted every 30 epochs to track the evolving $|\Psi|^2$).

All runs train for 1200 epochs. Evaluation uses a 20 000-sample VMC probe every 50 epochs, with the best-VMC checkpoint restored for a final 20 000-sample IS evaluation.

Main results

Table 7.5 presents the complete results of the 30-run campaign. For each $(\omega, \mathrm{recipe})$ combination, three seeds provide a mean, standard deviation, and worst-case error.

Several patterns are immediately apparent. First, the robust recipe dramatically improves the low-ω regimes: at ω=0.001 the mean error drops from +5.3% (baseline) to +0.28% (robust), a factor of 19 improvement. Second, the best single result at every ω is a robust run, achieving +0.013% at ω=1.0, +0.002% at ω=0.5, +0.263% at ω=0.1, +0.237% at ω=0.01, and +0.250% at ω=0.001. Third, the robust recipe exhibits higher seed-to-seed variance (CV ∼ 1.0) at high ω but dramatically lower variance at low ω, suggesting that its stabilisation mechanisms have the most impact precisely where sampling is hardest.

Table 7.5: Reliability campaign: relative energy errors for $N=6$ across five trap frequencies, scored against the reference column of Table 6.3—DMC at $\omega \geq 0.1$, and our own converged variational (PINN+CTNN) energy at $\omega \leq 0.01$, where no DMC value exists. The latter are therefore measures of internal consistency, not of absolute accuracy. Each cell shows the final IS-evaluated error for one (ω, recipe, seed) combination. Bold entries mark the best result at each ω. "Mean" and "CV" are the mean error and coefficient of variation across three seeds.

ω	Recipe	Seed 42	Seed 137	Seed 314	Mean	CV
1.0	Baseline	+0.073	+0.045	+0.060	0.059	0.24
	Robust	+0.032	+**0.013**	+0.105	0.050	0.97
0.5	Baseline	+0.106	+0.060	+0.108	0.091	0.30
	Robust	+**0.002**	+0.112	+0.047	0.054	1.04
0.1	Baseline	+0.419	+0.547	+0.427	0.464	0.15
	Robust	+0.325	+**0.263**	+0.270	0.286	0.12
0.01	Baseline	+0.500	+0.379	+0.746	0.542	0.34
	Robust	+0.289	+0.282	+**0.237**	0.269	0.10
0.001	Baseline	+3.876	+7.187	+4.733	5.265	0.33
	Robust	+**0.250**	+0.331	+0.263	0.281	0.15

The gradient-quality chain

The reliability campaign exposes a simple causal chain that governs the quality of collocation-trained wavefunctions:

$$\boxed{\begin{array}{l} \text{proposal mismatch} \xrightarrow{\text{causes}} \hat{k} \text{ inflation} \xrightarrow{\text{causes}} \text{gradient noise} \\ \qquad\qquad\qquad\quad \xrightarrow{\text{causes}} \text{optimisation stochasticity} \xrightarrow{\text{causes}} \text{run-to-run variance} \end{array}} \tag{7.3}$$

Each link in this chain can be verified quantitatively from the training logs. We now explain each component and the evidence supporting each causal link.

Step 1: proposal–target mismatch. As described in §4.5.1, the REINFORCE estimator in Eq. (4.8) requires samples from $|\Psi_\theta|^2$, but collocation draws from the analytic Gaussian mixture proposal $q(\mathbf{R})$ in Eq. (4.9) and reweights via importance sampling. The quality of this reweighting depends on how well q overlaps with $|\Psi_\theta|^2$.

For the harmonic quantum dot, $|\Psi|^2$ concentrates on a manifold whose geometry depends on ω: at ω=1.0, electrons occupy a compact cloud of radius of order the oscillator length $\ell = 1/\sqrt{\omega}$, and the three-component Gaussian mixture in Eq. (4.9) provides adequate coverage. At ω=0.001, the electron cloud extends to $\sim 100\, a_0^*$ (several oscillator lengths, $\ell \approx 31.6\, a_0^*$), the pair correlation function develops sharp shell structure (Section 6.2), and $|\Psi|^2$ concentrates on a thin correlated submanifold that the isotropic Gaussian proposal covers inefficiently.

The mismatch manifests directly in the importance weights. At ω=1.0, weights are relatively uniform: the effective sample size is typically ESS $\sim$ 5000–6000 out of $\sim$65k proposal samples (16× oversampling of 4096 points), and the Pareto shape parameter $\hat{k}$ of the weight tail is 0.4–0.5, indicating a well-behaved thin-tailed distribution. At ω=0.001, the ESS collapses to 1–6 (out of the same 65k proposal), and $\hat{k}$ rises above 1.0, entering the regime of infinite-variance importance weights.

Step 2: $\hat{k}$ inflation and gradient quality. The Pareto shape parameter $\hat{k}$ of the importance weight distribution, estimated via the PSIS diagnostic of Vehtari et al. [50], provides a precise measure of gradient reliability. When $\hat{k} < 0.7$, the importance-weighted gradient estimator has finite variance and the central limit theorem applies normally. When $\hat{k} > 1.0$, the variance of the estimator is infinite and the gradient signal is dominated by a few extreme samples.

Table 7.6 summarises the average $\hat{k}$ across the campaign. The correspondence between $\hat{k}$ and final accuracy is striking.

The robust recipe reduces $\hat{k}$ by approximately a factor of 2 at every ω. At ω=1.0 this takes $\hat{k}$ from 0.75 (borderline) to 0.50 (reliable), resulting in a modest improvement. At ω=0.1 it takes $\hat{k}$ from 1.48 (broken) to 0.70 (borderline), corresponding to the transition from mediocre (+0.46%) to marginal

Table 7.6: Average Pareto $\hat{k}$ of the importance weight distribution during training, by regime and recipe. Lower is better; $\hat{k} < 0.7$ is considered reliable, $\hat{k} > 1.0$ indicates infinite-variance weights.

	$\hat{k}$ (avg)		Mean error (%)	
ω	Baseline	Robust	Baseline	Robust
1.0	0.75	0.50	0.059	0.050
0.5	0.83	0.54	0.091	0.054
0.1	1.48	0.70	0.464	0.286
0.01	2.63	1.39	0.542	0.269
0.001	3.13	1.86	5.265	0.281

(+0.29%) accuracy. At ω=0.001 even the robust recipe has $\hat{k}$=1.86, but the adaptive proposal and rollback mechanisms prevent the catastrophic divergence seen in the baseline (+5.3%).

Step 3: from gradient noise to optimisation stochasticity. When $\hat{k}$ is large, most of the gradient signal comes from a few samples with extreme importance weights. Which samples are extreme depends on the random proposal draw at each epoch—a function of the random seed. Two seeds that happen to draw different extreme samples will follow different optimisation trajectories, even though they share the same architecture and hyperparameters.

This is not the usual seed dependence of stochastic gradient descent, where the law of large numbers ensures that gradient estimates are unbiased and the trajectory variance decreases with batch size. When $\hat{k} > 1$, the variance of the gradient estimator is *infinite*: increasing the batch size reduces the variance sublinearly (or not at all), and extreme samples can dominate the update indefinitely. The result is optimisation trajectories that diverge qualitatively across seeds, explaining the large run-to-run variance observed at low ω.

Step 4: the evaluation bottleneck. An additional subtlety is that the final energy evaluation itself uses importance sampling from the same proposal distribution. Even if the model has found an excellent wavefunction, a noisy evaluation can report a misleading energy.

The data expose this directly. Table 7.7 compares the VMC-best energy seen during training (a 20 000-sample Metropolis probe from the model's own $|\Psi|^2$, used for checkpoint selection) with the final IS evaluation that uses the Gaussian proposal. At high ω where IS is reliable, the two estimates agree closely. At low ω, the gap between them grows, and its sign is unpredictable—the IS evaluation can overestimate or underestimate the true energy.

Table 7.7: Gap between VMC-best checkpoint energy (Metropolis sampling from $|\Psi|^2$) and final IS evaluation, for selected runs. Positive gap means IS reports higher (worse) energy than the VMC probe saw.

ω	Recipe	Seed	VMC-best (%)	Final IS (%)	Gap (%)
1.0	Baseline	314	+0.001	+0.060	+0.059
1.0	Robust	137	+0.002	+0.013	+0.011
0.5	Robust	42	+0.037	+0.002	−0.035
0.1	Robust	137	+0.136	+0.263	+0.127
0.01	Baseline	42	+0.320	+0.500	+0.180
0.001	Baseline	314	+4.114	+4.733	+0.619

The first row is particularly instructive: this baseline run with seed 314 achieved a VMC-best energy of +0.001%—essentially exact—during training, but the IS evaluation reports +0.060%. The wavefunction is excellent; the evaluation is noisy. This demonstrates that some nominally "bad" runs may contain excellent wavefunctions measured poorly, and some nominally "good" results may benefit from evaluation noise in the favorable direction.

Summary of the chain. The causal chain in Eq. (7.3) provides a unified explanation for the full landscape of collocation results. At high ω, the proposal matches the target well ($\hat{k} \sim 0.5$), gradients are reliable, optimisation is stable, and all seeds converge to near-DMC accuracy. At low ω, the proposal-to-target mismatch grows, $\hat{k}$ inflates, gradients become noisy, optimisation becomes seed-dependent, and the evaluation itself becomes unreliable. The robust recipe acts on multiple links of this chain simultaneously: higher oversampling improves the raw ESS, reward normalisation stabilises the gradient scale, rollback prevents catastrophic divergence, and the adaptive proposal reduces the fundamental mismatch. No single mechanism is sufficient; each addresses a different link in the chain.

7.3.5 The memory wall at N=20

The reversal of the backflow advantage at N=20 highlights an important architectural trade-off. The copresheaf backflow network (`CTNNBackflowNet`) stores edge features of size $\mathcal{O}(N^2 \times h_{\text{bf}})$, which at N=20 with h_{bf}=128 requires $\sim$10$\times$ the memory of the Jastrow-only network. This forces a choice: reduce h_{bf} (losing expressivity), reduce the collocation budget (losing gradient quality), or reduce the oversampling factor (losing ESS). All three options degrade the result.

The Jastrow-only network, with $\sim$25k parameters and $\mathcal{O}(N^2)$ edge cost (no hidden-dimension multiplier), permits full-size collocation budgets (n_{coll}=4096, oversample=8–16) even at N=20. The resulting gradient is less noisy, and the simpler loss landscape (fewer parameters) makes optimisation more reliable.

This does not mean that backflow is unnecessary at N=20; it means that the current training pipeline cannot supply enough gradient signal to train backflow effectively at this scale. The implication is that scaling collocation to larger systems requires either (i) memory-efficient backflow architectures (e.g. sparse attention, low-rank message passing), or (ii) improved sampling that provides higher ESS at constant computational cost.

7.4 A proposal built from the physics, and the wall beyond it

The ESS collapse of §7.2 is a limitation of the *proposal*, not of collocation itself, and it can be substantially pushed back. This section reports a physics-informed proposal that extends MCMC-free collocation deep into the Wigner regime, the mechanism that limits it, and the wall that remains.

7.4.1 The origin-centred proposal misses the Wigner shell

Every proposal in the standard pipeline is a mixture of *origin-centred* Gaussians,

$$q(\mathbf{R}) \;=\; \frac{1}{K}\sum_{k}\mathcal{N}\big(\mathbf{R};\,0,\,\sigma^2_{f,k}/\omega\big)\,. \tag{7.4}$$

But as $\omega \to 0$ the electrons localise into a Wigner molecule on a *shell* far from the origin. The classical $N = 6$ minimum is the $(1,5)$ configuration: one central electron and five on a ring of radius $1.334\,\omega^{-2/3}$ [8, 9]. In oscillator units that ring sits at 1.96, 2.87 and 4.22 ℓ for $\omega = 0.1$, 0.01 and 0.001, moving outward as the trap weakens and past the reach of a width-2ℓ origin Gaussian. The proposal aims where $|\Psi|^2$ has no mass; hence the collapse.

7.4.2 The Wigner-ring proposal

We replace the origin mixture with a *Wigner-ring* proposal: a radial Gaussian at the shell radius times an angular distribution for the ring electrons, plus a central Gaussian. On trusted states this lifts the effective sample size by 15–47$\times$ over the origin mixture (Table 7.8), turning a collapsed sampler into a usable one—without any training, a cheap decisive test. Fitting the ring to $|\Psi|^2$ samples (the simplest adaptive form) doubles the ESS again; a $(1,5)+(0,6)$ mixture *hurts* at $N = 6$, since the state is $(1,5)$ and the extra channel dilutes the proposal mass.

7.4.3 Angular crystallisation is the deep-Wigner wall

Why does the ring work at $\omega = 0.01$ but fail at $\omega = 0.001$? The molecule *crystallises angularly* as the trap weakens. The nearest-neighbour angular-gap standard deviation of the five ring electrons is 40.8°/33.3°/25.5°/6.7° at $\omega = 1/0.1/0.01/0.001$: a "ring liquid" at moderate coupling becomes an angular crystal (electrons pinned near 72° apart) at $\omega = 0.001$. A proposal with independent, uniform angles

Table 7.8: ESS (of 4096) against a trusted $N = 6, \omega = 0.01$ state. The origin mixture is already collapsed; the ring proposal covers the shell.

proposal	ESS
origin-Gaussian mixture (standard)	1.5
Wigner ring $(1,5)$, hand-tuned	22.5
Wigner ring $(1,5)$, fitted to $\vert\Psi\vert^2$	47.5

almost never lands on the crystalline configuration—so the angular distribution must crystallise too. This is captured MCMC-free by modelling the five gaps (which sum to 2π) as Dirichlet(α): an exact tractable density whose single concentration α interpolates liquid (α small) to crystal (α large), fitted to the measured gap variance. This diagnosis *explains* the collocation wall that is otherwise only observed.

7.4.4 An MCMC-free cascade reaches $\omega = 0.0035$

Combining the ring proposal (radial Gaussian × Dirichlet gaps) with importance oversampling, a *continuous* adaptive re-fit of the proposal to the current importance-weighted batch (every few steps, no MCMC), and a gentle ω-cascade seeded from a converged $\omega = 0.01$ state, MCMC-free collocation produces states that lie on the Wigner energy-scaling curve down to $\omega = 0.0035$—four cascade rungs, with heavy-VMC energies within 1–2% of the scaling estimate (Table 7.9), well past where the origin proposal collapses (ESS$\approx$1.5 of 4096 at $\omega = 0.01$, Table 7.8). The essential ingredient is the continuous re-fit: it keeps ESS healthy ($\sim 10^2$) *while* the energy descends, where a static or slowly-updated proposal collapses to ESS$\sim$1 as the wavefunction moves away from it.

Table 7.9: MCMC-free Wigner-ring cascade, $N = 6$: heavy-VMC energy vs. the fitted Wigner scaling curve $E(\omega) \approx 0.690\,(\omega/0.01)^{0.689}$, whose exponent is close to the classical 2/3 but whose prefactor is anchored at $\omega = 0.01$, where no external reference exists. The agreement is therefore internal consistency, not validation. Good to $\omega = 0.0035$.

ω	E	scaling est.	ratio
0.0077	0.577	0.577	1.00
0.0059	0.481	0.480	1.00
0.0045	0.400	0.398	1.01
0.0035	0.340	0.335	1.02

7.4.5 The remaining wall: a fine-tuned Slater–Jastrow balance, not the sampler

Below $\omega = 0.0035$ the cascade becomes unstable, and—informatively—the instability is *non-monotonic* across proposal variants: wide-exploratory, tempered-tracking, and outward-anchored proposals each converge some rungs and diverge others. That pattern locates the obstacle not in the proposal but in the warm-start, and four transfer diagnostics (taking the converged $\omega = 0.0035$ rung down to $\omega = 0.0027$, a 23% step) pin down its mechanism precisely.

First, the frontier state is genuine: at its own $\omega = 0.0035$ the rung is a bona-fide Wigner molecule, with radial density peaked at 3.34ℓ (classical ring 3.42ℓ) and a *single* basin—initialising the sampler on the shell converges to the same 3.34ℓ. Transferred to $\omega = 0.0027$, however, the wavefunction collapses to $\sim 1\ell$, and two tests show *where* the collapse lives. It is not the sampler: initialising the Metropolis walkers directly on the classical shell (3.57ℓ) still relaxes inward to 1.1ℓ, so the collapsed density is a property of the ansatz, not of sampler initialisation. And it is not the backflow: its displacement is small ($|\Delta\mathbf{x}| \approx 0.13\ell$, radially inward) and essentially *preserved* across the transfer ($0.15\ell \to 0.13\ell$)—it scales correctly with ℓ but is far too small to bridge the 2.4ℓ gap between the Slater core ($\sim 1\ell$) and the Wigner shell.

The mechanism is a delicate cancellation. The bare harmonic-oscillator determinant suppresses the shell by $\exp\left(-\omega r^2/2\right) \sim \exp(-5.6)$ in amplitude ($\sim 10^{-5}$ in density); the Wigner peak exists only because the Jastrow amplifies it back by a comparable factor, with the backflow contributing a small inward nudge. A 23% change in ω perturbs both the Slater confinement and the Jastrow correlation length, the fine-tuned balance tips, and the density falls back to the non-interacting shell—from which the collocation re-fit cannot reliably climb out (hence the non-monotonic proposal behaviour). This rules out the natural fix of an ω-robust warm-start by coordinate rescaling: because the ansatz relaxes inward even when handed shell configurations, no coordinate transform can preserve the state. Reaching $\omega < 0.0035$ would require per-ω re-optimisation against a shell-*amplitude*-anchored objective, or a genuinely ℓ-covariant parametrisation—a distinct problem left to future work. The contribution here is concrete: a physics-informed proposal that carries MCMC-free collocation from the origin-proposal wall down to a verified Wigner molecule at $\omega = 0.0035$, and a precise account of the angular-crystallisation and Slater–Jastrow-balance effects that bound it.

7.5 A gauge caveat, and what survives it

Which of the numbers above can be leaned on is not obvious, and the answer is not flattering to all of them.

Component-level quantities—the backflow's rank, its displacement magnitude $|\Delta\mathbf{x}|$, the per-component d_{eff}—are **gauge-like**. The Jastrow and the backflow are substitutes. How correlation is divided between them is a nearly flat, path-dependent direction of the loss, so those quantities record the route the training took and not the state it reached. The worry is not hypothetical. Every claim that reversed during this project was of this kind—the d_{eff} inversion at the crystal that did not survive seeding, a mislabelled backflow, an artefactual rank collapse produced by training both components jointly from scratch.

Four measurements are invariant to that gauge, and the conclusions of this chapter rest only on them. *Energy* and *ablation*—delete a component and price it—carry the architecture comparison. $\mathrm{Var}(E_L)$ separates the two families where energy cannot and where d_{eff} has stopped trying: at matched protocol and production capacity it is lower for the message-passing ansatz at every confinement, by factors of two to seven (Table 7.1), and seed-cleanly so at the crystal. And *state overlap* is what makes the comparison between training paradigms a claim about states rather than about numbers that happen to agree.

Rank is not on that list. It runs through this chapter as the clearest *signature* of the mechanism—a field collapsing onto a single mode is easy to see. What carries the finding is the half percent of energy that collapses with it.

Part V

Discussion

Convictions are more dangerous enemies of truth than lies.

FRIEDRICH NIETZSCHE
Human, All Too Human

Chapter 8

Discussion

This chapter asks what the preceding results mean, which is a different question from what they are. It does not re-report them.

The three threads of this thesis turned out to be one geometry. Architecture sets the shape of the variational tangent space—message passing compresses it in the correlator and keeps it from collapsing in the nodes. The optimiser inverts the metric on that space, and is worth its cost exactly when the metric is badly conditioned and still measurable. The training paradigm chooses which measure defines the metric in the first place: $|\Psi|^2$ conditions itself and stays on target, while a fixed covering proposal loses first the conditioning and then the state.

Four things remain to be said. What the energies do and do not establish. How far the internal numbers can be pushed, given that several of them are gauge-dependent. Where the boundary of Markov-chain-free training actually lies, and why it lies there. And what the picture predicts for systems larger than these.

8.1 What the energies establish

The energies are the least interesting result in this thesis and the most necessary one. Everything downstream—every claim about what the network learned, every structural diagnostic read off the trained state—is worthless if they do not hold. So the question worth pressing is not whether the numbers are good. It is what having good numbers actually buys.

Less than it appears at one end, and more than it appears at the other.

Less, at the top end. Agreement with a fixed-node DMC calculation is agreement with another approximation, not with the truth. Where our message-passing ansatz sits *below* a published DMC value—N=12 at ω=1, for instance—the natural reading is not that something has gone wrong but that a better nodal surface has been found; the variational principle bounds the energy from below by the exact ground state, not by the nodal constraint of the reference calculation. That reading is plausible and we adopt it, though establishing it would require the DMC trial nodes and a matched uncertainty analysis, which we have not done. What can be said without qualification is that at $\omega \geq 0.1$ two independent methods agree to a few hundredths of a percent. That is the strongest external statement available.

More, at the bottom end—but only by inheritance, and the inheritance has to be argued rather than assumed. Below $\omega = 0.1$ no published value exists for $N \geq 6$. What transfers there is not accuracy but *conditioning*: the same architecture, the same protocol, the same trap-unit scaling and soft-core features and analytic cusp, applied across three decades of ω without adjustment. A method that had to be retuned at each confinement would carry nothing across the boundary. This one does not, and its energies join the benchmarked regime smoothly and follow the classical scaling exponent once they are past it. That is a reason to take the deep-Wigner numbers seriously as variational predictions. It is not a reason to call them validated, and we do not.

8.2 How far the internal numbers can be pushed

There is no guarantee that a trained network organises itself in a way anyone can interpret. High-dimensional function approximators are entitled to be incomprehensible, and usually are. The surprise here is not that the internal organisation is interpretable but that it is *small*, and small in a way that follows the physics rather than the parameterisation: a real-space transition involving shells, ring occupancies and angular stiffness, held on one to three effective directions whose identity—not whose number—changes across the crossover.

It is tempting to read that as a discovery about neural wavefunctions. It cannot quite be one, for a reason that reaches well past this thesis.

The Jastrow and the backflow are partial substitutes. Any measurement made on one of them alone—its rank, its displacement magnitude, its share of the energy—records a division of labour that the loss barely constrains, and that a different training route could have made differently while reaching the same state. Such quantities are gauge-like. Reporting them is not wrong; treating them as properties of the architecture is. Every claim that reversed during this project was of that kind.

What survives is what can be measured on the state or by deletion: the energy, the local-energy variance, the price of removing a component, and the overlap between two states. And on those, the controlled comparison in Chapter 7 does support a genuine architectural statement—one sharper than the dimension count that first suggested it. The low-rank geometry says *how many* directions the correlator uses. The tangent-kernel analysis says *which*, and that turns out to be the substantive question. The message-passing correlator's dominant directions are the Hamiltonian's own collective modes—breathing, correlation hole, relative excitation—to $R^2 \approx 1$ across ω and up to $N{=}12$, while a separable network of matched or larger capacity leaves nearly half its tangent budget where no compact physical dictionary finds it. A direction outside the dictionary need not be unphysical; the robust claim is that one tangent space is far more completely accounted for by a small set of physical operators than the other. That gap is present at every confinement. Strong confinement merely disguises it, because correlation is a small slice of the energy there and the two networks agree to four figures.

The practical consequence is a design rule rather than a theorem. A compact, smooth correlator plus a moderately wide backflow is a good default for parabolic dots: the correlator behaves like a learned set of collective coordinates, and the backflow like a local correction channel that switches on where near-field physics is delicate and falls quiet in the Wigner limit. But the *kind* of backflow keeps mattering after its energy contribution has gone. A conventional single-round network collapses onto one collective mode exactly where the copresheaf network does not, and pays half a percent for it. Falling quiet and falling over are not the same thing.

8.3 Where Markov-chain-free training stops, and why

Of everything here, the collocation programme is the part whose claim most needs its boundaries drawn before it is stated. "An accurate many-body wavefunction trained with no Markov chain" is the kind of sentence that travels further than its evidence. It is also the part that consumed the most time and produced the most failures, and the two facts are related: the method works inside a regime whose edges had to be found by walking off them.

What "MCMC-free" does and does not mean. Three distinctions are easily conflated. (i) The pipeline is *Markov-chain-free*: no configuration is ever drawn from a Markov chain during training—samples come from an analytic proposal $q(\mathbf{R})$ and are importance-reweighted. It is not free of Monte Carlo in the broader sense; importance sampling and multinomial resampling are Monte-Carlo operations themselves. (ii) It is not, in general, *reference-free*. Where a DMC energy exists ($\omega \geq 0.1$), the residual pretraining anneals its target toward E_{DMC} and the instability rollback monitors the relative error against it, so the DMC value enters the training signal. Those runs are trained without a Markov chain; they are not an unsupervised solution of the Schrödinger equation. (iii) The deep-Wigner cascade is the one regime that is both Markov-chain-free *and* reference-free: below $\omega = 0.1$ no DMC value exists, so the cascade is guided by the variational objective alone and cross-checked against the fitted Wigner scaling curve, never against an external benchmark.

There are two walls, and they are different. Most of the failures lie on a single chain, and it is worth stating in order. Proposal–target mismatch inflates the tail of the importance weights; the inflated tail destroys the gradient estimator; a destroyed estimator makes the optimisation seed-dependent and the empirical metric unreliable; and once the metric is unreliable, inverting it—which is what natural-gradient preconditioning does—amplifies the noise instead of the signal. Langevin refinement joins that chain at the first link, by breaking the proposal the weights are computed against. CG-SR joins it at the last: it fails at low ω not because the metric is ill-conditioned but because it can no longer be estimated. The N=20 reversal, where a Jastrow-only ansatz beats the backflow it should in principle beat, reaches the same chain through memory rather than through confinement, the edge tensor forcing the collocation budget down until the gradient degrades. That chain is a sampling-quality chain, and a better proposal moves it: rebuilding the proposal from the Wigner shell pushes the wall from $\omega \approx 0.01$ down to $\omega = 0.0035$.

Below 0.0035 something else takes over, and conflating the two would be a mistake. There the obstacle is not the sampler. Metropolis walkers initialised directly on the classical shell relax inward regardless; the backflow displacement transfers correctly and is simply far too small to bridge the gap; and the collapse is non-monotonic across proposal variants in a way that a sampling failure would not be. What fails is the warm start. The Wigner peak at these confinements exists only because the Jastrow amplifies an exponentially suppressed determinant back up by a comparable factor, and a 23% change in ω perturbs both sides of that cancellation. The balance tips and the density falls back to the non-interacting shell. This is a representational limitation—the ansatz is not covariant enough in ℓ—not a statistical one, and no improvement to the proposal will remove it.

Two walls, then, of different kinds. The first is statistical and yields to a better measure. The second is representational and will not. Distinguishing them matters practically: it says where effort should go. And the general lessons survive the distinction. Any importance-sampled scheme applied to a weakly confined many-body system will meet the first chain in that order, and any second-order method applied to one must certify that the metric it inverts is statistically resolved at the sample sizes in use. Neither statement is about quantum dots.

Where the two routes actually stand. Compared directly against the SR route, collocation is competitive for $N \lesssim 12$ and $\omega \gtrsim 0.1$, and loses ground below that. At N=6, ω=1 the two are indistinguishable, both inside DMC uncertainty. At N=12, ω=0.1 collocation is a few times worse. At N=20 it is not competitive at all. The SR pipeline has its own scaling problems—mixing times grow with N and with correlation strength—but they arrive later, because an MCMC sampler adapts to $|\Psi|^2$ throughout training while a fixed proposal cannot. The natural reading is that the two are complementary rather than rival: collocation for fast, parallel, chain-free exploration and pretraining; SR for final accuracy at large N or weak confinement.

8.4 What the picture predicts

If the argument of this thesis is right, the same three statements should hold well outside the quantum dot.

Up to the frontier, the proposal is the bottleneck rather than the objective. That makes the next move a better sampler rather than a better loss—normalising flows trained against $|\Psi|^2$, or the physics-informed proposals of §7.4 generalised beyond ring geometries. Rebuilding the proposal from the Wigner shell lifted the effective sample size by 15–47× without any training at all. Past the frontier the target changes: what is needed there is an ℓ-covariant parametrisation, or an objective anchored on the shell *amplitude*, so that a converged state survives being carried to a weaker trap.

Relational structure will matter more, not less, with N. The tangent-space gap between message-passing and separable correlators widens with particle number in the range we can measure—$\lesssim 0.02$ against ≈ 0.77 dictionary-orthogonal mass at N=12. If that continues, the architectures that scale are the ones that maintain a transported edge representation rather than pooling pairwise features and discarding them.

The quadratic backflow is the first thing that has to change. The $\mathcal{O}(N^2 h_{\mathrm{bf}})$ edge tensor is what stops the method at twenty electrons, and it stops it for reasons of memory rather than of physics. Sparse or low-rank message passing, or a local backflow that avoids all-to-all edge features, would extend the reach without touching anything else in the pipeline—and, on the evidence of Chapter 7, without giving up the nodal correction that makes the backflow worth having at all.

Finally, a caution that applies to this thesis more than to any of its predictions. The architecture comparison is cleanest at $N=6$, where it is seeded; the $N=12$ contrast rests on fewer trained models and the $N=20$ analysis on a single production state. The tangent-space picture is therefore established at one size and indicated at the others. Extending it is not a matter of interpretation but of compute.

Part VI

Conclusion

All things are implicated with one another, and the bond is holy; and there is hardly anything unconnected with any other thing.

MARCUS AURELIUS
Meditations

Chapter 9

Conclusions

The question this thesis was assigned was where to start. The answer it arrived at has a shape, and the shape is a loop.

We built a wavefunction and checked it against every external number that exists. The checked wavefunction then became an instrument, and what it showed us was a Wigner molecule—rings at definite occupancies, angular order that stiffens as the trap weakens, a shell radius growing as $\omega^{-2/3}$. Those are facts about electrons. They are also, it turned out, exactly the facts a sampler needs in order to aim at the right part of configuration space. Building the proposal out of the measured shell geometry carried Markov-chain-free training from $\omega \approx 0.01$ down to $\omega = 0.0035$, into states the original method could not reach. The physics we learned from the solution improved the solver, and the improved solver produced solutions we could not otherwise have learned from.

That loop is the thesis. Everything below is an account of how far around it we got, and where it broke.

9.1 What was asked, and what came back

Architecture. What does message passing buy that a separable network cannot? A relational channel—and, measured through the tangent kernel, a specific one (§7.1). A message-passing correlator reaches the same ground state along half as many effective directions as a separable network of matched or larger capacity, and those directions are the collective modes of the Hamiltonian rather than an arbitrary basis. The separable network spends nearly half its tangent budget where no compact physical dictionary finds it, matches the energy to four figures anyway, and carries a higher local-energy variance for it at every confinement. In the nodes the same channel shows up inverted: what message passing supplies at the Wigner crystal is not a contribution but the absence of a collapse. A conventional backflow falls onto one mode and costs half a percent; the copresheaf backflow, at that confinement, could be deleted for 0.017%. Both channels give way at the same ω, and a better correlator postpones both.

Optimiser. When does natural-gradient preconditioning earn its cost? When the tangent metric is anisotropic *and still estimable* (§7.2). That is a narrower claim than the usual one and it has three regimes rather than two. Under $|\Psi|^2$ sampling the metric is well enough conditioned that Adam already suffices. On the collocation objective it is badly conditioned and stochastic reconfiguration is what makes the optimisation work at all. And once the sampling that builds the metric collapses, inverting it amplifies the noise rather than the signal, and the method that was decisive becomes harmful. Ill-conditioning is not the criterion. Estimability is.

Paradigm. Can an accurate wavefunction be trained without a Markov chain? Yes, inside a regime whose edges we had to find by walking off them (§7.3). With no chain anywhere in the gradient, the weak-form objective reaches DMC accuracy at $N \in \{6, 12\}$ under strong confinement and stays within a few tenths of a percent of our own converged energies well into the correlated regime. It fails where the importance sampling loses the target, and it fails through one chain of causes—proposal mismatch, weight-tail inflation, gradient noise, an unestimable metric—which a better proposal moves rather than removes.

Physics. Can the trained state serve as an instrument? It can, and the evidence is that six independent diagnostics agree when they need not have (§6.2). Shell topologies match the classical Coulomb-cluster predictions at every size we studied, with the competing sectors carrying the weight a rotating molecule should give them. The reconstructions of $g(r)$ from shell geometry alone reach cosine similarity 0.989–1.000. At N=20 a three-shell molecule collapses toward a single shell within one decade in ω. None of this was put into the ansatz, and none of it is visible in the density that one would naturally plot.

9.2 What is established, what is consistent, and what is indicated

The preface promised to be wrong on the record. That obliges a ledger, because these three are not the same kind of claim and the difference is easy to lose.

Established against something outside this thesis. The energies for $N \leq 12$ at $\omega \geq 0.1$, and for N=2 down to $\omega = 0.01$. Two independent methods agree there to a few hundredths of a percent. That is the whole of the external ground truth, and every relative error quoted anywhere else in this work depends on a reference we computed ourselves.

Internally consistent, and consistent in ways that could have failed. The deep-Wigner energies, the structural picture, and the architecture comparison. Nothing here is validated from outside, but the internal agreements are not cheap: independent diagnostics that measure different things converge on the same crossover; the fitted energy exponent lands near the classical 2/3 that nothing in the ansatz knows about; the shell topologies reproduce classical ground states nobody trained the network to reproduce; and the architecture result survives seeding, a change of probe measure, and a capacity ladder that runs against it.

Indicated, and no more. The size-scaling of the tangent-space picture. The architecture comparison is seeded at N=6, rests on fewer models at N=12, and on a single production state at N=20. The trend is in the right direction and widens with particle number, which is what the mechanism predicts. It is not settled, and settling it is a question of compute rather than of interpretation.

To that ledger belongs one further entry, which is methodological rather than physical. Quantities defined on a single component of the ansatz—the backflow's rank, its displacement magnitude, its share of the energy—are gauge-like. The Jastrow and the backflow are substitutes, so the division of labour between them is a nearly flat direction of the loss that the training route settles and the state does not. Every claim that reversed during this project was of that kind. What survives is what can be measured on the state or by deletion, and the conclusions above rest only on those.

9.3 Where it stops, and what would move it

Two walls bound the Markov-chain-free route, and they are different in kind. The first is statistical: the proposal loses the target, and a better proposal moves the wall—from $\omega \approx 0.01$ to $\omega = 0.0035$, in our case, by building the proposal out of the physics. The second is representational and does not yield to sampling at all. Below $\omega = 0.0035$ a converged state does not survive transfer to a weaker trap, because the Wigner peak exists only through a near-cancellation between an exponentially suppressed determinant and a Jastrow that amplifies it back, and a 23% change in ω tips that balance. Walkers placed directly on the classical shell relax inward regardless. No proposal can repair that. What is needed is an ℓ-covariant parametrisation, or an objective anchored on the shell amplitude rather than on the energy alone.

Three other things would move the picture, in rough order of how much they would move it. A backflow that is not quadratic in particle number: the $\mathcal{O}(N^2 h)$ edge tensor is what stops the method at twenty electrons, and it stops it for reasons of memory rather than of physics. A proposal that adapts on its own—a normalising flow, or the shell-informed construction of §7.4 generalised past ring geometries. And the same architecture study repeated at N=12 and N=20 with the seeding the N=6 comparison already has, which would turn the last entry of the ledger from indicated into established.

9.4 A closing thought

The three axes of this thesis—what the network is, how it is optimised, and what measure defines the loss—looked at the outset like three separate engineering choices. They are one geometry. Architecture sets the shape of the variational tangent space. The optimiser inverts the metric on that space. The sampling measure decides whether the metric can be estimated at all. Read that way, the recurring finding is not about neural networks in particular: what matters is not capacity but alignment with the low-dimensional, collective structure the problem actually has—which, in real space, *is* the Wigner molecule.

The most durable results here came from the places where the attempt failed, and they came only because the failures had been made measurable: a rank collapse that survived seeding and a dimension inversion that did not; a sampler blamed for a wall the ansatz had built; an importance-sampling bug whose signature—accuracy falling away at larger N and weaker ω—was exactly what an inadequate architecture would have produced, and which no amount of thinking about architectures would ever have found. That is what the record was for.

Part VII

Appendices

Chapter A

Laplacian conditioning

A.1 Residual training for Schrödinger

We consider the stationary many-body Schrödinger equation on configuration space $\mathbf{R} = (\mathbf{r}_1, \dots, \mathbf{r}_N) \in \Omega \subset \mathbb{R}^D$ with $D = dN$ (for our quantum dots, $d = 2$):

$$\hat{H}\Psi(\mathbf{R}) = E\Psi(\mathbf{R}), \qquad \hat{H} = \hat{T} + \hat{V}, \qquad \hat{T} = -\frac{1}{2}\sum_{i=1}^{N} \Delta_i, \tag{A.1}$$

where $\hat{V}$ collects external and interaction potentials.

In *strong residual minimization* one introduces

$$L := \hat{H} - E \quad \Rightarrow \quad L\Psi = 0, \tag{A.2}$$

and minimizes a squared residual loss of the schematic form

$$\mathcal{L}_{\text{res}}(\theta) = \frac{1}{2}\|L\Psi_\theta\|^2_{L^2(\Omega;w)} = \frac{1}{2}\int_\Omega |(L\Psi_\theta)(\mathbf{R})|^2\, w(\mathbf{R})\, d\mathbf{R}, \tag{A.3}$$

for a nonnegative weight w induced by the sampling strategy. A key message of the numerical-analysis viewpoint is that strong residual losses can be *severely ill-conditioned*, and training speed is tightly linked to spectral properties of an operator/matrix induced by L [42].

A.2 Linearized geometry and the matrix A

To expose the mechanism, follow the standard linearization argument used in conditioning analyses [42]. Fix an expansion point θ_0 and define tangent (feature) functions

$$q_j(\mathbf{R}) := \partial_{\theta_j}\Psi_\theta(\mathbf{R})\big|_{\theta=\theta_0}, \qquad j = 1, \dots, P. \tag{A.4}$$

For small parameter increments $\delta\theta$, we have the first-order model

$$\Psi_{\theta_0+\delta\theta}(\mathbf{R}) \approx \Psi_0(\mathbf{R}) + \sum_{j=1}^{P} q_j(\mathbf{R})\,\delta\theta_j, \qquad \Psi_0 := \Psi_{\theta_0}. \tag{A.5}$$

Applying the residual operator gives

$$(L\Psi_{\theta_0+\delta\theta})(\mathbf{R}) \approx (L\Psi_0)(\mathbf{R}) + \sum_{j=1}^{P}(Lq_j)(\mathbf{R})\,\delta\theta_j. \tag{A.6}$$

Introduce the weighted inner product (complex-valued wavefunctions allowed)

$$\langle f, g\rangle_w := \int_\Omega f(\mathbf{R})^*\, g(\mathbf{R})\, w(\mathbf{R})\, d\mathbf{R}. \tag{A.7}$$

Let the Jacobian J be the linear map whose jth column is Lq_j. Then the loss (A.3) becomes (to second order in $\delta\theta$)

$$\mathcal{L}_{\text{res}}(\theta_0 + \delta\theta) \approx \frac{1}{2}\|r_0 + J\,\delta\theta\|_w^2, \qquad r_0 := L\Psi_0. \tag{A.8}$$

with normal-equation matrix

$$A := J^*J, \qquad A_{ij} = \langle Lq_i,\, Lq_j\rangle_w. \tag{A.9}$$

Here A is Hermitian positive semidefinite; in practice it can be close to singular, so one often effectively works with a damped variant $A + \lambda I$ (implicit or explicit).

Now consider gradient descent on the quadratic model (A.8) (with step size η). The error in parameter space evolves as

$$\delta\theta^{(t+1)} - \delta\theta^\star = (I - \eta A)\left(\delta\theta^{(t)} - \delta\theta^\star\right), \tag{A.10}$$

so each eigen-direction of A contracts by a factor $(1 - \eta\lambda)$. Hence convergence speed is governed by the eigenvalue spread of A, typically summarized by the (effective) condition number

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \tag{A.11}$$

(over the nonzero/regularized spectrum). Large $\kappa(A)$ implies stiffness: a step size small enough to be stable for $\lambda_{\max}$ yields slow progress along directions associated with $\lambda_{\min}$. This is precisely the conditioning bottleneck emphasized in physics-informed training analyses [42].

A.3 Bi-Laplacian scaling

The defining feature of Schrödinger is that L contains a second-order differential operator. To isolate the kinetic contribution, consider the regime where the residual is dominated by the Laplacian part,

$$L \approx -\frac{1}{2}\Delta_{\mathbf{R}}. \tag{A.12}$$

From L to L^*L. The matrix A in (A.9) is a Gram matrix in the feature space $\{Lq_j\}$. The corresponding operator-level object is the *Hermitian square* L^*L (with adjoint taken in the $L^2(\Omega; w)$ inner product) [42]. For general non-constant w, the adjoint L^* differs from the formal differential adjoint by weight-induced lower-order terms; the cleanest picture is obtained for constant w and standard boundary conditions (periodic/Dirichlet), where the Laplacian is self-adjoint.

In the Laplacian-dominant regime (A.12), one has (up to a constant factor)

$$L \approx -\tfrac{1}{2}\Delta \quad \Rightarrow \quad L^*L \approx \tfrac{1}{4}\,\Delta^2, \tag{A.13}$$

i.e. a bi-Laplacian scaling. The prefactor does not affect conditioning ratios.

Fourier/eigenmode derivation of k^4. On a periodic domain (or, more generally, in a Laplacian eigenbasis) $-\Delta\phi_{\mathbf{k}} = |\mathbf{k}|^2\phi_{\mathbf{k}}$, so $\Delta^2\phi_{\mathbf{k}} = |\mathbf{k}|^4\phi_{\mathbf{k}}$. If an error $\delta\Psi$ expands as $\delta\Psi = \sum_{\mathbf{k}} c_{\mathbf{k}}\phi_{\mathbf{k}}$ and $L \approx -\frac{1}{2}\Delta$, then

$$\|L\,\delta\Psi\|_{L^2}^2 \approx \frac{1}{4}\sum_{\mathbf{k}} |\mathbf{k}|^4 |c_{\mathbf{k}}|^2. \tag{A.14}$$

Thus the squared residual penalizes high-frequency components with a *quartic* weight $|\mathbf{k}|^4$. If the relevant spectrum spans $|\mathbf{k}| \in [k_{\min}, k_{\max}]$, the induced eigenvalue ratio scales like

$$\kappa \sim \left(\frac{k_{\max}}{k_{\min}}\right)^4, \tag{A.15}$$

which is a sharp mathematical expression of "Laplacian-driven stiffness". This quartic growth is consistent with the Laplacian/Poisson toy examples and frequency-dependent conditioning effects emphasized in [42].

Length-scale interpretation. Let ℓ denote the smallest length scale that must be represented. Then $k_{\max} \sim 1/\ell$ and (A.15) implies

$$\kappa \text{ grows rapidly as } \ell \downarrow 0. \tag{A.16}$$

In many-body Schrödinger problems, small ℓ arises from nodal structure, strong localization, and short-range correlation features; in all such regimes, the Laplacian makes the strong residual extremely curvature-sensitive.

A.4 Barron/spectral perspective: the Laplacian

The conditioning story above can be rephrased in approximation terms. In spectral Barron-type viewpoints [39], function classes are controlled by norms that weight the Fourier transform by powers of frequency (or related spectral weights). A key implication is that applying derivatives systematically increases the importance of high-frequency tails [42]. Since Δ multiplies a Fourier mode by $|\mathbf{k}|^2$, and the squared residual effectively involves L^*L (hence Δ^2 in Laplacian-dominant regimes), the residual can require substantially more "spectral budget" than the wavefunction itself. In practical terms: even if Ψ is approximable in a given network class, *approximating its strong residual* can be much harder, because the loss implicitly demands accurate high-frequency curvature (cf. (A.14)). This is one of the motivations, in numerical analysis of PINNs, for considering weak/variational formulations that reduce derivative order when the strong form is too stiff [42].

A.5 Implications for Stochastic Reconfiguration

Stochastic reconfiguration (SR) updates parameters using a natural-gradient-like preconditioner [29, 31, 54]. Writing $O_i = \partial_{\theta_i} \log \Psi_\theta$, one common SR update is

$$S(\theta)\,\delta\theta = -g(\theta), \qquad S_{ij} = \mathrm{Cov}_{p_\theta}[O_i, O_j], \qquad g_i = 2\,\mathrm{Cov}_{p_\theta}[E_L, O_i], \tag{A.17}$$

where $p_\theta \propto |\Psi_\theta|^2$ and $E_L = (\hat{H}\Psi_\theta)/\Psi_\theta$. SR improves conditioning in parameter space by using the (quantum) information geometry of the variational family [29, 54]. However, the Laplacian-induced stiffness described above is an *operator-level* phenomenon: strong residuals magnify curvature errors via Δ and Δ^2 (cf. (A.14)), which remains a fundamental challenge even when parameter updates are geometrically preconditioned.

Chapter B

Coulomb conditioning

B.1 Coulomb structure and singularity

For our two-dimensional electronic systems ($\mathbf{r}_i \in \mathbb{R}^2$), the interaction contains Coulomb terms of the form

$$\hat{V}_{\mathrm{C}}(\mathbf{R}) = \sum_{1 \le i < j \le N} \frac{1}{r_{ij}}, \qquad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|. \tag{B.1}$$

This is a *singular multiplication operator*: it diverges at particle coalescence $r_{ij} \to 0$. The exact eigenfunction remains finite (or vanishes by antisymmetry) but develops non-smooth short-distance structure (a cusp) so that divergences in kinetic and potential contributions cancel in physically meaningful quantities such as the local energy [31].

The central point for learning is:

> Coulomb does not merely add "another term" to the residual; it introduces *singular short-range physics* that forces delicate cancellations with the Laplacian. Strong residual objectives square these effects and become extremely sensitive to tiny cusp errors.

B.2 2D cusp conditions: antiparallel vs parallel spins

We present the 2D cusp logic in a way that separates the two spin channels. The cleanest statement (especially for practical ansätze) is in terms of the short-distance behaviour of a *Slater–Jastrow* factorization

$$\Psi(\mathbf{R}) = D(\mathbf{R})\, J(\mathbf{R}), \qquad J(\mathbf{R}) = \exp\Big(\sum_{i<j} u(r_{ij})\Big), \tag{B.2}$$

because (i) the Slater part D enforces antisymmetry and determines whether Ψ vanishes at coalescence, and (ii) the Jastrow slope $u'(0)$ is what cancels the $1/r$ Coulomb singularity in the local energy [31].

Fix a pair (i, j) and introduce relative/centre-of-mass coordinates in 2D:

$$\mathbf{R}_{ij} := \frac{\mathbf{r}_i + \mathbf{r}_j}{2}, \qquad \mathbf{r} := \mathbf{r}_i - \mathbf{r}_j, \qquad r := |\mathbf{r}|. \tag{B.3}$$

A direct computation gives, in any dimension,

$$\Delta_i + \Delta_j = \frac{1}{2}\Delta_{\mathbf{R}_{ij}} + 2\,\Delta_{\mathbf{r}}, \tag{B.4}$$

hence the singular short-distance kinetic behaviour is controlled by the relative term $-\Delta_{\mathbf{r}}$.

A key 2D identity is

$$\Delta_{\mathbf{r}} r = \frac{1}{r} \qquad (r > 0,\ \mathbf{r} \in \mathbb{R}^2). \tag{B.5}$$

Antiparallel spins (distinguishable in the Slater part). If the two electrons have opposite spin, they enter different determinants in D and generically D does *not* vanish at $\mathbf{r} \to 0$. Then $\Psi(0) \neq 0$ and a direct cusp expansion in 2D is legitimate:

$$\Psi(\mathbf{r}, \ldots) = \Psi(0, \ldots)\Big(1 + a\, r + \mathcal{O}(r^2)\Big), \qquad r \to 0. \tag{B.6}$$

Using (B.5), the singular part of the relative Laplacian is

$$\Delta_{\mathbf{r}}\Psi \approx \Psi(0)\, a\, \Delta_{\mathbf{r}} r = \Psi(0)\, \frac{a}{r}, \qquad \frac{\Delta_{\mathbf{r}}\Psi}{\Psi} \approx \frac{a}{r}. \tag{B.7}$$

Since the pair kinetic contribution contains $-\Delta_{\mathbf{r}}$, the singular part of the local energy is

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi}{\Psi} \supset \left(-\frac{a}{r} + \frac{1}{r}\right) = \frac{1-a}{r}. \tag{B.8}$$

Thus finiteness of E_L at coalescence requires

$$a = 1, \qquad \text{i.e.} \qquad \partial_r \log \Psi|_{r=0} = 1 \quad \text{(2D, antiparallel spins)}. \tag{B.9}$$

Equivalently, in a Slater–Jastrow form with $D(0) \neq 0$, this is the same as $\partial_r \log J|_0 = 1$ for the ij-pair.

Parallel spins (identical fermions, node at coalescence). For same-spin electrons, antisymmetry forces D (and hence Ψ) to vanish as $\mathbf{r} \to 0$. Generically one has a *linear* zero:

$$D(\mathbf{r}, \ldots) \;=\; \boldsymbol{\eta}(\ldots) \cdot \mathbf{r} \;+\; \mathcal{O}(r^2), \qquad r \to 0, \tag{B.10}$$

with some nonzero vector $\boldsymbol{\eta}$ depending on the other coordinates. In this case, the cancellation of the Coulomb singularity is controlled by the Jastrow factor through cross terms in $-\Delta_{\mathbf{r}}(DJ)/(DJ)$. A standard short-distance analysis (carried out explicitly for 2D quantum dots) shows that the divergent kinetic contribution behaves like

$$-\frac{\Delta_{\mathbf{r}}(DJ)}{DJ} \supset -\frac{(2k+d-1)}{r}\, \partial_r \log J \qquad \text{with} \qquad d = 2, \;\; k = 1 \text{ (identical particles)}, \tag{B.11}$$

while the Coulomb potential contributes $+1/r$ to E_L as before. Hence finiteness of the local energy requires

$$\partial_r \log J|_{r=0} = \frac{1}{2k+d-1} = \frac{1}{3} \quad \text{(2D, parallel spins)}. \tag{B.12}$$

This 1 (antiparallel) versus 1/3 (parallel) 2D cusp statement is the standard quantum-dot/Jastrow–Slater cusp condition [55].

What to remember. In 2D:

$$\text{antiparallel spins:} \quad \partial_r \log \Psi|_0 = 1 \quad \Longleftrightarrow \quad \partial_r \log J|_0 = 1, \tag{B.13}$$

$$\text{parallel spins:} \quad \Psi(0) = 0 \text{ (linear node)}, \quad \partial_r \log J|_0 = \frac{1}{3}. \tag{B.14}$$

B.3 Why Coulomb is ill-conditioned (2D emphasis)

The cusp conditions above show *how* Coulomb creates stiffness: it forces kinetic and potential contributions to be individually large and singular, while their *sum* is finite only under a short-distance constraint. Strong residual minimization is therefore a *cancellation problem*.

Sensitivity amplification in the local energy. If the learned antiparallel-spin cusp slope is $a_\theta = 1 + \delta a$, then (B.8) implies

$$E_L(\mathbf{R}) \supset -\frac{\delta a}{r}, \tag{B.15}$$

so an arbitrarily small cusp mismatch produces an arbitrarily large local-energy spike as $r \to 0$.

For parallel spins, if the learned Jastrow slope is $\partial_r \log J|_0 = \frac{1}{3} + \delta b$, then the same short-distance analysis leading to (B.12) implies

$$E_L(\mathbf{R}) \supset -\frac{3\,\delta b}{r}. \tag{B.16}$$

Thus both channels generate $1/r$ spikes unless the cusp is satisfied.

Why squaring is especially dangerous in 2D. A $1/r$ defect in E_L (or in $(\hat{H}-E)\Psi$) is concentrated near coalescence. But in 2D the squared singularity is not integrable—it diverges logarithmically:

$$\int_{r<\varepsilon}\left(\frac{1}{r}\right)^2 d^2\mathbf{r} = \int_0^\varepsilon \frac{1}{r^2}\,(2\pi r\,dr) = 2\pi\int_0^\varepsilon \frac{dr}{r} = +\infty.$$

Therefore, without effectively resolving the cusp, *strong* squared-residual objectives can be dominated (or even made ill-defined in the continuum limit) by the coalescence region. In Monte Carlo practice, this manifests as heavy-tailed fluctuations and extreme sensitivity to rare close-pair samples.

Residual expansion and cancellation structure. Writing $L = \hat{T} + (\hat{V} - E)$ and expanding the squared residual norm:

$$\|L\Psi\|_w^2 = \|\hat{T}\Psi\|_w^2 + \|(\hat{V}-E)\Psi\|_w^2 + 2\,\mathrm{Re}\langle\hat{T}\Psi,\ (\hat{V}-E)\Psi\rangle_w, \tag{B.17}$$

we see that near coalescence both $\hat{T}\Psi$ and $\hat{V}_{\mathrm{C}}\Psi$ are large. The exact solution relies on cancellation between these contributions (cf. (B.8) and (B.12)), which means (B.17) demands high *relative* accuracy in the cusp region: if either term is off by a small fraction, the net residual can be large in absolute magnitude. This is a classic recipe for ill-conditioning: the loss landscape contains directions where small parameter changes cause huge residual changes in a small region.

Operator-level picture (L^*L). From the same viewpoint as Appendix A, strong residual training induces matrices/operators tied to L^*L [42]. For $L = \hat{T} + (\hat{V} - E)$, one has (formally)

$$L^*L = \hat{T}^2 + (\hat{V}-E)^2 + \hat{T}(\hat{V}-E) + (\hat{V}-E)\hat{T}. \tag{B.18}$$

Coulomb enters here in two destabilizing ways:

1. the positive term $(\hat{V}-E)^2$ involves a $1/r^2$-type singular weight;
2. the mixed terms contain derivatives of V through product rules, e.g. $\hat{T}(V\Psi)$ generates $\nabla V\cdot\nabla\Psi$ and ΔV contributions, which are even more singular/distributional at coalescence.

This formal expansion explains why the Coulomb singularity can severely inflate the effective spectral spread of the training operator, aligning with the operator-conditioning viewpoint [42].

B.4 Barron/spectral perspective: the Coulomb singularity

Coulomb forces the *true* 2D solution to have cusp-like, non-smooth short-range structure, so approximating the *strong residual* requires representing high-frequency content that is not well captured by globally smooth/low-complexity function classes.

In spectral/Barron-style approximation theories [39], approximation quality is governed by spectral decay and norms that weight the Fourier tail [42]. A cusp (or a linear node times a cusp in J) implies slower spectral decay than a globally smooth function. Strong residual training then *further* amplifies the high-frequency tail because it applies Δ (and effectively L^*L), which multiplies Fourier modes by powers of $|\mathbf{k}|$ (Appendix A). Hence, even if Ψ can be approximated reasonably in function value, the strong residual can remain large unless the cusp structure is represented accurately. This is exactly the type of regularity mismatch that motivates weak/variational formulations in numerical analysis of PINN-type methods, since weak forms reduce the derivative order demanded of the approximation [42].

Practically, QMC wavefunctions therefore incorporate explicit short-range correlation structure (e.g. cusp-enforcing Jastrow factors), precisely to remove these singular residual pathologies and reduce the variance of local quantities [31, 55].

B.5 Implications for Stochastic Reconfiguration

SR performs a natural-gradient step using a (quantum) Fisher/metric matrix [29, 31, 54]. In one common formulation,

$$S_{ij} = \mathrm{Cov}_{p_\theta}[O_i, O_j], \qquad g_i = 2\,\mathrm{Cov}_{p_\theta}[E_L, O_i], \qquad S\,\delta\theta = -g, \tag{B.19}$$

with $O_i = \partial_{\theta_i} \log \Psi_\theta$.

Coulomb-induced cusp mismatch produces E_L spikes of the form (B.15)–(B.16), which has two consequences:

1. **Heavy-tailed Monte Carlo forces.** The estimator of $g_i = \mathrm{Cov}[E_L, O_i]$ becomes noisy when E_L has rare but extreme spikes (short-range events). This increases variance and can destabilize the linear solve unless one uses damping/regularization, as standard in SR practice [31].

2. **Coupling to curvature sensitivity.** Even if SR corrects anisotropy in parameter space (natural gradient) [29, 54], it does not remove the underlying operator-driven stiffness: the spikes originate from imperfect kinetic–potential cancellation enforced by Coulomb cusp physics.

Bottom line. Laplacian dominance creates stiffness through k^4-type amplification of curvature errors; Coulomb creates stiffness by enforcing cusp constraints (in 2D: 1 vs 1/3 depending on spin channel) and cancellation against the Laplacian. Both mechanisms worsen conditioning of strong residual objectives and increase statistical difficulty of SR through heavy-tailed local-energy fluctuations [31, 42, 55].

Chapter C

The collocation–backflow catch-22

This appendix formalizes a structural difficulty encountered when training backflow transformations under the *strong-form* collocation objective—the one that minimises the variance of the pointwise local energy E_L—and presents extensive numerical evidence for the $N{=}6$, $\omega{=}0.5$ quantum dot. The core observation is a *catch-22*: backflow's learning signal is concentrated near the nodal surface, but the local energy diverges there for any imperfect wavefunction, and any sampling strategy that includes these points produces gradients with unbounded variance. The result is an irreducible accuracy floor—roughly 0.3%, set by the Jastrow alone—that is absent under VMC + SR training of the same ansatz.

Two things about the scope of this result should be stated before the details, so the floor is not misread. First, it is a property of the *strong form* specifically: the weak-form (score-function / REINFORCE) collocation objective used for the results in the main text never differentiates through the Laplacian and does not inherit this pathology, which is why it trains the same ansatz to far below the 0.3% floor. The value here is therefore a diagnosis of one objective's limits, not a measure of what MCMC-free training can achieve. Second, the analysis is what motivated the shift to the weak form in the first place; it is presented as a worked negative result—the kind that redirects a research programme—rather than as a headline number.

C.1 Setup and notation

We consider a Slater–Jastrow–backflow ansätz of the form

$$\Psi_\theta(\mathbf{R}) = \det \tilde{S}_\uparrow(\mathbf{R};\theta_{\mathrm{BF}})\ \det \tilde{S}_\downarrow(\mathbf{R};\theta_{\mathrm{BF}})\ \exp\bigl(J(\mathbf{R};\theta_J)\bigr), \tag{C.1}$$

where $\tilde{S}_\sigma$ is the Slater matrix evaluated at backflow-shifted coordinates $\tilde{\mathbf{r}}_i = \mathbf{r}_i + \Delta\mathbf{x}_i(\mathbf{R};\theta_{\mathrm{BF}})$, J is a neural Jastrow factor (PINN), and $\theta = (\theta_J, \theta_{\mathrm{BF}})$ are trainable parameters. A CTNN message-passing network produces the backflow shifts $\Delta\mathbf{x}_i$.

We train by minimizing the *screened collocation* objective

$$\mathcal{L}(\theta) = \mathrm{Var}_{\mathcal{S}}\bigl[E_L(\mathbf{R};\theta)\bigr], \qquad E_L = \frac{\hat{H}\Psi_\theta}{\Psi_\theta}, \tag{C.2}$$

where the collocation set $\mathcal{S}$ is obtained by importance-sampling from a Gaussian proposal q and retaining the top-K points ranked by $|\Psi_\theta|^2/q$ ("screened collocation"). This top-K selection naturally suppresses near-node configurations where $|\Psi|^2 \approx 0$.

C.2 The divergence of E_L near nodes

The local energy decomposes as

$$E_L = -\frac{1}{2}\frac{\nabla^2\Psi}{\Psi} + V = -\frac{1}{2}\Bigl[\nabla^2 J + |\nabla J|^2 + 2\,\nabla J\cdot\frac{\nabla D}{D} + \frac{\nabla^2 D}{D}\Bigr] + V, \tag{C.3}$$

where $D := \det\tilde{S}_\uparrow \det\tilde{S}_\downarrow$ and the last term $\nabla^2 D/D$ contains second derivatives of the backflow shifts $\nabla^2\Delta\mathbf{x}_i$ amplified by the inverse determinant.

For the exact wavefunction, E_L is constant. If Ψ is an eigenstate, $E_L = E$ everywhere, and the apparent D^{-1} singularity is cancelled by the numerator vanishing at the same rate as the denominator. Concretely, for a smooth wavefunction with a simple node at surface $\mathcal{N}$, both D and $\nabla^2 D$ vanish linearly in the signed distance d from $\mathcal{N}$:

$$D(\mathbf{R}) = \mathcal{O}(d), \qquad \nabla^2 D(\mathbf{R}) = \mathcal{O}(d), \qquad \frac{\nabla^2 D}{D} \text{ is finite on } \mathcal{N}. \tag{C.4}$$

For an approximate wavefunction, the cancellation generically fails. During training, the backflow parameters θ_{BF} are imperfect: the *learned* nodal surface $\mathcal{N}_\theta$ (where $D(\cdot;\theta) = 0$) does not coincide with the *true* nodal surface. Let d_θ denote the signed distance from the learned node and d^* that from the true node. Near a point where $d_\theta \approx 0$ but $d^* \neq 0$, the determinant $D \sim d_\theta \to 0$ while the kinetic contributions from the Jastrow and potential terms remain finite:

$$E_L(\mathbf{R}) \supset \frac{\nabla^2 D(\mathbf{R};\theta)}{D(\mathbf{R};\theta)} \sim \frac{c}{d_\theta} \to \pm\infty \qquad \text{as } d_\theta \to 0,\ d^* \neq 0, \tag{C.5}$$

for some configuration-dependent constant $c \neq 0$. Conversely, near a point on the true node ($d^* = 0$) but off the learned node ($d_\theta \neq 0$), $\Psi \sim d^* \to 0$ and the wavefunction assigns vanishing weight. The pathology (C.5) is therefore not an artifact of the nodal structure itself—it is a property of *mismatched* nodes, and is worst precisely when backflow still has something to learn.

C.3 The geometric mismatch argument

We now show that the variance of the local energy (and hence of the training gradients) generically diverges for imperfect backflow parameters.

Consider the contribution to $\mathrm{Var}[E_L]$ from a neighborhood U_ε of the learned nodal surface:

$$\mathrm{Var}[E_L] \geq \int_{U_\varepsilon} |\Psi(\mathbf{R};\theta)|^2 \left(E_L(\mathbf{R};\theta) - \bar{E}\right)^2 d\mathbf{R}. \tag{C.6}$$

When the node is "simple" (D vanishes linearly), we have

$$|\Psi|^2 \sim d_\theta^2, \qquad\qquad E_L^2 \sim d_\theta^{-2\alpha}, \tag{C.7}$$

where α characterises the severity of the mismatch. For the generic case where the numerator $\nabla^2 D$ does *not* vanish on $\mathcal{N}_\theta$ (because $\mathcal{N}_\theta \neq \mathcal{N}^*$), one has $\alpha = 1$ so that $E_L^2 \sim d_\theta^{-2}$.

In D-dimensional configuration space, the volume element in a tubular neighborhood of a codimension-1 surface scales as $dV \sim d_\theta^0\, d\sigma\, dd_\theta$. Thus the integrand in (C.6) behaves as

$$|\Psi|^2 \cdot E_L^2 \cdot dV \sim d_\theta^2 \cdot d_\theta^{-2} \cdot dd_\theta = dd_\theta, \tag{C.8}$$

which integrates to $\mathcal{O}(\varepsilon)$ and is therefore *finite*: a simple node does not, by itself, make $\mathrm{Var}[E_L]$ diverge under the $|\Psi|^2$ measure. The instability enters one derivative later, in the *parameter gradient.* But the key observation is that this is the *best case.* In practice, the mismatch between $\mathcal{N}_\theta$ and $\mathcal{N}^*$ is not uniform: there exist regions where the surfaces cross at angles, creating pockets where $|d^*|/|d_\theta|$ is large (the true wavefunction is non-negligible while the learned determinant is near-zero), and there the E_L^2 spike is not compensated by $|\Psi|^2$ suppression.

More concretely, the gradient $\partial\mathcal{L}/\partial\theta_{\mathrm{BF}}$ involves terms of the form

$$\frac{\partial E_L}{\partial \theta_{\mathrm{BF}}} \supset \frac{1}{D}\frac{\partial(\nabla^2 D)}{\partial \theta_{\mathrm{BF}}} - \frac{\nabla^2 D}{D^2}\frac{\partial D}{\partial \theta_{\mathrm{BF}}}, \tag{C.9}$$

both of which diverge as $D \to 0$. Even in the marginal case where $\mathrm{Var}[E_L]$ is finite, the *gradient variance* (which involves these higher-order poles) generically diverges, making gradient-based optimisation impossible at points near the learned node.

C.4 The catch-22

The preceding analysis reveals a three-sided dilemma:

1. **Backflow learns at the node.** The purpose of backflow is to reshape the nodal surface of the Slater determinant. The gradient signal $\partial E_L/\partial\theta_{\mathrm{BF}}$ is largest where the backflow shifts actually change which side of the node a configuration point lies on, i.e., where $D \approx 0$.
2. **E_L diverges at the node for imperfect θ.** As shown in §C.2–C.3, the local energy and its parameter gradient contain poles of the form D^{-1} and D^{-2} whenever the learned node does not coincide with the exact node. The resulting gradient variance is unbounded.
3. **Good collocation sampling must avoid the node.** Under screened collocation, the top-K selection ranks points by $|\Psi|^2/q \propto D^2 e^{2J}/q$. Since $D^2 \to 0$ at the node, near-node points are systematically excluded from the training batch. This is *protective*—it prevents gradient explosion—but it also removes the points where backflow's learning signal lives.

A "catch-22" emerges:

> *Including near-node points gives divergent, noisy gradients and backflow shrinks defensively or diverges. Excluding them gives stable training but removes backflow's learning signal. In either regime, collocation cannot effectively train backflow beyond an accuracy floor set by the Jastrow (PINN) quality alone.*

C.5 Why VMC + SR escapes

Under VMC, the sampling distribution *is* $|\Psi_\theta|^2$, which assigns vanishing measure to the nodal surface. Near-node configurations are strongly *suppressed*—for a simple node the probability of lying within a distance ϵ scales as ϵ^3—rather than strictly absent, which is enough to keep the D^{-1} singularity out of the gradient estimator in practice. However, VMC *still optimises the node position* through a subtle mechanism:

Indirect nodal information through bulk correlations. The variational energy $\langle E_L\rangle_{|\Psi|^2}$ is sensitive to nodal position because shifting the node changes the wavefunction *everywhere* (through normalisation and through the functional form of D). The SR force vector $g_i = \mathrm{Cov}_{|\Psi|^2}[E_L, O_i]$ (where $O_i = \partial_{\theta_i} \log\Psi$) receives contributions from the *bulk*—regions where both $|\Psi|^2$ and E_L are well-behaved—that are *correlated with* nodal position but do not require evaluating E_L at the node.

SR amplifies the signal. The Fisher metric $S_{ij} = \mathrm{Cov}[O_i, O_j]$ captures how sensitive $\log\Psi$ is to each parameter direction. By solving $(S + \varepsilon I)\,\delta\theta = -g$, SR amplifies exactly those subtle bulk signals that encode nodal information, compensating for the fact that the node itself has zero weight under $|\Psi|^2$.

Core asymmetry. VMC obtains nodal information *indirectly* through a well-conditioned integral (the covariance g_i is finite because E_L is sampled away from the node); collocation tries to obtain it *directly* through a divergent pointwise quantity (E_L evaluated near the node). None of the sampling, loss, or preconditioning strategies we tested (§C.6) overcame this asymmetry within the strong-form collocation framework; we present it as a structural obstruction of the tested pipeline rather than a theorem that no collocation method can ever avoid it.

C.6 Experimental evidence

We present a systematic study for N=6 electrons in a 2D harmonic trap with ω=0.5 (DMC reference: $E_{\mathrm{DMC}} = 11.78484(6)$ [16]). The ansätz uses a PINN Jastrow (∼15,000 parameters) and a CTNN backflow network (∼8,600 parameters) with bf_scale = 0.7. All energies are evaluated via MCMC with 15 000 samples after burn-in.

C.6.1 Baseline: PINN alone vs. PINN + CTNN

Table C.1: Baseline collocation results for N=6, ω=0.5. "PINN only" trains without backflow; "PINN + CTNN joint" trains both networks from scratch for 300 epochs with bf_scale = 0.7. The backflow provides only a marginal improvement of ∼0.1 percentage points.

Configuration	E (Ha)	σ_E	err (%)
PINN only (300 ep)	11.834	0.003	0.42
PINN + CTNN joint (300 ep)	11.823	0.003	0.33
PINN + CTNN separate opt	11.822	0.003	0.32
DMC reference	11.78484	—	0.00

The backflow reduces the error from 0.42% to 0.32–0.33%, far short of the $< 0.01\%$ accuracy achievable with VMC + SR on the same ansatz (Table 6.2).

C.6.2 Sampling strategies

We tested multiple strategies to bring near-node information into the collocation batch, hoping to give the backflow a stronger learning signal.

Table C.2: Effect of sampling strategy on collocation accuracy. All experiments use 50-epoch fine-tuning from the best collocation checkpoint (0.35%), except "hard-smooth (scratch)" which trains 300 epochs from random initialisation.

Method	Description	err (%)	Outcome
Top-K baseline	standard screened collocation	0.35	stable
Hard mining + Huber	top-K of residuals	0.41	neutral
Hard mining + smoothness	+ $\|\nabla^2 \Delta x\|^2$ penalty	0.38	slight help
SIR (single proposal)	$\lvert\Psi\rvert^2$-reweighted resampling	0.47	worse
SIR (mixture)	Gaussian mixture proposal	0.44	worse
Gumbel top-K	stochastic top-K selection	5.00	catastrophic

Every strategy that includes more near-node points (SIR, Gumbel) produced *worse* results—consistent with the catch-22: the additional points carry divergent E_L values that corrupt the gradient. Gumbel top-K (which replaces deterministic selection with stochastic Gumbel-softmax) was catastrophic, with early stopping triggered by rapid energy divergence.

C.6.3 Loss modifications

Table C.3: Effect of loss function modifications. The hybrid loss adds a mean-energy (variational) term; the det filter discards points with small $|\det \tilde{S}|$.

Method	Description	err (%)	Outcome
Variance loss (baseline)	Huber on $E_L - \bar{E}$	0.33	best collocation
Hybrid anneal	$\beta \cdot \mathrm{Var} + (1-\beta) \cdot \mathrm{Mean}$, $\beta : 0.7 \to 0.15$	2.59	catastrophic
Gumbel + hybrid	stochastic top-K + hybrid	5.00	catastrophic
Hybrid + det filter	β=0.2 fixed + $\lvert\det\rvert$ filter	24.75	broken

The mean-energy gradient $\nabla_\theta \langle E_L \rangle_{\mathcal{S}}$ under biased collocation sampling does *not* equal the VMC gradient—the sampling distribution is not $|\Psi|^2$—so the variational principle does not hold, and the backflow drifts to large, unphysical shifts ($|\Delta \mathbf{x}| \approx 0.83$, versus ∼0.23 under variance-only training). The variance loss's implicit constraint on backflow magnitude is *protective*, not pathological.

Table C.4: Regularisation of the E_L singularity (50-epoch fine-tune from 0.35% checkpoint). Each modification smooths or clips one of the divergent contributions to E_L.

Method	Modification	err (%)	Outcome
Soft Coulomb	$1/r \to 1/\sqrt{r^2+\varepsilon^2}$, ε=0.2	1.41	much worse
Regularised det	$\log\lvert D\rvert \to \text{logaddexp}(\log\lvert D\rvert, \log\delta)$	0.49	worse
T clipping	$\max(\lvert T\rvert, T_{\max})$, $T_{\max}$=30	0.43	neutral

C.6.4 Gradient regularisation

We attacked the divergence sources in E_L directly:

Soft Coulomb trains on wrong physics (V is no longer the true potential). Regularising the determinant distorts the $\log\Psi$ landscape. Kinetic energy clipping is neutral because the clipped points are exactly those that carried the (noisy) backflow signal. None of the regularisations improve upon the baseline.

C.6.5 Gradient preconditioning

Table C.5: Gradient preconditioning experiments (300-epoch scratch training unless noted). K-FAC uses block-diagonal Kronecker-factored Fisher on the backflow network only.

Method	err (%)	Outcome
Vanilla Adam	0.33	baseline
K-FAC on CTNN	0.34	no improvement
K-FAC + smoothness	0.51	worse
Variance-minimising pretrain	0.54	worse
Adam fine-tune (30 ep, lr=1e-4)	0.34	no improvement
Adam burst (15 ep, lr=3e-4)	0.56	worse

K-FAC provides a block-diagonal approximation to the Fisher matrix and should, in principle, help with the ill-conditioned parameter-space geometry. In practice, it does not: the conditioning problem is *operator-level* (Appendices A–B), not merely parameter-space anisotropy, and K-FAC cannot address the fundamental divergence in E_L.

C.6.6 Summary of all interventions

The pattern is striking:

- Every intervention that *includes* more near-node points (SIR, Gumbel) *increases* the error—backflow shrinks defensively under gradient noise or diverges entirely.
- Every intervention that *changes* the loss to tolerate near-node contributions (hybrid mean-energy, det filter) produces catastrophic results because the collocation sampling is not $|\Psi|^2$ and the variational principle does not apply.
- The smoothness penalty $\lambda\|\nabla^2\Delta\mathbf{x}\|^2$ helps modestly (from 0.59% to 0.38% without it vs. with it) by damping the $\nabla^2 D/D$ poles, but cannot eliminate the fundamental divergence.
- Ironically, the only method that trains backflow effectively is MCMC-based VMC + SR—which sidesteps the entire collocation framework.

C.7 Connection to Appendices A and B

The catch-22 is a specific manifestation of the general conditioning difficulties analysed in Appendices A–B:

Table C.6: Complete summary of all collocation-based backflow experiments for N=6, ω=0.5. No intervention breaks the ∼0.32% accuracy floor. For comparison, the same PINN + CTNN ansatz achieves < 0.01% under VMC + SR (Table 6.2).

Experiment	err (%)	Category
PINN + CTNN sep opt (best)	0.32	baseline
PINN + CTNN joint 300 ep	0.33	baseline
K-FAC on CTNN	0.34	preconditioner
Adam fine-tune 30 ep	0.34	optimiser
Top-K baseline (retrained)	0.35	sampling
Hard mining + smoothness	0.38	sampling
Hard mining + Huber	0.41	sampling
PINN only	0.42	no backflow
T clipping	0.43	regularisation
SIR (mixture)	0.44	sampling
SIR (single)	0.47	sampling
Regularised det	0.49	regularisation
K-FAC + smoothness	0.51	preconditioner
Var-min pretrain	0.54	pretrain
Adam burst lr=3e-4	0.56	optimiser
Soft Coulomb	1.41	regularisation
Hybrid loss (annealed)	2.59	loss
Gumbel top-K + hybrid	5.00	sampling + loss
Hybrid + det filter	24.75	loss + filter
Same ansatz, VMC + SR	**<0.01**	**variational**

Laplacian stiffness (Appendix A). The k^4-type amplification of curvature errors (§A.3) is compounded by backflow, because $\nabla^2(\det \tilde{S})$ introduces second derivatives of $\Delta\mathbf{x}$ that are not present in a fixed-node ansatz. These additional high-frequency modes inflate the spectral spread of the training operator L^*L.

Coulomb singularity (Appendix B). The $1/r$ Coulomb potential (§B.1) adds a singular multiplication operator to $\hat{H}$. Near both the coalescence point ($r_{ij} \to 0$) and the node ($D \to 0$), E_L contains products of these two singularities, and the cusp cancellation (§B.2) must be maintained simultaneously with the nodal cancellation (C.4). In the strong residual squared $\|LE_L\|^2$, these compound singularities create an extremely stiff landscape.

The backflow-specific pathology. What distinguishes the backflow catch-22 from the general conditioning problem is the *self-referential* nature of the singularity: the backflow parameters θ_{BF} determine *where* the node is, so updating θ_{BF} moves the singularity in configuration space. Under the collocation loss, the gradient pushes θ_{BF} in a direction determined by E_L values at the current collocation points, but the resulting node shift invalidates those very points. VMC+SR avoids this because it samples from the *current* $|\Psi_\theta|^2$ at every step, so the sampling distribution co-adapts with the node position.

Chapter D

Post-catch-22 experimental history and philosophy

This appendix documents the full experimental program executed after the backflow catch-22 analysis in Appendix C. The goal here is not to present a single best number, but to make explicit the strategy, the failed branches, the partial recoveries, and the current status for difficult regimes ($\omega \in \{0.01, 0.001\}$ and larger N).

D.1 The experimental programme

After the catch-22 analysis the work shifted from searching for a single better architecture to engineering a training pipeline that makes a physically strong ansatz trainable across regimes. Four principles organised it: keep the Slater–Jastrow–backflow ansatz but treat sampling and preconditioning as first-class objects; replace one-shot "hero" runs with staged continuation (solve easy regimes, warm-start hard ones); separate claims by evidence tier (online probe, restored VMC-best, heavy exact VMC); and record negative results, because many theoretically plausible variants fail on gradient-noise geometry.

What was varied. We varied the *sampling* (Gaussian and Gaussian-mixture proposals scaled by $1/\sqrt{\omega}$; oversample-then-resample with ESS diagnostics; ESS-adaptive oversampling; weight tempering and log-weight clipping), the *objective* (finite-difference collocation residuals; a hybrid weak form; and pure REINFORCE, which became the reliable default, with the Laplacian kept in the reward path but never differentiated), the *preconditioner* (Adam; diagonal-Fisher natural gradient; and full SR in Woodbury/CG modes with damping anneals and trust-region caps), and the *training style* (single sweeps; wave-based cascades with checkpoint inheritance; asynchronous campaigns that relaunch from the best available checkpoint). Two empirical conclusions were robust. First, optimisation quality dominated architectural novelty: moderately sized BF+Jastrow models with stable training beat more ambitious variants with weaker optimisation geometry. Second, at low ω regularising the resampling distribution (ESS control, top-mass reduction) helped while hard cusp gating alone did not—a controlled $2\times2\times3$ matrix (hard gate on/off, resampling-regularisation on/off, three seeds) confirmed both, with no measurable gain from gating and a consistent gain from resampling regularisation.

Outcomes. SR-style preconditioning in CG mode, warm-started along the confinement cascade ω : $1.0 \to 0.5 \to 0.1 \to 0.01 \to 0.001$, was what carried performance from solved regimes into harder ones; cold-start SR at low ω did not converge. The full set of best energies is in §7.3.2; representative cascade successes are N=6 at ω=1.0 ($E = 20.16167 \pm 0.00181$, $+0.012\%$), ω=0.5 ($E = 11.78466 \pm 0.00199$, -0.002%) and ω=0.1 ($E = 3.56525 \pm 0.00046$, $+0.32\%$), and N=12 at ω=1.0 ($E = 65.70699 \pm 0.00595$, $+0.010\%$). For N=2 the collocation route now covers the whole confinement range cleanly—$+0.012\%$, $+0.023\%$, -0.004%, $+0.002\%$ and $+0.154\%$ at $\omega = 1, 0.5, 0.1, 0.01, 0.001$ (zero-variance extrapolated)—so the paradigm is DMC-accurate wherever the importance sampling stays healthy and degrades only in the deep-Wigner limit.

D.2 Current frontier: low-ω and high-N stabilisation

The current unresolved front remains the combination of extreme correlation regimes and larger particle count:

- $\omega \in \{0.01, 0.001\}$: transfer helps, but reliability and reproducibility are not yet at the same confidence level as $\omega \in \{1.0, 0.5, 0.1\}$.
- $N \geq 20$: memory pressure, architecture-capacity constraints, and higher gradient noise make training significantly less stable than $N \in \{6, 12\}$.

Dedicated stabilisation runs introduced stricter SR trust constraints, heavier damping, ESS-adaptive sampling, rollback controls, and explicit process-level relaunch hygiene for GPU memory contention. These changes improved operational stability but have not yet closed the full accuracy gap for the hardest targets.

The N=20 collocation results. For completeness we record here the N=20 collocation energies referred to in §7.3.2; they are the first collocation-trained wavefunctions at this size, and they show the memory bottleneck in the clearest possible form. The backflow's edge-feature tensor scales as $\mathcal{O}(N^2 \times h_{\text{bf}})$, and at N=20 fitting it in memory forces the hidden dimension down to h_{bf}=48–64. The best backflow run at ω=1.0 then reached only $\sim$+18% error, while a Jastrow-only ansatz—freed of that tensor, and so able to afford a larger collocation budget and higher oversampling—reached +1.32% ($E = 157.934 \pm 0.018$). The architecture that should in principle be the more expressive is beaten by the simpler one, purely because the memory-forced capacity cut degrades its gradients more than its extra flexibility repays. Polishing the Jastrow checkpoints at reduced learning rate improved the ω=1.0 error from $\sim$+2.6% to +1.32% over $\sim$400 epochs and the ω=0.5 error from $\sim$+7.0% to +2.74% ($E = 96.447 \pm 0.015$); the best ω=0.1 result was +5.53% ($E = 31.637 \pm 0.008$). The gap to the sub-percent accuracy reached at N=6, 12 follows from the memory-forced capacity reduction and the gradient noise it brings, not from the collocation objective; closing it is a question of a more memory-frugal backflow, not of a different method.

D.3 Interpretive summary

The post-catch-22 phase supports six robust conclusions.

1. The collocation program is not a single recipe but a coupled system: sampling, objective, and preconditioning must be tuned jointly.
2. Transfer/cascade is not optional for difficult regimes; it is the primary mechanism by which low-ω performance becomes attainable.
3. SR-style preconditioning is central for stable progress beyond early Adam-only basins.
4. Diagnostic metrics (ESS, top-mass concentration, rollback counts, probe-final gaps) are essential; scalar final energy alone is insufficient for method development.
5. Several intuitively attractive interventions (e.g., hard cusp gating in this tested range) did not deliver gains, and documenting these negatives was necessary to avoid circular search.
6. The method now demonstrates near-DMC quality in multiple nontrivial targets, but the strongest claims must still be bounded to regimes where repeated campaign evidence exists.

Chapter E

Structural diagnostics in full

Chapter 6 reports the two radial quantities the argument turns on—r_{mode} and the width-to-scale ratio γ. The complete diagnostics are collected here: the mean radius, width, FWHM and decile quantiles for N=2, and the ring-angle spread and pair Lindemann ratio for N=6. Estimators and stability protocol are defined in §5.1. All lengths are in Bohr; angles in radians.

Table E.1: Radial diagnostics for N=2 across ω (Bohr units).

ω	r_{mode}	$\langle r \rangle$	σ_r	FWHM	q_{10}	q_{50}	q_{90}	γ
1.00	1.435	1.632	0.705	1.743	0.735	1.581	2.581	0.491
0.50	2.303	2.475	1.007	2.466	1.209	2.409	3.829	0.437
0.10	3.601	3.830	1.728	4.449	1.640	3.714	6.127	0.480
0.01	14.191	14.593	5.615	13.602	7.390	14.322	22.038	0.396
0.001	64.292	63.953	18.479	43.977	40.392	63.814	87.714	0.287

Table E.2: Radial and angular diagnostics for N=6 across ω (Bohr; angles in radians).

ω	r_{mode}	σ_r	γ	σ(ring angles)	$\gamma_{r_{ij}}$
1.00	2.0907	0.9249	0.4424	0.7046	0.3538
0.50	3.0702	1.3648	0.4445	0.6943	0.3397
0.10	8.1686	3.5025	0.4288	0.6842	0.3149
0.01	33.1538	13.9094	0.4195	0.6497	0.2889
0.001	113.5542	47.8076	0.4210	0.6368	0.2664

Chapter F

Declaration on the Use of Artificial Intelligence

The collocation approach in this thesis was developed over several months of failed experiments. Its central difficulty was not merely that a network did not converge, but that the parts of the Schrödinger residual that made the method physically meaningful also made its gradients unstable near the wavefunction's nodes. During that period, AI systems were used as sounding boards: to test interpretations of unexpected results, to formulate possible explanations, and to decide which experiment should come next. The exchange below is a short sample from one longer conversation at a point where the negative evidence had begun to make abandoning collocation seem plausible.

Author

I am trying to develop a PINN to find ground-state solutions of the Schrödinger equation. I have tried multiple approaches, but I am not succeeding. ...What do you think about this? What should we do from here?

AI response 1

...Anything that injects near-node points makes gradients explode or become uselessly noisy. ...

⋮

AI response 2

...This asymmetry is fundamental and cannot be resolved by sampling tricks, loss modifications, or gradient preconditioning within the collocation framework. ...

⋮

AI response 3

...Stick with SR + energy (or energy+variance) VMC. ...

Artificial-intelligence systems were therefore used for discussion, planning, and review, rather than as a source of results. They helped turn observations into hypotheses that could be checked, suggested alternative diagnostics when a run failed, and reviewed the final prose for clarity and mistakes. The code, the experiments, the scientific judgments, and the text of this thesis were written by the author. In particular, the recommendation shown above was not accepted as a methodological conclusion: it became one claim among several to investigate. The work that followed led to the collocation analysis reported in this thesis.

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